! ACK-212
No leakage for cell 'rf1'.

DESCRIPTION
Leakage not present for the cell.

SEVERITY & IMPACT
Major warning. The leakage wouldn't be used in analysis.

ACTIONS
Please verify the APL characterization to identify what went wrong.

! ACK-213
Only C0 and R0 for cell '<name>'.

DESCRIPTION
Only the ESR/ESC values for certain state is present for the cell.Rest are missing.

SEVERITY & IMPACT
Major warning. The missing values won't be used in analysis.

ACTIONS
Please verify the APL characterization to identify what went wrong.

! ACK-229
No header in file '.apache/cell.cdev'

DESCRIPTION
The header for the APL output file for device capacitance
consisting of information regarding the data version, tool name,
apl version, released date, signature etc. is missing.

SEVERITY & IMPACT
Warning generated during setup design. The APL files will
not be read and this will impact dynamic analysis results.

ACTIONS
Provide the apl file with proper header format.

! ANA-W01
No region specified. Dump the current for entire chip.

DESCRIPTION
Whenever user plots the current using the "-region" option, he is supposed

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to specify the lower left and upper right coordinates of the region. Else, tool will issue out this warning and generate the current waveform for the entire chip.

SEVERITY AND IMPACT
Result interpretation will be wrong if the current waveform is for fullchip and user assumes it for a region.

ACTION
While using plot current -region, specify the lower left and upper right coordinates as follows:
plot current -region <llx> <lly> <urx> <ury>

! ANL-008
Dynamic analysis in the 1-th run ended unsuccessfully.

DESCRIPTION
The dynamic simulation failed due to a variety of reasons. The actual problem is reported as SIM-XXX, before ANL-008.

SEVERITY & IMPACT
Critical error. Dynamic results are not generated.

ACTIONS
Please fix the error messages reported by the simulator (SIM-XXX).

! ANL-012
Simulation time <sim_time_value>(ps) is large, please expect long CPU time.

DESCRIPTION
This message warns the user about the dynamic simulation time which is set in the gsr. If it is too high the run time will increase accordingly. Ideally, for Dynamic drop analysis, user needs to simulate only for 1 or 2 cycles.

SEVERITY & IMPACT
Critical message as this setting can increase the run time of RH drastically.

ACTIONS
Check gsr DYNAMIC_SIMULATION_TIME and decrease it to the required value

----
! ANL-032
Instance <inst> is not controlled by a power gate but specified to be POWER-UP through GSC.

DESCRIPTION
GSC file specifies the status of any instances or blocks in the design where power switching occurs. The mentioned instance is in the always on domain. It is not controlled by any switches.

SEVERITY & IMPACT
Since the instance is in the always on domain the status mentioned in the GSC file for the instance will be ignored .

ACTIONS
Check whether the instance is supposed to be controlled by switches,
If not remove it from GSC.

! ANL-200
Invalid voltage is encountered and results may be unreliable

DESCRIPTION
RedHawk encounters an invalid voltage during simulation making the simulation results unreliabe .

SEVERITY & IMPACT
Simulation results are unreliable

ACTIONS
Please check the disk space, if there is enough disk space then report the error to Apache

----
! ANL-303
Instance <inst_name> is not a switch but specified to be ON through GSC.

DESCRIPTION
The ON keyword in GSC file is supposed to be used only for switches, so that the intances connected through the switch will be powered up. The status of the instances which are not specified in the GSC file will be determined by Redhawk according to the vectorless algorithm

SEVERITY & IMPACT
It is not a critical message, it warns the user to check whether you specified instances instead of switches to be in ON state

ACTIONS
Remove the instances in GSC file which are specified to be in ON state.
Instead of ON, user can give HIGH, LOW,STANDBY or DISABLE

! ANL-430
Frequency for the instance is over set limit 5e+10(Hz)

DESCRIPTION
RedHawk errors out if the Frequecy of the any instance is above the set limit of 5e+10(Hz) .

SEVERITY & IMPACT

Corresponding instances will not be accounted for switching and simlation results are unreliable

ACTIONS
Please make sure the frequency specified for the instance is correct in the timing file and within the limits

----
! APL-002
Slew normalization setting mis-match in APL current file

DESCRIPTION
APL data characterized using latest version of RedHawk contains slew values normalized from 0 to 100%. If you have performed the characterization using some very old version of RedHawk, you will see slews which are not normalized to 0-100% range. In such cases, RedHawk will interpret the slew values wrongly in the analysis.

SEVERITY & IMPACT
This will have impact on accuracy of dynamic analysis.

ACTIONS
Specify "SLEW_NORMALIZATION 0" in GSR and rerun the analysis

! APL-005
APL interpolation failed for xxx.

DESCRIPTION
Some APL interpolation failed common reasons are:

1) Incorrect DD sample file .apache/apache.apl;
2) Incorrect DI data 'xxxx' - run aplchk on 'xxxx';
3) Insufficient DI samples in data 'xxxx' - need to re-run apl;

For more detail, please refer to interpolation log file 'adsRpt/Log/aplint.log'.

SEVERITY & IMPACT
Error, It will cause import APL failure.

ACTIONS
Fix the error

! APL-006
APL merge failed on 'xxxx'.

DESCRIPTION
Some APL merge failed common reasons are:

1) Incorrect interpolated apl data;

2) Inconsistent apl data (e.g. different process corner);
For more detail, please refer to merge log file
'adsRpt/Log/aplmerge.log'.

SEVERITY & IMPACT
Error, It will cause import APL failure.

ACTIONS
Fix the error

! AVM-208
High Leakage_I value- 'leakage_value' A in cell 'cell_name'

DESCRIPTION
This warns the user to check the leakage value given in the avm config file,
since the value given is too high

SEVERITY & IMPACT
Its not a critical message, it is a warning to check the leakge value
specified is right or wrong

ACTIONS
Check the leakage value and correct if it is needed

! AVM-211
Cannot find required 'PROCESS' keyword in the config file.

DESCRIPTION
To comply with the MSDF flow, each characterized <cell>.current/cdev file
needs to provide the process corner name for RedHawk simulation. Although
this information is usually available from the Spice device library (corner
name), the meaning of the library names may not be clear to the timing tool,
so this specification is required .

SEVERITY & IMPACT
It is important to give PROCESS keyword in avm config file.If the PROCESS
corner is not specified it will not be present in the header section of
the <cell>.current/cdev file .

ACTIONS
Include PROCESS keyword in avm config file

#----------
! AVM-361
'Cpd_standby' value higher than 'Cpd_read' value

DESCRIPTION
'Cpd_standby' value is higher than 'Cpd_read' value for the corresponding
cell wherein ideally 'Cpd_standby' should be less than 'Cpd_read'

SEVERITY & IMPACT
Current profile taken for the cell might not be accurate thus impacting the results

ACTIONS
Please make sure all the inputs used in the AVM config file have correct values

----
! BPA-005
No connection point for 'gport1 VSS', 0.166667A current not assigned.

DESCRIPTION
Redhawk was not able to find nodes on specified region/block and layer to assign specified current.

SEVERITY & IMPACT
Specified current would be lost and analysis results would be optimistic.

ACTIONS
Please check if given block/region and layer has vias on upper or lower layer so that nodes are created to assign power. OR in order to force redhawk to create nodes and distribute power uniformly irrespective of via existence, use GSR keyword BPA_CURRENT_DENSITY and rerun.

! BPA-006
Power assignment (xyz W) too small for '<region/block name> <net>', need to be abc W or greater.

DESCRIPTION
Power assigned to given region/block in BLOCK_POWER_ASSIGNMENT construct is smaller than the sum of power assigned to sub-regions/sub-blocks under it.

SEVERITY & IMPACT
In hierarchical power assignment, redhawk subtracts sum of sub-block/region power from power assigned to block/region and assigns remaining power. If block/region power is less than sum of sub-blk/sub-region power, redhawk will not assign any power to the block/region.

ACTIONS
Power specified for region/block should be greater than sum total power for its sub-block/region.

! BPA-007
Current assignment (xyz A) too small for '<region/block name> <net>', need to be abc A or greater.

DESCRIPTION

Current assigned to given region/block in BLOCK_POWER_ASSIGNMENT construct is smaller than the sum of current assigned to sub-regions/sub-blocks under it.

SEVERITY & IMPACT

In hierarchical power assignment, redhawk subtracts sum of sub-block/region power from power assigned to block/region and assigns remaining power. If block/region power is less than sum of sub-blk/sub-region power, redhawk will not assign any power to the block/region.

ACTIONS

Current specified for region/block should be greater than sum total current for its sub-block/region.

! BPA-105

Incorrect power assignment due to overlap of BPA regions and instances

DESCRIPTION

RedHawk expects the BPA regions to have zero overlap with other regions and instances. If there is any overlap between such regions then this may lead to double counting of power for regions with overlaps .Please check whether there is any overlap between different regions by looking at the co-ordinates of their bounding boxes .

SEVERITY & IMPACT

RedHawk may miscalculate higher power for the overlapping regions .

ACTIONS

Correct the co-ordinates of the bounding boxes of BPA region

! CCL-004

Cannot add net<> to cell<>

DESCRIPTION

This error message reports a failure of adding the net to the cell. One possibility is the specified net already exists in the cell and the duplication caused the confusion.

SEVERITY & IMPACT

Error. The net and its associated attributes may get lost.

ACTIONS

Users should remove the duplication if exists.

! CCL-014

Mismatched power (value vs. value) in <file>:<line>

DESCRIPTION

RedHawk performs internal data integrity checking againts the specified power file vs. RedHawk database. This message reports some inconsistency

in between.

SEVERITY & IMPACT
This warning message indicates the power recognized by RedHawk will be different from the power reported by the power file. If the differences are small, users may ignore it.

ACTIONS
If the power file was prepared by users, either manually or by scripts, users need to fix the power file. If the power file was created by RedHawk's power calculation tool, this warning message very likely indicate some bug in the code. Please submit the data to Apache Design Solutions for further investigation.

! CCL-015
Flattened inst<> should not consume power <file>:<line>

DESCRIPTION
Any flattened instances should not consume any power. Instead, the power must be distributed to its components. This message indicates a flattened instance still takes power.

SEVERITY & IMPACT
The power associated with the flattened instance will be ignored.

ACTIONS
Users need to submit the data to Apache Design Solutions for detailed investigation.

! CCL-016
inst<> in power file<> does not exist.

DESCRIPTION
The instance specified in the power file cannot be found in the design. This most likely happens when importing an old power file which does not match the new design.

SEVERITY & IMPACT
The associated power will be ignored due to the unfound instance. This warning could be critical if the instance is a memory or IP which consumes significant amount of power.

ACTIONS
Users may want to use RedHawk to recalculate the power using the new design. If the warning persists, please submit the data to Apache Design Solutions for further investigation.

! CCL-017
Some of the instance P/G pins present in imported power database does

not exist in the design

DESCRIPTION
When the adsPower directory from another RedHawk run is imported in the analysis, RedHawk will try to annotate the power into the design elements. This error is issued when power/ground pin specified for the instance does not exist in the design. This can happen when there is a mis-match in the design database between the runs where the power database is created and the present run.

SEVERITY & IMPACT
Power for these instances will not be used in the analysis

ACTIONS
Resolve the mismatch and rerun the analysis

! CCL-028
There are # instances in design not connnected to VDD and GND network.

DESCRIPTION
These instance are not connected to P/G network. It could be their pins do not have correct layout geometry or there are routes missed in P/G network to connect their pins.

SEVERITY & IMPACT
WARNING. These instances won't be analyzed by RH. All the power and capacitance associated with these instances will be drop in RH simualtion.

ACTIONS
Check the layout by highlighting disconnected wire pull down menu to find what metal or via missed, and check the layout geometries of instances' pins, then fix the DEF/LEF files.

! CCL-030
There are # instances in design only connnected to VDD but not GND network.

DESCRIPTION
These instance are only connected to Power network. They are not connected to Ground network. It could be that their pins do not have correct layout geometry or there are routes missed in Ground network to connect their pins.

SEVERITY & IMPACT
WARNING. These instances won't be analyzed in RH dynamic analysis and ground network's static analysis. All the power and capacitance associated with these instances will be droped in RH dynamic simualtion and ground network's static analysis. No impact to power network's static analysis.

ACTIONS
Check the layout by highlighting disconnected wire pull down menu to find

what metal or via missed, and check the layout geometries of instances' pins, then fix the DEF/LEF files.

! CCL-032
There are # instances in design only connnected to GND but not VDD network.

DESCRIPTION
These instance are only connected to Ground network. They are not connected to power network. It could be that their pins do not have correct layout geometry or there are routes missed in Power network to connect their pins.

SEVERITY & IMPACT
WARNING. These instances won't be analyzed in RH dynamic analysis and power network's static analysis. All the power and capacitance associated with these instances will be droped in RH dynamic simualtion and power network's static analysis. No impact to ground network's static analysis.

ACTIONS
Check the layout by highlighting disconnected wire pull down menu to find what metal or via missed, and check the layout geometries of instances' pins, then fix the DEF/LEF files.

! CCL-034
#instances in design are missing timing info

DESCRIPTION
This message reports the number of instances belonging to the analyzed P/G network that
1) have no timing window (zero frequency), and
2) positive static power consumption.

These inactive instances are listed in adsRpt/<design>.noTiming.

SEVERITY & IMPACT
The consequence of this error is that these instances will not switch during dynamic simulation and so the associated power will be dropped. The severity of this error depends on the scenarios of these instances - whether they should switch, whether they are high-powered instances, and whether they are clustered together so their inactiveness would have accumulated impact to the final dynamic voltage drops.

ACTIONS
Users should verify the completion of their STA files that these instances indeed do not switch. If wrong, the STA files must be fixed before users can proceed to check the dynamic results.

! CCL-054
.ploc : Pad Pin <name> (type:P|G) at (x y) is short with (type:P|G) wires.

DESCRIPTION
The pad pin reported is of type VDD and is detected overlap in the same layer with a wire of type GND, vice versa.

SEVERITY & IMPACT
Error. Power and ground net short usually is serious problem unless one of the net is a dummy net (a net with no driver).

ACTIONS
A user should check the .ploc file to make sure the reported pad's location is correct. Or, check the DEF file to correct the wires around the reported pad location.

! CCL-056
.ploc : Pad pin <name> (type:P|G) at [x,y] is floating

DESCRIPTION
A pad pin's port rectangles do not touch any wires on any layer.

SEVERITY & IMPACT
Error. The net logically connected to the pad may be without driver due to this floating pad.

ACTIONS
A user should check the .ploc file to make sure the reported pad's location is correct. Or, check the DEF file to make sure there exist wires connecting to the reported pad.

! CCL-060
File `.ploc' : Pad Pin <name> (type:P|G) at (x y) is short with (net:name) wires!

DESCRIPTION
The pad pin reported is of type VDD and is found to overlap a net in the same layer but with different type, vice versa.

SEVERITY & IMPACT
Error. Power and ground net short usually is serious problem unless one of the net is a dummy net (a net with no driver).

ACTIONS
Check the .ploc file to make sure the reported pad's location is correct. Or, check the DEF file to correct the wires around the reported pad location.

! CCL-064
Instance is connected to multiple VDD domains.

DESCRIPTION
This message reports the instances that are connected to multiple VDD

domains. This is not handled in RedHawk 5.1 but will be supported in 5.2.

SEVERITY & IMPACT

Warning. The biggest challenge in handling the instances that belong to multiple VDD domains is how to correctly distribute the instance power to each domain. RedHawk 5.2 will get this info from Synopsys liberty files if available. Otherwise, RedHawk will do the following (as 5.1).

(1) For static analysis: The instance power is evenly distributed to all the domains specified in GSR (VDD_NETS). If only one domain is specified in GSR, all the power will be assigned to that domain.

(2) For dynamic analysis: The instance power is assigned to the primary VDD domain specified in GSR. For example, given an instance which is connected to VDD2 and VDD5, assuming VDD2 is ranked 2nd and VDD5 5th in VDD_NETS, the primary domain of the instance is VDD2 and all the instance power will be assigned to VDD2.

ACTIONS

Users need to be aware of the implication of the results. Also, if desired, users may manipulate VDD_NETS in GSR to control the power assignments. In order to use the new multiple VDD suport in RedHawk 5.2, users' .lib must have multi-rail info.

! CCL-065

Pad Pin<name> of instance<name> (type:P|G) at (x y) does not touch any wire of net<name>!

DESCRIPTION

A pad instance's P|G pin is not physically connected to the net<name>.

SEVERITY & IMPACT

Error unless this pin is not used but the connectivity information is not available at thsi stage.

ACTIONS

Check the DEF file to make sure the pad instance's location is correct.

! CCL-066

Pad Pin<name> of instance<name> (type:P|G) at (x y) is short with (type:P|G) wires!

DESCRIPTION

A pad instance's VDD pin is shorted with a GND wire, vice versa.

SEVERITY & IMPACT

Error unless this pad insatnce's pin is not used or the shorted wire is a dangling wire.

ACTIONS

Check the DEF file to make sure the pad instance's location is correct.

! CCL-071
Inconsistent inst <inst_129538/adsU1> property in field<32> line<30325>
file<adsPower/GENERIC.nets.adsLib.power>.

DESCRIPTION
It indicates inconsistency in the imported power file.

SEVERITY AND IMPACT
Power values can get affected if there is a major inconsistency

ACTION
Check the imported pwoer file for inconsistent data.

! CCL-075
Inconsistent data (3968 vs 3972) in file <file_name>

DESCRIPTION
This message is issued anytime the tool finds an inconsistency while writing
or reading a file. This could be caused by a network or disk access error

SEVERITY & IMPACT
It is a critical message and results may get affected

ACTIONS
The user should rerun the tool if this error appears, after making sure that
there is sufficient disk space available and no network issues exist

----
! CCL-080
Pad Pin<name> of instance<name> (type) at (x y) shorted with wires (type:P|G).

DESCRIPTION
This warning appears if a signal pin is physically shorted with power/ground
nets at the location indicated.

SEVERITY & IMPACT
Minor warning.Correct the short in the P&R tool for more accurate analysis.

ACTIONS
Please correct the short for accurate analysis.

! CCL-112
The power file is not multi-rail ready!

DESCRIPTION
Power file that is input into RedHawk is not multi-rail ready as it is
created with an older RH version which do not support multi-rail cell

SEVERITY & IMPACT
Power calculated might be inaccurate and the results are impacted

ACTIONS
Please create the power file with the latest version and redo the anlysis.

! CDV-416
Cell pin <cell123:vdd2> in file <cell.cdev> doesn't exist.

DESCRIPTION
The LEF for the cell <cell123> does not have a Power/Ground pin named "vdd2", but the APL cdev for the cell has that pin.

SEVERITY & IMPACT
The APL cdev data for that pin "vdd2" will be ignored and not used in dynamic simulation.

ACTIONS
Please make sure Power/Ground pins in APL are consistent with Power/Ground pins in LEF. If spice PG pins are not consistent with PG pins in LEF, use SPICE2LEF_PIN_MAPPING in APL config file during APL characterization.

! CEL-000
Cannot add vdd/vss pin due to name conflict

DESCRIPTION
RedHawk is not able to add a ploc since its name conflicts with other ploc name or the net name

SEVERITY & IMPACT
ploc will not be created for such pins and the results are impacted

ACTIONS
Add pin names which are unique and different from the net/domain names and rerun the analysis

----
! CIS-004
Instance not connected to any power/ground rail

DESCRIPTION
You will get this warning when a particular instance is not connected to any power or ground rail. This can happen due to many reasons, such as due to issues with top level connecting nets to the instance or pin definition of the instance itself. These instances should figure in the adsRpt/*unconnect reports.

SEVERITY & IMPACT
Warning. Since they are not connected, they will not be part of analysis

ACTIONS
Review the connectivity of the instance and do the necessary modifications.
Check in GUI if top level power rails are overlapping correctly with
instance's power pins

! CIS-005
Instance is not drawing power from top level power/ground nets

DESCRIPTION
You will get this warning when a particular instance is not drawing power
from top level power/ground nets. This can happen if the instance is not
connected properly. Connectivity issues can be due to many reasons, such
as due to issues with top level connecting nets to the instance or pin
definition of the instance itself. If CIS-004 warnings are issued for these
instances, fix them first.

SEVERITY & IMPACT
Warning.

ACTIONS
Review the connectivity of the instance and do the necessary modifications.
Check in GUI if top level power rails are overlapping correctly with
instance's power pins

! CLK-451
ClockRoot specified in GSR not found in the design.

DESCRIPTION
The design object specified in ClockRoot does not exist in the design.
The possible causes include name case mismatch, incorrect/missing
hierarchical path, hierarchical divider or pin/bus delimiters mismatch,
incomplete DEF files, or simply a typo.

SEVERITY & IMPACT
Important to fix or this clock root will be ignored. The direct impact
is to a wrong clock network extraction and frequency propagation.

ACTIONS
Do the following checking:
(1) any typo?
(2) the specified design object does exist in DEF
(3) the name case is consistent to DEF. Also, check if
NAME_CASE_SENSITIVITY is set to 0 in GSR. In this case, all design
object names are remapped to uppercase.
(4) correct hierarchical path name with consistent hierarchical divider.

(5) consistent pin delimiter and bus delimiter, if the clock root is
a pin or pinInst.

! CLK-452
ClockRoot in GSR is not connected to a net.

DESCRIPTION
The clock root is dangling and no tracing will be performed from this
clock root.

SEVERITY & IMPACT
Warning. It is okay if the clock root is expected to be dangling.
Otherwise this warning is critical and possibly indicates the DEF files
are not complete.

ACTIONS
Users need to verify the clock root is indeed dangling or fix the
incomplete DEF files.

! CLK-803
ClockRoot in GSR cannot be found in the design.

DESCRIPTION
This message indicate the clock root specified in GSR does not exist
in the design. This could be due to name mapping issues or simply a typo
when preparing the GSR file. Another possibility is due to missing DEFs
when importing designs if the clock root belongs to a sub-block.

SEVERITY & IMPACT
This clock root will be ignored.

ACTIONS
Users need to verify the clock root is correctly specified in GSR and
the DEFs do contain this clock root.

! CLK-804
ClockRoot in GSR is not connected to a net.

DESCRIPTION
The identified clock root is dangling and not driving a clock network.

SEVERITY & IMPACT
This clock root is ignored.

ACTIONS
Users need to verify the dangling clock root is expected.

! CLK-812

Cell not defined in .lib

DESCRIPTION

You will get this warning when a particular cell present in the design is not defined in .lib files. Redhawk uses timing arcs present in .lib for tracing paths. It tries to trace the clock tree from the clock root defined in gsr file. However if it comes across a cell which is not defined in .lib, it is not able to trace forward and gives this warning. Redhawk expects all cells in the clock path to be defined in the supplied .libs

SEVERITY & IMPACT

Warning. It blocks forward clock tracing from such cells.

ACTIONS

Supply the .lib with the particular cell defined in it.

! CLK-821

STA file is not available for the design.

DESCRIPTION

The power for each net is calculated from the transition times ("slew") of each instance's input/output pins, using the clock frequency domain for all the instances that are specified in the STA output file.If this file is not available then the GSR keyword CLOCK_ROOTS traces the nets from a specified clock root and finds the respective clock domain for all the nets and instances.Specifies the names of the clock roots(either net name or pin name) and the frequency.

SEVERITY & IMPACT

Warning generated during setup design. With neither STA and CLOCK_ROOTS option given, the slew will be the default slew and frequency will be the default GSR frequency. Also TW will be assumed 0 for all instances.Hence, the results will be inaccurate. More important with respect to dynamic analysis.

ACTIONS

Include the keyword CLOCK_ROOTS in the GSR file:

Syntax:

```
CLOCK_ROOTS
{
   <clock_root_name> <freq in Hz>
   ...
}
```

Or else provide the STA file.

! CMM-109

Issues seen due to multiple VCD imports from CMM_CELLS and also through GSR:VCD_FILE

DESCRIPTION
RedHawk is finding issues due to multiple imports of VCD, one from CMM models and other through GSR:VCD_FILE which is not supported.

SEVERITY & IMPACT
Important error, VCD is not properly imported and the QOR will be impacted

ACTIONS
Input the VCD either only through CMM or GSR:VCD_FILE but not both and rerun the analysis

----
! CON-105
Pin of cell is connected to the PG network but not used as input pad.

DESCRIPTION
Pin declared in DEF/LEF is connected to the PG network, but it is not declared to be used as input pad in RH.

SEVERITY & IMPACT
WARNING. Pin won't be served as input pad, which will affect simulation result.

ACTIONS
Check design to see if this is what you want.
See manual to use this pin as input pad.

! CON-106
P/G resistor network is not fully connected to input pad.

DESCRIPTION
Part of the extracted resistor network is not connected to input pad. This may result from part of the routes of the net is not physically connected to input pads.

SEVERITY & IMPACT
WARNING. Isolated parts are droped in RH simulation. Final result is affected.

ACTIONS
Check the layout by highlighting disconnected wire pull down menu to find what metal or via missed, then fix the DEF files.

! CON-109
Net1 is shorted to net2 on layer at location.

DESCRIPTION

This error reports the short between two nets. The exact short location
and layer are also reported so users can use GUI to verify the short.

SEVERITY & IMPACT
RedHawk honors the short and will merge these two nets together for
analysis. If net1 is VDD and net2 is GND, the final worst VDD drop
and GND bounce could be severe.

ACTIONS
Users need to verify and fix all the shorts in the design. If the shorts
are expected and users want to proceed the analysis by ignoring the shorts,
`IGNORE_SHORT 1` can be set in GSR and CON-110 will be issued as a warning
instead and the shorts will be separated during analysis.

! CON-110
Ignoring short between net1 and net2 on layer at location.

DESCRIPTION
This error reports the short between two nets. The exact short location
and layer are also reported so users can use GUI to verify the short.
Different from CON-109, the short is ignored during analysis.

SEVERITY & IMPACT
RedHawk recognizes the short but will ignore the short during analysis
(net1 and net2 are treated separately). The impact from the short will
not be reflected in the final analysis.

ACTIONS
Users need to verify all the shorts in the design and also ensure this
behavior (setting `IGNORE_SHORT 1` in GSR) is desired.

! CON-111
A short between nets <net1_name> and <net2_name> is detected on layer
<layer_name> at ((x1,y1) (x2,y2)) by wire and wire.

DESCRIPTION
A short exists between two different nets.

SEVERITY & IMPACT
Warning.Since <net2_name> is not extracted,the R network of <net1_name>
is not effected. Redhawk ignores the shorts and proceeds with the analysis.

ACTIONS
Fix the design.
#----------
! DEF-005
Syntax error during DEF parsing.

DESCRIPTION
Non-fatal syntax error in DEF input file during the DEF parsing stage.

SEVERITY & IMPACT
Redhawk continues the analysis after giving out the DEF syntax warning

ACTIONS
Please correct the DEF syntax and rerun analysis

! DEF-016
Instance does not exist in cell.

DESCRIPTION
The specified instance doesn't exist in the DEF file that is currently being processed.

SEVERITY & IMPACT
Warning. Error in net connectivity.

ACTIONS
Users should fix the COMPONENTS section in the DEF file.

! DEF-019
Un-defined via model in DEF File

DESCRIPTION
VIA models present inside the geometry section of the DEF file should be pre-defined in the technology LEF or the VIA section in the def.

SEVERITY & IMPACT
RedHawk will ignore these vias in the analysis

ACTIONS
Include the via model in technology LEF file or VIA section in the DEF and rerun the analysis

----
! DEF-022
Missing cut layer definition for the specified via.

DESCRIPTION
The cut layer for the specified via is missing.

SEVERITY & IMPACT
Error.

ACTIONS
Please add cut layer definition.

! DEF-023

No NETS section in this DEF file?

DESCRIPTION
There are no nets defined in the DEF file that is currently being processed.

SEVERITY & IMPACT
Warning.

ACTIONS
Users should verify whether this is a false alarm.

! DEF-024
No block power pin and no block ground pin defined in the PINS section.

DESCRIPTION
There are no power pins and no ground pins defined in the PINS section of the DEF file that is currently being processed.

SEVERITY & IMPACT
Fatal. Block without power and ground pins will cause RedHawk to exit. It is important for blocks to have power and ground pins defined to ensure correct logical connections. If it is expected to have blocks without power and ground pins, add "IGNORE_DEF_ERROR 1" to the gsr file to allow RedHawk to proceed.

ACTIONS
Users should verify the use of the pins in PINS section.

! DEF-025
No block ground pins defined in the PINS section.

DESCRIPTION
There are no ground pins defined in the PINS section of the DEF file that is currently being processed.

SEVERITY & IMPACT
May not be critical for the top block. Important for sub-blocks to ensure correct logical connection.

ACTIONS
Users should verify the use of the pins in PINS section.

! DEF-026
Design's TOP DIEAREA is not defined. Computed as (x1,y1 x2,y2).

DESCRIPTION
This Warning comes when you dont define the TOP DIEAREA in the def file.

SEVERITY & IMPACT
Its not a critical warning. As long as the RH calculated value matches with the actual value, you can ignore this.

ACTIONS
You ignore this as long as the RH calculated value matches with the design's area. Otherwise, specify the DIEAREA keyword in the top def and avoid this warning message

----
! DEF-027
No cell instances defined in the COMPONENT section.

DESCRIPTION
There are no components defined in the DEF file that is currently being processed.

SEVERITY & IMPACT
Warning.

ACTIONS
Users should verify whether this is a false alarm.

! DEF-028
LEF and DEF pin mismatch.

DESCRIPTION
Failed to connect the specified pin to the specified net because the specified instance and its master LEF cell don't have the pin.

SEVERITY & IMPACT
Error. The specified DEF instance may not be completely hooked up to the specified net.

ACTIONS
Users should verify the mismatch between the specified LEF cells and DEF instances.

! DEF-030
Ignoring instance <name> which has no placement location.
DESCRIPTION
The instance would be skipped in RH analysis as it is not placed in DEF.

SEVERITY & IMPACT
RedHawk will ignore these instances in the analysis. Minor warning, provided the unplaced instance doesn't need to be included in analysis

ACTIONS
Please ensure that the instance is not important w.r.t to analysis.

! DEF-031
Numbers of DEF nets with types assigned based on their
names: PWR(0) GND(0) CLK(287)

DESCRIPTION
Redhawk by default assumes the type of DEF nets based on the name of the nets.
If a name includes strings like 'vdd', 'VDD', 'VCC', 'vcc', 'gnd', 'GND', 'vss',
'VSS' etc, Redhawk treats them as P/G nets no matter what the type is specified
in DEF. This can be disabled with the GSR keyword PGNET_HONOR_DEF_TYPE set to 1.
The default value of the keyword is 0.

SEVERITY & IMPACT
Redhawk will by default assume the net type from net name and give out the warning.

ACTIONS
If there are net names in DEF which have names different from their type,
example CLK1 which is not clock net, then set the gsr keyword
PGNET_HONOR_DEF_TYPE to 1 and run the analysis.

! DEF-036
Design top level has no power pins. Power pin locations will be
imported from .ploc file.

DESCRIPTION
Design top level power pins will be imported from .ploc file.

SEVERITY & IMPACT
Information message.

ACTIONS

! DEF-037
Design top level has no ground pins. Ground pin locations will be
imported from .ploc file.

DESCRIPTION
Design top level ground pins will be imported from .ploc file.

SEVERITY & IMPACT
Information message.

ACTIONS

! DEF-038
The specified pin doesn't exist in the specified cell instance.

DESCRIPTION
The specified pin may not be defined in the instance master cell.

SEVERITY & IMPACT
Warning.

ACTIONS
Fix the discrepancy in LEF and DEF pins.

! DEF-039
DEF file being imported more than once.

DESCRIPTION
Warning message for duplicated DEF files in .defs file.

SEVERITY & IMPACT
Warning. Duplicated DEF files are ignored.

ACTIONS
Remove duplicated DEF files from .def file.

! DEF-040
The number of vias, pins, components, special nets, and nets declared is inconsistent with the number of vias, pins, components, special nets, and nets defined in DEF file, respectively.

DESCRIPTION
The number of vias, pins, components, special nets, and nets specified in VIAS, PINS, COMPONENTS, SPECIALNETS, and NETS sections is not the same as the number of vias, pins, components, special nets, and nets defined in DEF file, respectively.

SEVERITY & IMPACT
Warning.

ACTIONS
Fix DEF data.

! DEF-052
Decap cell cannot be a block but it must be a leaf cell

DESCRIPTION
RedHawk is seeing that the decap cell provided under DECAP_CELLS keyword is not a leaf cell but a block name

SEVERITY & IMPACT
Decap block will be ignored in the analysis.

ACTIONS

Please provide the corresponding decap leaf cells under DECAP_CELLS
keyword in GSR and rerun the analysis

! DEF-053
Components in the DEF are lying outside the die area

DESCRIPTION
If any instance lies outside the die area , this warning will be flagged.

SEVERITY & IMPACT
This is a warning.The instance will be placed at the given coordinates.
The placement location may be wrong.

ACTIONS
Check your DEF for this mismatch

! DEF-104
BUSBITCHARS is not defined in DEF file.

DESCRIPTION
BUSBITCHARS specifies the characters to enclose bus bits. If BUSBITCHARS
is not defined, "[]" are the default special characters for bus bits.

SEVERITY & IMPACT
Warning. If BUSBITCHARS is not defined and bus bits are not enclosed
by "[]", bus names may be handled incorrectly.

ACTIONS
Define BUSBITCHARS in DEF files.

! DEF-105
DIVIDERCHAR is not defined in DEF file.

DESCRIPTION
DIVIDERCHAR specifies the character to express hierarchy.
If DIVIDERCHAR is not defined, "/" is the default hierarchical divider.

SEVERITY & IMPACT
Warning. If DIVIDERCHAR is not defined and hierarchical divider is
different from the default, DEF names may be handled incorrectly.

ACTIONS
Define DIVIDERCHAR in DEF files.

! DEF-107
Ignore invalid ViaModel '<name>' due to 0-sized rects on '' layer!

DESCRIPTION

The viamodel will be skipped due to null-sized rectangular geometries.

SEVERITY & IMPACT
RedHawk will skip the specified viamodels during analysis.

ACTIONS
Please correct the viaModel definition in DEF.

! DEF-108
No SITE <unit> in LEF (or gsr FAO_ROW_SITE) for DEF/ROW creation

DESCRIPTION
FAO_ROW_SITE specifies a particular set of “sites” (a placement grid)by its name in DEF, to avoid conflicting and overlapping decap insertions from multiple library row definitions.LEF “sites”, also called “unit-tiles”, are defined in LEF and used in the DEF/ROW section to create “row boxes” to allow P/R tools to create multiple clusters using different cell grid libraries, and thus may create overlapping ROW boxes. By default,without using FAO_ROW_SITE, FAO takes all ROW grids defined in DEF to create placeable decap areas.
Optional; default: all defined DEF row grids.

Example:
For a DEF file ROW section containing several row definitions as follows:
ROW CORE_ROW_37549 bonuscoreaxg 3547600 2680600 N DO 7793 BY 1 STEP 690 0;
ROW CORE_ROW_52855 core 8927530 2680600 N DO 2054 BY 1 STEP 230 0 ;
Using ‘FAO_ROW_SITE core’, only rows defined by the ‘core’ site name are created for FAO decap placement.This warning is generated when no SITE isspecified in the LEF (or gsr FAO_ROW_SITE) keyword is not used.

SEVERITY & IMPACT
Warning.FAO won't consider row!

ACTIONS
Define SITE in LEF.
Syntax:
```
Site
SITE siteName
CLASS {PAD | CORE} ;
[SYMMETRY {X | Y | R90} ... ;]
SIZE width BY height ;
END siteName
```
And then define the following in GSR:
Syntax:
```
FAO_ROW_SITE {
  <LEF_site_name>
  ...
}
```

! DEF-109
Def viaModel<name> - already defined! Ignore it

DESCRIPTION
The mentioned via model is defined more than once in the DEF.

SEVERITY & IMPACT
When the via model is defined more than once first definition
will be taken and the others will be ignored.

ACTIONS
Remove duplicated via model definitions to avoid this warning

----
! DSG-131
VDD_NET in GSR cannot be found in design.

DESCRIPTION
This message indicate the vdd net specified in GSR does not exist
in the design. This is usually due to name mapping issues or simply a typo
when preparing the GSR file. Another possibility is due to incomplete DEF
files.

SEVERITY & IMPACT
This vdd net will be ignored from extraction and analysis.

ACTIONS
Users need to verify the vdd net is correctly specified in GSR and also
the DEFs are complete.

! DSG-133
GND_NET in GSR cannot be found in design.

DESCRIPTION
This message indicate the gnd net specified in GSR does not exist
in the design. This is usually due to name mapping issues or simply a typo
when preparing the GSR file. Another possibility is due to incomplete DEF
files.

SEVERITY & IMPACT
This gnd net will be ignored from extraction and analysis.

ACTIONS
Users need to verify the gnd net is correctly specified in GSR and also
the DEFs are complete.

! DSG-185
Connectivity issues identified in package netlist hookup

DESCRIPTION
RedHawk is finding issues in hookup between package and die where the package node which is associated with a particular domain of certain voltage is connected to a different volatge domain in the package.
Please look into adsRpt/pkg_check.rpt for details.

SEVERITY & IMPACT
QoR will be impacted due to connectivity issues between package and die

ACTIONS
Resolve the connectivity issues and rerun the analysis

! DSG-186
Pkg node xyz disconnected to all P/G domains.

DESCRIPTION
RedHawk found that that the given package node is not electrically connected to voltage source/battery through package RLCK network.
Please look into adsRpt/pkg_check.rpt for details.

SEVERITY & IMPACT
QoR will be impacted due to connectivity issues between package and die

ACTIONS
Resolve the connectivity issues and rerun the analysis

! DSG-187
Connectivity issues identified in package netlist hookup

DESCRIPTION
RedHawk is finding issues in hookup between package and die.
Please look into adsRpt/pkg_check.rpt for details.

SEVERITY & IMPACT
QoR will be impacted due to connectivity issues between package and die

ACTIONS
Resolve the connectivity issues and rerun the analysis

! DSG-188
Connectivity issues identified in package netlist hookup

DESCRIPTION
RedHawk is finding issues in hookup between package and die where same package node is connected to mulitple domains.
Please look into adsRpt/pkg_check.rpt for details.

SEVERITY & IMPACT
QoR will be impacted due to connectivity issues between package and die

ACTIONS
Resolve the connectivity issues and rerun the analysis

! ECO-000
Adding eco_via1 at (x, y) will short two different nets

DESCRIPTION
Adding a via at specified location will cause short between two power/ground nets.

SEVERITY & IMPACT
Redhawk will not be able to add the via or account for it in analysis.

ACTIONS
Do not add via at specified location. Check in layout why adding a via at specified location causes short.

! FAO-134
[Missing Via Report] Ignore GDS block <name>

DESCRIPTION
The GDS blocks are ignored by default for missing vias check by Redhawk. In order to get the missing via report for GDS blocks, type the following command :

```
mesh vias -toplayer <top_layer_name> -bottomlayer <bottom_layer_name> -gds
```

To exclude specific instances from missing vias report, the following gsr setting can be used:

```
MISSING_VIAS_CHECK_IGNORE_CELLS {
<cell_name 1>
<cell_name 2>
....
}
```

SEVERITY & IMPACT
Redhawk performs analysis for the design and gives warning regarding the missing vias.

ACTIONS
Add the missing vias for GDS blocks and rerun the analysis

! FAO-139
Voltage @(xlocation1,ylocation1) on <metal_layer_name> layer is not available in database, get closest via voltage(value) from <via_name> @(xlocation2 ylocation2) for instead.

DESCRIPTION
The metal layer might be floating layer in stacked via design, so Redhawk gets voltage from the nearest via that is properly connected and uses this value for analysis.

SEVERITY & IMPACT
The design might have floating geometries at the specified location for which redhawk will continue analysis with a warning

ACTIONS
Please check if the metal layer is floating.

! FAO-178
not set -fao_layer, using GSR defined FAO_LAYERS as layer constrains

DESCRIPTION
RedHawk allows user to specify layers on which he wishes to do the FAO operation using the -fao_layer option. If user doesn't use this option, then layers specified using the GSR keyword FAO_LAYERS will be considered for FAO operations.

SEVERITY AND IMPACT
Less critical error, which is mainly to remind the user to provide the fao_layer, in case he wants to restrict his FAO operations to certain layers.

ACTION
While executing the FAO commands, user can use the option "-fao_layer".

! FLW-001
Package parasitics are not annotated in the analysis

DESCRIPTION
RedHawk expects package parasitic information annotated in the design before performing voltage drop analysis. Especially in dynamic analysis, It is very important to include package parasitics to capture the L*di/dt impact on the overall voltage drop. Also, you will be able to accurately analyze the decap effectiveness in the design only if you include package parasitics in the analysis.

SEVERITY & IMPACT
Important error, especially for dynamic analysis. Voltage drop results will be optimistic if package parasitics are not included

ACTIONS

There are multiple methods available in RedHawk to annotate the package parasitics. Package spice netlist can be generated from package Extraction tools like SN-PSI or PakSi-E in the form of S-parameter or RLCK netlist. This package spice netlist can be included in GSR using PACKAGE_SPICE_SUBCKT keyword.

If detailed package spice netlist is not available for analysis, user can also annotate lumped RLC values for package/bond-wire/pad-wire using "setup package/wirebond/pad" command.

! GEN-025
Failed to open file to read.

DESCRIPTION
The file that is specified to read does not exist or the user has no permission to read.

SEVERITY & IMPACT
Error. The severity of missing a file depends on the importance of the file. Usually this is critical.

ACTIONS
Users should check the existence of the file as well as the permission to read.

! GEN-026
Failed to open file to write.

DESCRIPTION
The file that is specified cannot be created or overwritten.

SEVERITY & IMPACT
Error. The impact is critical if the file is an internal file created/consumed by RedHawk or the DB files exported from RedHawk. For a reporting file, users may ignore this error as long as the file is not crucial.

ACTIONS
Users should check the existence of the directory that the file is being created, or the permission to write if it already exists.

! GEN-031
Failed to copy file "name" into directory "name"!.

DESCRIPTION
Clear from the message.

SEVERITY & IMPACT
Error. Redhawk may not be able to continue depending on what file copying failed.

ACTIONS
Check file's and directory's existences and permissions.

! GEN-032
Failed to find file "name" in directory "name"!

DESCRIPTION
Clear from the message.

SEVERITY & IMPACT
Error. Redhawk may not be able to continue depending on what file is missing.

ACTIONS
Check file's existence first. If the file is created by a previous command, check the log file to see if something is wrong with that command.

! GEN-039
Error or warning during file integrity checking.

DESCRIPTION
Redhawk pre-parses the command file and various infered files needed for the run for file existence checking and limited file syntax checking. This error implies that the required file doesn't exist, can't be accessed, or doesn't seem like the required data file (lef, def, lib, tech, etc.).

SEVERITY & IMPACT
Error if file doesn't exist or can't be accessed. Warning is issued if the file syntax doesn't seem correct. It is possible for the pre-parser to give warning for correct data file because it only does limited keyword checking without really parsing the data file to make sure pre-parse is fast. Redhawk continues on pre-parse warnings.

ACTIONS
Users should check the existence of the refered file, or the syntax of the mentioned file. If it is a bug of the pre-parser to wrongly exit on correct data files, please let us fix it and user can bypass pre-parser by a GSR option "SKIP_FILE_PREPARSE 1".

! GSC-014
No POWERUP mode block specified in GSC file

DESCRIPTION
When user sets the analysis_mode as "lowpower", RedHawk expects him to specify POWERUP for blocks/domain in the GSC file. If POWERUP mode is not specified in the GSC file, then tool issues out this warning.

SEVERITY AND IMPACT

Results will be inaccurate, as the tool expects to do a POWERUP analysis but none of the blocks will power up.

ACTION

In the GSC file, specify POWERUP mode in the following way:

[<blockName> | <instanceName>] ? <domain_name>? POWERUP

! GSC-017

Non power gated instances assigned wrong GSC state

DESCRIPTION

If any instance which is not controlled by a power gate is assigned POWERUP or POWERDOWN state in GSC , this warning will be issued

SEVERITY & IMPACT

Its a warning message.The instance will be disabled in the power gated analysis

ACTIONS

Check the GSC file and correct it

! GSR-031

GSR keyword DYNAMIC_MOVIE is obsolete.

DESCRIPTION

DYNAMIC_MOVIE set to 1, saves data necessary to create an instance based (or transistor-based for MMX designs) dynamic voltage drop “movie” on a selected sample basis. You can specify a sample time as a multiple of the time step, or an optimum sample time is selected automatically. A large amount of disk space is required, depending on the size of the design. Optional. Default: 0 (Off).This keyword is obsolete now.

SEVERITY & IMPACT

Warning. It is ignored without affecting results.

ACTIONS

The GUI Results drop down Menu can be used instead for generating the movie. Results->Movie->Make/Show

----

! GSR-123

VCD_FILE/FRAME_SIZE very different from 1/gsr_freq, please verify

DESCRIPTION

The frame_size setting under vcd_file section in GSR should be set to close to 1/FREQ value in GSR.

SEVERITY & IMPACT

frame_size setting in GSR mainly affects power computation and cycle selection.

It will not affect accuracy of VCD dynamic analysis results.

ACTIONS
Change frame_size setting under vcd_file section in GSR to be equal or close to 1/FREQ value in GSR for accurate power computation/cycle selection purpose.

! INT-000
Unexpected corner case identified.

DESCRIPTION
This code reports that an unexpected corner case was hit and all the following operations could produce invalid results.

SEVERITY & IMPACT
Usually this error is severe and should not be ignored.

ACTIONS
Please report the error, together with adsRpt/apache.log file, to Apache Design Solutions.

! ITG-001
Report cells referenced in design that are not defined in LIB.

DESCRIPTION
The reported cells are listed in <adsRpt/apache.refCell.noLib>.

SEVERITY & IMPACT
Warning. The reported cells and their pins may have wrong attributes.

ACTIONS
Add LIB definitions for the reported cells.

! ITG-002
Report cells referenced in design that are not defined in LEF.

DESCRIPTION
The reported referenced cells are listed in <adsRpt/apache.refCell.noLef>.

SEVERITY & IMPACT
Warning. The reported cells and their pins may have wrong attributes.

ACTIONS
Add LEF definitions for the reported cells.

! ITG-003
Report cells referenced in design that are not defined in LEF and LIB.

DESCRIPTION

The reported cells are listed in <adsRpt/apache.refCell.noLefLib>.

SEVERITY & IMPACT
Warning. The reported cells and their pins may have wrong attributes.

ACTIONS
Add LEF and LIB definitions for the reported cells.

! ITG-004
Report cells referenced in design that have no internal power defined in LIB.

DESCRIPTION
The missing internal power cells are listed in <adsRpt/apache.refCell.noPwr>.

SEVERITY & IMPACT
Warning. The instances of the reported cells will only have switching power in lib-based power calculation. Missing internal power may affect power, ir, and em results.

ACTIONS
Define internal power in LIB for the reported cells.

! ITG-005
Check the existence of LEF and LIB definitions for gdsmem cells and their parent cells.

DESCRIPTION
LEF and LIB definitions are required for gdsmem cells and their parent cells.

SEVERITY & IMPACT
Fatal. RedHawk will exit if there are gdsmem data errors.

ACTIONS
To fix gdsmem data errors, please
1. reference log files to add missing LEF and/or LIB cell definitions, or
2. add missing LIB cell definitions to custom lib file if the type of errors is missing LIB cell definitions.

GSR option "IGNORE_GDSMEM_ERROR 1" allows RedHawk to continue with errors, but may result in incorrect analysis results.

! ITG-006
RedHawk failed due to gdsmem data errors.

DESCRIPTION
Report the reason that causes RedHawk to stop. For gdsmem cells and their parent cells, both LEF and LIB definitions are required.
See also ITG-005.

SEVERITY & IMPACT
Fatal. Gdsmem data error must be fixed.

ACTIONS
Please reference apache log files in adsRpt directory to fix the
gdsmem data errors. GSR option "IGNORE_GDSMEM_ERROR 1" allows RedHawk
to continue with errors, but may result in incorrect analysis result.

! ITG-007
Report that RedHawk proceeds with gdsmem data errors.

DESCRIPTION
If gsr option IGNORE_GDSMEM_ERROR is set, RedHawk continues with errors.
See also ITG-005.

SEVERITY & IMPACT
Error. The analysis result may be incorrect if there are gdsmem data errors.

ACTIONS

! ITG-008
Report cells referenced in design that have triggered edge unidentified.

DESCRIPTION
The reported cells are listed in <adsRpt/apache.refCell.noTriggerEdge>.
The default triggered edge is rising edge.

SEVERITY & IMPACT
Warning. Instance triggering conditions may be affected if triggered edges
are not identified.

ACTIONS
Define cell triggered edges in custom lib file.

! ITG-009
Report cells referenced in design that have no power or ground pins.

DESCRIPTION
The reported cells are listed in <adsRpt/apache.refCell.noPwrGndPins>.

SEVERITY & IMPACT
Warning. The reported cell instances won't be connected to power or ground grids.

ACTIONS
Fix data if necessary.

! ITG-010

Pin delimiter ":" used in LEF macro <xyz> pin "abc:def".
Change the pin delimiter to ";".

DESCRIPTION
Redhawk found ":" character in pin names. As a result, it changed the default pin delimiter from ":" to ";". One application of pin delimter is to identify instance and pin name from clock root for clock tree tracing. It is only required when STA file is not available and user wants to use clock tree tracing for initial flow flush. In such cases, the clock roots with pin names will not be mapped to the design.

SEVERITY & IMPACT
Impacts analysis only when user is using clock tree tracing due to unavailability of STA file.

ACTIONS
If you are performing clock tree tracing, make sure to use ";" or some special character as pin delimiter in clock root and specify the same in GSR. Alternatively, avoid using ":" character in pin names.

! ITG-012
Via definition conflict in multiple LEF files

DESCRIPTION
The mentioned via is defined in multiple LEF files imported in RedHawk and the definitions looks conflicting between the files

SEVERITY & IMPACT
RedHawk will take the via definition present in the first LEF file read in the analysis. This will have accuracy impact in analysis.

ACTIONS
Resolve the conflict in via definition across all LEF files and rerun the analysis

----
! ITG-015
Net <net_name> connected to instance vdd pin <instance_name:pin_name> is not a power net

DESCRIPTION
The net name in the DEF might be specified as the wrong type. The net will be specified as signal or ground instead of power in the DEF. That is, the issue is caused due to mismatch of pin type and net type to which the pin is connected.

SEVERITY & IMPACT
If the net is specified as signal instead of power, then Redhawk will ignore the corresponding net for power analysis. The extraction of the net will not be done by Redhawk. This will affect the power values obtained.

ACTIONS
Please check the "USE POWER" section of the net in DEF and change it to the required type, that is, "USE GROUND" or "USE SIGNAL".

! ITG-016
Missing PG-arc definitions for some cells

DESCRIPTION
For cells having multiple POWER and GROUND pins (mVmG cells), RedHawk require PG Arc information which defines the current flow between the POWER pin and GROUND pin. For example, if a cell has 2 POWER pins(VDD1 and VDD2) and 2 GROUND pins (VSS1 and VSS2), you should associate each power pin with the associated ground pin. (Example: VDD1->VSS1 & VDD2->VSS2). This information is passed to RedHawk as a custom lib file in LIB_FILES section.

SEVERITY & IMPACT
Critical error. Voltage drop results will be impacted for mVmG cells.

ACTIONS
Create a custom lib file containing PG ARC information like given below:

```
cell vdds18hvt {
    pgarc {

        biasfet bus_vsss
        bus_vdd vss
        vdds bus_vsss
    }
}
```

The above file needs to be included in LIB_FILES GSR with a custom attribute.

```
LIB_FILES  {
  <other lib files>
  custom.lib custom
}
```

! ITG-017
Inconsistent power/ground pin decap (value vs value) in cell <name>.

DESCRIPTION
The decap values for PG pins are drastically different, which is unexpected.

SEVERITY & IMPACT
Major warning. This might be a problem in settings used in APL characterization

ACTIONS
Please verify the APL characterization to identify the problem.

! ITG-018
Out of date gdsmem data

DESCRIPTION
gdsmem data provided in the GSR is out of data. Please recreate
the gdsmem data with latest RedHawk/Totem version.

SEVERITY & IMPACT
Results may be impacted.

ACTIONS
Recreate gds2def data with latest RedHawk/Totem version.

! LEF-003
Cannot handle port path iterate geometries for the specified pin.

DESCRIPTION
No support for port geometries defined by PATH ITERATE statements.

SEVERITY & IMPACT
Warning. Port geometries are ignored.

ACTIONS
Please specify port geometries without path iterate.

! LEF-005
Warning message generated by LEF parser.

DESCRIPTION
This message reports the warning messages generated by LEF parser.

SEVERITY & IMPACT
Warning. The severity depends on the content.

ACTIONS
Users should take actions based on the warning message.

! LEF-014
Macro is redefined with less pins. Ignore this redundant definition.

DESCRIPTION
If a macro is redefined and the latter definition has fewer pins,
then the latter definition is discarded.

SEVERITY & IMPACT
Warning.

ACTIONS
Users should resolve the reported duplicated cell problem.

! LEF-015
Macro has been redefined and be replaced with the newer definition.

DESCRIPTION
If a LEF macro is defined multiple times, the last definition overrides the previous ones.

SEVERITY & IMPACT
Warning.

ACTIONS
Users should resolve the reported duplicated cell problem.

! LEF-017
Failed to open the first LEF file.

DESCRIPTION
The specified file doesn't exist or the user has no permission to open it.

SEVERITY & IMPACT
Error.

ACTIONS

! LEF-019
Support for adsmem cells will become obsolete in next Redhawk release!

DESCRIPTION
Previously, the adsmem-based gds2def -m program generated LEF, DEF, and LIB files. (adsmem-based cells are now not supported in RedHawk power calculation). Pin based gds2def -m generates LEF, DEF, and *.pratio files.The warning indicates that adsmem cells would no longer be supported from next release.

SEVERITY & IMPACT
Minor Warning. Mainly to be noted for next releases.

ACTIONS
Please move to newer gds2def flow to prevent issues with new releases. You must use the GDS_CELLS GSR keyword to import gds2def -m cells

----
! LEF-023
The number layers of the specified generated via rule is not 3.

DESCRIPTION
Generated vias should consist of shapes on three layers.

SEVERITY & IMPACT
Warning.

ACTIONS
Fix the generated via rule at the specified line in LEF file.

! LEF-024
Un-defined via model in LEF file

DESCRIPTION
VIA models present inside the geometry section of the LEF file should be pre-defined in the technology LEF.

SEVERITY & IMPACT
RedHawk will ignore these vias in the analysis

ACTIONS
Include the via model in Tech Lef and rerun the analysis

! LEF-025
BUSBITCHARS is not defined in LEF file.

DESCRIPTION
BUSBITCHARS specifies the characters to enclose bus bits. If BUSBITCHARS is not defined, "[]" are the default special characters for bus bits.

SEVERITY & IMPACT
Warning. If BUSBITCHARS is not defined and bus bits are not enclosed by "[]", bus names may be handled incorrectly.

ACTIONS
Define BUSBITCHARS in LEF files.

! LEF-026
DIVIDERCHAR is not defined in LEF file.

DESCRIPTION
DIVIDERCHAR specifies the character to express hierarchy. If DIVIDERCHAR is not defined, "/" is the default hierarchical divider.

SEVERITY & IMPACT
Warning. If DIVIDERCHAR is not defined, LEF names may be handled incorrectly.

ACTIONS
Define DIVIDERCHAR in LEF files.

! LEF-030
LEF file has already been read.

DESCRIPTION
Warning message for a LEF file being imported more than once.

SEVERITY & IMPACT
Warning. The duplicated files are skipped.

ACTIONS
Remove duplicated file names from .lefs file.

! LEF-042
Layer<layer_name> of viaModel <via_lef_model_name> is not used

DESCRIPTION
The specified via model is not used in the design, although it is specified in the LEF. That is, the via model is not used in DEF. The model might a contact below Metal layer 1, which is not considered during Redhawk analysis.

SEVERITY & IMPACT
The redhawk results will not be affected as the via model is not present in DEF. If the model is for contact below Metal layer 1, then the data is not required for Redhawk analysis.

ACTIONS
Please check if the via model is required for analysis.

! LEF-043
Layer<layer_name> : type conflict - tech (type ROUTING) vs LEF (type MASTERSLICE)! LEF type ignored

DESCRIPTION
Masterslice layers are typically polysilicon layers and are needed only if cell MACRO's have pins on polysilicon layer. The LEF tech file given as input to Redahwk, will have the type of layer <layer_name> as MASTERSLICE. But the technology file will have the same layer specified as ROUTING. The layer is specified as metal layer in the technology file, which is treated as ROUTING type by Redhawk.

SEVERITY & IMPACT
Redhawk will consider the layer as of ROUTING type as it honours technology file and give out the warning.

ACTIONS
Redhawk does not need the Masterslice layers unless cell MACRO's have pins on them. So they can be removed from tech file.

! LEF-080
Layers inside GENERATE VIARULE definition is not defined inside the tech file

DESCRIPTION

All layers present inside the VIARULE definition section should be predefined
inside RedHawk tech file

SEVERITY & IMPACT
RedHawk will ignore these vias in the analysis

ACTIONS
Correct the viarule definition inside the LEF and rerun.

! LEF-084
GENERATE VIARULE must have 1 CutLayer

DESCRIPTION
RedHawk expects at least one cut layer in a generated via rule.
Please check whether any one of the layers defined inside the specified viarule
is defined as a cut layer in the tech file

SEVERITY & IMPACT
RedHawk will ignore these vias in the analysis

ACTIONS
Correct the viarule definition inside the LEF and rerun.

! LEF-102
The layer name referenced in PIN statement is not defined.

DESCRIPTION
Pin geometries are defined on invalid layers. Pin geometries are ignored.

SEVERITY & IMPACT
Error.

ACTIONS
Fix the undefined layer referenced in the specified pin geometries.

! LEF-107
Ignore invalid ViaModel <viamodel> due to 0-sized rects on <layername> layer

DESCRIPTION
The viamodel defnition is incorrect because it has 0 sized rects on
top and bottom layers.

SEVERITY & IMPACT
Invalid viamodels will be ignored and will be missing in the RH run.
This can cause unconnects in the design because of missing vias.

ACTIONS

Check the lef and provide correct viamodels.

! LEF-108
Site name <site_name> repeated.

DESCRIPTION
Multiple definitions for a SITE exist. Eg. While reading multiple LEF files in a design, the same SITE has been defined in more than one files.

SEVERITY & IMPACT
Warning generated during setup design while importing LEF_FILES. The last definition is considered overwriting the previous ones.

ACTIONS
Make sure that the SITE <site_name> is defined only once.

! LEF-109
Decap cells have been specified twice.

DESCRIPTION
Redhawk will give this warning if some decap cells have been specified twice or more times. Redhawk expects each decap cell to be specified only once. Consequent declarations are ignored.

SEVERITY & IMPACT
Warning. Consequent declarations are ignored.

ACTIONS
Check if the decap cells have been specified twice in GSR or in LEF.

! LIB-121
Duplicated cells in lib. Please check the redundancy in LEF.

DESCRIPTION
Duplicated cells in design library.

SEVERITY & IMPACT
Error. Duplicated cells may cause inconsistent DEF pin instance connection.

ACTIONS
Users should resolve the reported duplicated cell problem.

! LIB-143
External and Internal pins of a switch tied to the same net

DESCRIPTION

If the same net is tied to the internal and external pins of a switch,
Redhawk issues this warning message

SEVERITY & IMPACT
Its a warning during setup design.Redhawk will ignore this warning and proceed
with setup design.The switch will be ignored in analysis.

ACTIONS
Check the DEF file for logical connectivity and correct it

! LIB-145
'CTRL_PIN' is not defined for cell <xyz> in the switch model file.

DESCRIPTION
Power switch model file does not have CTRL_PIN defined. This is must for
rush current analysis.

SEVERITY & IMPACT
Rush current or power-up analysis may error out.

ACTIONS
Check the switch model file and how they have been created. Switch model
file created through utility aplsw should have it correctly captured.
The switch model file could have been hand-created.

! LIB-150
Clock pin missing for non-combinational cells

DESCRIPTION
RedHawk is not able to identify the clock pin from the ".lib" for the
specified cell. This cell is identified as a non-combinational cell by
RedHawk using the library attributes like ff(), latch() etc. Or, this
could be a CMM cell generated from Totem.

SEVERITY & IMPACT
This cell may not switch in dynamic simulation. Results will get impacted.

ACTIONS
Correct the library attributes and rerun the analysis

! LIB-250
Clock pin missing for non-combinational cells

DESCRIPTION
RedHawk is not able to identify the clock pin from the ".lib" for the
specified cell. This cell is identified as a non-combinational cell by
RedHawk using the library attributes like ff(), latch() etc. Or, this
could be a CMM cell generated from Totem.

SEVERITY & IMPACT
This cell may not switch in dynamic simulation. Results will get impacted.

ACTIONS
Correct the library attributes and rerun the analysis

! LIB-401
The cells in .lib are not defined in LEF. Same as SLB-041.

DESCRIPTION
This message reports LIB cells that have no LEF definitions.

SEVERITY & IMPACT
Warning. Since only cells with physical geometry are considered
by RedHawk, the reported cells are ignored.

ACTIONS
Users can define the reported cells in LEF.
The missing LEF cells are listed in adsRpt/apache.noLEF.

! LIB-402
The cells in .lef are not defined in LIB. Same as SLB-042.

DESCRIPTION
This message reports LEF cells that have no LIB definitions.

SEVERITY & IMPACT
Warning. The reported cells and their pins may have incorrect attributes.

ACTIONS
Users can define the reported cells in LIB or in custom lib file.
The missing LIB cells are listed in adsRpt/apache.noLIB.

! LNW-029
Simulation Pin/Instance/Net missing in DEF

DESCRIPTION
You will get this warning when a particular pin or net present in simulation input (VCD/FSDB) is not found in the DEFs for that design. This can happen when simulation file is generated from a different version of the design than what is present in DEF. This can also happen if complete DEF of the designs have not been read into Redhawk.

SEVERITY & IMPACT
Warning. If many such warning are present, it will affect analysis accuracy.

ACTIONS

Investigate whether the simulation and DEF inputs are generated from the
same database version.

! LNW-045
Insert propagationOrigin for the <pin>

DESCRIPTION
State Propagation sets toggle activities at the state points (input,
ff/latch outputs) and assumes all nets in the design are covered by the
state propagation from these state points. It checks any dangling inputs,
assigns zero toggle rates, and issues the warning message of
"Insert propagationOrigin..." to confirm with the user.

SEVERITY & IMPACT
This warning is not critical if these points are kept dangling by the user.
Zero toggle rate will be assigned to these pins.

ACTIONS
Check whether the pins are supposed to be dangling or else correct
the pin connections

! LNW-049
In VCD mode, GATED_HIGH_PERCENTAGE/GATED_ON_PERCENTAGE keyword
in STATE_PROPAGATION is ignored.

DESCRIPTION
In VCD mode, RedHawk won't take the GATED_HIGH_PERCENTAGE/GATED_ON_PERCENTAGE
settings as that information is read in from the VCD.

SEVERITY & IMPACT
RedHawk ignores this setting in VCD mode.

ACTIONS
User can correct the GSR setting and remove it in the case of reading a VCD file.

#----------
! LNW-051
Toggle rate is assigned multiple times for ob_core/FecOOB/n484
(orig 9.06033e-05 < 0.00765238).

DESCRIPTION
This warnings lets the user know that there are multiple sources of Toggle Rate
for an instance from different VCD files.

SEVERITY & IMPACT
If the two toggle rates have a difference greater than 0.0001, RedHawk prints
this warning to remind the user to check the consistency issue from the
multiple VCD files.

ACTIONS
Please verify your VCD settings and make sure that the data for an instance is read in only from one VCD.

#----------
! LNW-055
GATED_ON_PERCENTAGE and GATED_HIGH_PERCENTAGE are obsolete

DESCRIPTION
Keyword GATED_HIGH_PERCENTAGE is now obsolete for all gated clock analysis of the design. GATED_ON_PERCENTAGE is obsolete for multi-cycle clock gating analysis. Instead use the GATED_ON_FRACTION keyword in GSR. Refer to "Gated Clock Dynamic Analysis" section in the Redhawk manual for its syntax and application.

SEVERITY & IMPACT
Warning. Constraints specified with it will not work in future.

ACTIONS
Please use GATED_ON_FRACTION instead. Refer to "Gated Clock Dynamic Analysis" section in the Redhawk manual for further understanding.

! LNW-056
Simulation mode not DYNAMIC. GATED_CLOCK_TRANSIENT ignored

DESCRIPTION
In case of Redhawk Static simulation, if the tool encounters any GATED_CLOCK_TRANSIENT settings, it will ignore them as it is to be run only in Dynamic mode.

SEVERITY & IMPACT
Please note that GATED_CLOCK_TRNASIENT has been replaced with POWER_TRANSIENT from 10.1 version onwards.

ACTIONS
Check if it's in the correct mode of simulation, and then ignore this error.

#----------
! LNW-058
Please check <adsRpt/state_propagation.GCControls.errMsgs> for more details.

DESCRIPTION
Redhawk State Propagation engine lists out possible incorrect user settings in this file. They could be a typo, or invalid clock gating setting.

ACTIONS
The user can discard after the review if the errors are minor.

#----------

! MOD-001
Via array defined in lef/def contains different-sized via cuts and cannot be splitted into smaller via arrays.

DESCRIPTION
To improve the simulation accuracy, RH will cut large via array defined in lef/def into smaller ones. For a via array appeared in this warning message, it contains via cuts of different sizes and thereby cannot be splitted into smaller via arrays. This may reduce the accuracy of the Redhawk static/dynamic simulation.

SEVERITY & IMPACT
Warning.

ACTIONS
No action is needed.

! MOD-005
Coupling capacitance between two wires on the same layer is 0.

DESCRIPTION
This message warns the dielectric constant and/or thickness of the dielectric above or below the metal are not set correctly in technology file.
The coupling capacitance of two neighboring wires on the same layer is set to 0.

SEVERITY & IMPACT
Warning. It could affect dynamic simulation results if C-extraction is performed.

ACTIONS
Please check the dielelectric parameter and metal parameters in tech file for this layer. The key parameters to be checked are: thickness and dielectric constant of the dielectric layer above and below the metal, and metal thickness.

! MOD-006
Metal layer's EM limit is 0.

DESCRIPTION
The EM limit parameter of metal layer is not set correctly in the technology file.
The default value is 0.

SEVERITY & IMPACT
This error may cause numerous false alarms in EM violation reports.

ACTIONS
Correct metal EM limit in the technology file.

! MOD-009

Cannot extract C for layer <layer_name>.

DESCRIPTION
The dielectric is not defined at this layer and the layers below.

SEVERITY & IMPACT
Warning generated while setting up database for design.

ACTIONS
Please specify the dielectric in the tech file.

! MOD-010
WARNING(MOD-010): Can not extract C for layer <M5> and layer below it!

DESCRIPTION
RedHawk gives this error when it does not find any dielectrics defined for a metal layer in the Tech file.

SEVERITY & IMPACT
When RedHawk gives this warning, it will not extract Capacitance for the particular metal layer, and only R will be extracted. Hence the Dynamic analysis results may not be reliable.

ACTIONS
Please check the tech file to confim that the particular metal layer has a dielectric layer defined above and below it.

#----------
! MWY-03
Failed to read the specified cell.

DESCRIPTION
The specified cell doesn't exist in milkyway database.

SEVERITY & IMPACT
Error. Data may be incomplete.

ACTIONS
Please check original directory structure of reference library.

! MWY-39
The instance which is connected to the specified net doesn't exist in the specified cell.

DESCRIPTION
Milkyway database inconsistence.

SEVERITY & IMPACT
Error. The connectivity of the undefined instance is ignored.

ACTIONS
Check milkyway database.

! NET-111
PinInst of net has inconsistent type as the net.

DESCRIPTION
This error reports a POWER pin is connected to a GROUND net or vice versa.

SEVERITY & IMPACT
This usually indicates data errors which will cause wrong connectivity
and false results.

ACTIONS
Users need to verify/fix the LEF definition of the cell or DEF definition
of the net connectivity.

! NET-121
PinInst<name:name> (x1,y1 x2,y2) of net<name> is
out of cell<name>'s boundary(x1,y1 x2,y2)

DESCRIPTION
Error. The setup before extraction detects that an instance's pin
(as reported) is outside the top cell's boundary. This out-of-bound
pin is still considered during the extraction.

SEVERITY & IMPACT
An instance's pin out-of-bound may be due to an incorrect instance location
or orientation. This in turn may cause the pin to become unconnected and
the extraction results may be wrong.

ACTIONS
A user should check the DEF files to make sure the reported instance is
correctly placed.

! NET-124
Pin<name> (x1,y1 x2,y2) is out of cell<name>'s boundary(x1,y1 x2,y2)

DESCRIPTION
Error. The setup before extraction detects that a top cell's pin
is outside the top cell's boundary. This out-of-bound pin is still
considered during the extraction.

SEVERITY & IMPACT
A top cell's pin out-of-bound may be due to an incorrect pin location
in the top cell's DEF. This pin may become unconnected and extraction
results may be wrong.

ACTIONS
A user should check the top cell's DEF file to make sure the reported pin is correctly placed.

! NET-141
Net has # of via collisions.

DESCRIPTION
On same net, there are vias of the same layer with viaholes overlapping each other. The collided vias are reported in adsRpt/<design>.collided_vias.

SEVERITY & IMPACT
WARNING. For the collided vias only one is kept.

ACTIONS
Users need to verify the design using adsRpt/<design>.collided_vias and make sure there are no different vias (of the same layer) at the same location.

! NET-142
Fail to find any current path from a pin instance in a block to any via in top level cell.

DESCRIPTION
Only happens in RH hierarchical simulation flow. This pin instance is logically connected to PG network in DEF. However, RH can't find a current path from this pin instance in the sub block to any via in top level cell.
In general, pin instance should have a path containing some vias to PG pad.
This is a indication that the pin instance is not connected to any PG pad.

SEVERITY & IMPACT
WARNING. The pin instance will be drop from the PG network. So power associated to this instance through this pin will not be used in the simulation.

ACTIONS
Check DEF files to find out weather this pin instance should be connected to PG network or not. If yes, make sure there is routing to connect this pin instance to PG pad.

! NET-145
Net<name> in top cell<name> is not driven by any PAD.

DESCRIPTION
Error. The extraction checking finds out that a top cell's P/G net does not have a driver.

SEVERITY & IMPACT
If the P/G net reported is used in the design, the simulation will fail.

ACTIONS

A) A user should check the .ploc/.pad/.pcell file to make sure the pad locations for the reported net is specified and correctly placed. Seach log file to see if any (CCL-056) or (CCL-065) messages exist. These messages points to pads which are detached and are potential source of this problem.

B) If power gates are specified for a design and this is an internal net, check DEF/LEF for physical and logical connectivity.

! NET-150

Failed in building connectivity for net<name>.

DESCRIPTION

Power or ground nets (specified in the power or ground net sections of the GSR) that are present in the design but are not connected to the global network that is driven by VDD or GND power supply sources are considered dangling wires. Their presence lead to a failure in building up a fully connected power or ground network for the specified power or ground nets. This can happen due to lack of either physical or logical connectivity of these portions of the network to the rest of power or ground networks.

The extraction checking finds that there exist disconnet from a top cell's P/G pin to all instance pins connected by the same net.

SEVERITY & IMPACT

If the P/G net reported is used in the design, the simulation will fail.

ACTIONS

This should be investigated to ensure data correctness. Inside RedHawk it is possible to debug this error message by using the "View->Highlight unconnected wires/vias" feature following a static or dynamic analysis. This will highlight the portions of the power or ground networks that are dangling. User may check the .ploc/.pad/.pcell file to make sure the pad locations for the reported net is specified and correctly placed. Search log file to see if any (CCL-056) or (CCL-065) messages exist. These messages points to pads which are detached and are potential source of this problem. Also, search for (NET-121) messages to check instance pins.

! NET-403

Ignoring the pin instance which does not have any physical geometry.

DESCRIPTION

Pin instance does not have any layout geometry. This pin instance could not physically connected to the net.

SEVERITY & IMPACT

WARNING. The pin instance will be drop from the PG network. So power associated to this instance through this pin will not be used in the

simulation.

ACTIONS
Correct LEF file by adding layout geometry to pin instance's master pin.

! NET-404
Ignoring the pin which does not have any physical geometry.

DESCRIPTION
Pin does not have any layout geometry. This pin could not physically connected to the net.

SEVERITY & IMPACT
WARNING. The pin will be drop from the PG network.

ACTIONS
Correct LEF file by adding layout geometry to this pin.

! NET-405
Pin is not physically (only logically) connected to net.

DESCRIPTION
This error identifies a pin that is logically connected to a net but physically not. This is usually due that (1) the net has incomplete broken routing geometries in DEF, or (2) the pin does not have physical geometries defined.

SEVERITY & IMPACT
If the disconnected pin is a pad, it will not serve as the driver to the net. If the extraction is performed on a sub-block, any external connection to this pin will not be connected to the internal net.

ACTIONS
Users need to confirm it is okay to disconnect the pin from the net.

! NET-412
Leafcell<> not defined in .lib and pin<> is treated as a bi-directional.

DESCRIPTION
RedHawk internally calculates C1/R/C2 for each driver of a net using the RC networks specified in SPF files and annotate the C1/R/C2 to the corresponding drivers. In order to decide if a pinInst of a net is a driver or not, RedHawk relies on the pin direction from .lib files.
This warning message happens when RedHawk may fail to decide if a pinInst is a driver because the cell is not in .lib and no pin direction was given.

SEVERITY & IMPACT
The pinInst is treated as a bi-directional pin.

ACTIONS
Users may choose to either add the missing cells to the .lib files or explicitly specify the pin directions in a CUSTOM_LIB_FILE.

! NET-413
The SPF file does not cover specified net.

DESCRIPTION
This message reports a net that is not covered in SPF files. This is due to the incomplete SPF files.

SEVERITY & IMPACT
Warning. The net will get the estimated parasitics using Steiner tree model.

ACTIONS
Users need to check the completeness of the SPF files.

! NET-450
There are # wires not physically connected to the net.

DESCRIPTION
There are wires defined as routes of the net in DEF, but RH finds these wires are not physically connected to the net's input pad, so electrically, these wires are disconnected from the net. These disconnected wires are listed in adsRpt/<design>.Wire.unconnect.

SEVERITY & IMPACT
WARNING. These wires won't be used in RH simulation.

ACTIONS
Check the layout by highlighting disconnected wire pull down menu to find what metal or via missed, then fix the DEF files.

! NET-451
There are # vias not physically connected to the net.

DESCRIPTION
There are vias defined as routes of the net in DEF, but RH finds these vias are not physically connected to the net's input pad, so electrically, these vias are disconnected from the net. These disconnected vias are listed in adsRpt/<design>.Via.unconnect.

SEVERITY & IMPACT
WARNING. These vias won't be used in RH simulation.

ACTIONS
Check the layout by highlighting disconnected via pull down menu to find what metal or via missed, then fix the DEF files.

! NET-452
There are # pins not physically connected to the net.

DESCRIPTION
There are pins logically connected to the net in DEF, but RH finds these pins are not physically connected to net's input pad, so electrically, these pins are disconnected from the net. One common scenario is the pins do not have layout geometries.

These disconnected pins are listed in adsRpt/<design>.Pin.unconnect.

SEVERITY & IMPACT
WARNING. These pins won't be used in RH simulation.

ACTIONS
Verify the pin geometries in DEF. Users may ignore this warning if the pin is intended to be a virtual pin with no physical geometries.

! NET-453
There are # pin instances not physically connected to the net.

DESCRIPTION
There are pin instances logically connected to the net in DEF, but RH finds these pin instances are not physically connected to net's input pad, so electrically, these pin instances are disconnected from the net. There are two kind of possibilties causing the disconnectivity:

1) pin instance does not have correct layout geometry, or
2) there are metal or via missed to route the pin instance to input pad.

These pin instances are listed in adsRpt/<design>.PinInst.unconnect.

SEVERITY & IMPACT
WARNING. These pin instances won't be used in RH simulation, so the power associated with these pin instances will be droped.

ACTIONS
Check the layout by highlighting disconnected metal/via to trace the route from the disconnected pin instance to find the cause. Then fix the cause in DEF/LEF.

! PAD-002
Pad file should end with .ploc or .pcell or .pad!

DESCRIPTION
Redhawk only accepts file names ending with ".ploc", ".pcell" and ".pad" as legal files for specifying design power/ground pads. Each of these 3 file extension indicates a different format and meaning of its content.

SEVERITY & IMPACT
Error. There is no way for Redhawk to input correctly the P/G pad information unless the appropriate file extension is used. Redhawk exits since chip P/G cannot be specified.

ACTIONS
A user should rename the file name with the correct file extension and make sure that the format of the content match the file extension used. Please consult the User Manual for details of the acceptable formats.

! PAD-004
Pad Instance <name> does not exist in DEF.

DESCRIPTION
A pad instance name specified in the .pad file is not included in the input DEF files.

SEVERITY & IMPACT
Warning. It is fine if this pad instance is not used to drive any P/G net in the design. But may cause serious problem if it is used but missing from the design.

ACTIONS
User should check the .defs file to make sure the DEF file containing the missing pad instance is included.

! PAD-005
LEF is absent for the pad cell mentioned

DESCRIPTION
LEF files contains physical description of the cells . RedHawk is not able to find the LEF file for the given cell .

SEVERITY & IMPACT
This warning is important and user must specify LEF files for pad cells. The given pad cell will not be treated as a pad if LEF file is missing

ACTIONS
Provide LEF file for the cell to remove this warning

! PAD-007
line (#) in file<name>: Pad <name> (x y) is outside top cell's boundary!

DESCRIPTION
A pad location specified in .ploc file (line# and file name are shown) is outside the top cell's boundary.

SEVERITY & IMPACT

Warning. It is fine if this pad is not used to drive any P/G net in
the design. If it is used, the pad may become unconnected to the net
and causes serious problem.

ACTIONS
A user should check the .ploc file and make sure the specified location
is correct. Or, double check the top DEF to make sure the top cell boundary
is correct.

! PAD-011
Voltage source with name 'Vdd' connected to GROUND

DESCRIPTION
Voltage source with 'Vdd' in its name has been declared as type GROUND in
the ploc file. It can possibly be a typo. It is a bad practice to have
string 'Vdd' as part of pad name which connects to ground rail on the chip.

SEVERITY & IMPACT
Warning.

ACTIONS
Review the ploc file and check whether there is a typo.

! PAD-012
line (354) in .ploc file<adsRpt/DEF.ploc>: source named
<vss_pllclkout_1.extra1_Def> is connected to POWER ?

DESCRIPTION
This Warning reminds the user to check that a Ploc with "VSS" in its name
is connected to POWER.

SEVERITY & IMPACT
RedHawk still takes in this Ploc as Power as per the connection mentioned in
the PLOC file.

ACTIONS
Check and confirm the Ploc connection once and then ignore this Warning.

! PAD-032
Pad<name> in .ploc shares package .subckt node<name> with other pads.

DESCRIPTION
This warning appears if multiple bumps are connected to the same package.
While this may be expected for some packages, in some cases this may be
due to wrong connection in the ploc file.

SEVERITY & IMPACT

Minor warning. You may skip this warning if such a connection is expected.
Please correct this if the ploc connection is incorrect.

ACTIONS
Please correct ploc connection if necessary.

! PAD-038
Incorrect package definition in the PLOC file

DESCRIPTION
Package parasitics per bump can be defined in the ploc file
Please see the syntax below :
<pin_name> <x_loc> <y_loc> <layer> <power|ground> <L in pH>
<R in Ohms> <C in pF>
For the above mentioned line in the ploc file, check ploc file
syntax.

SEVERITY & IMPACT
Package RLC values provided will be ignored and assume default
default package values.

ACTIONS
Correct ploc file and rerun RedHawk.

! PAD-050
POWER and GROUND sources are listed in file `adsRpt/PG.ploc' (in .ploc format).

DESCRIPTION
An information telling users the VDD/GND pads used inside Redhawk.

SEVERITY & IMPACT
No impact.

ACTIONS
Check file `adsRpt/PG.ploc' to see if Redhawk's pad assignments are correct.

! PIN-011
Pin<pin_name> is not connected to any net.

DESCRIPTION
The pin <pin_name> does not have any connection specified to any net in the DEF.

SEVERITY & IMPACT
Warning. May cause some nets' resistor network disconnected!

ACTIONS
Check DEF for connectivity.

! PIN-013
P/G source at (19.500 242.000) M6 does not touch any P/G pin! Ignored.

DESCRIPTION
RedHawk issues this warning when it cannot find any metal pin at a defined ploc/pcell PG-source location.

SEVERITY & IMPACT
Redhawk will ignore this PG-source location from the analysis.

ACTIONS
Please check the PAD_FILES section in the GSR and correct the PG-source location coordinates in the ploc/pcell file.

#----------
! PIS-604
POWER/GROUND type mismatch between instance pin and Net

DESCRIPTION
There is a logical connectivity issue exist in the design. RedHawk is finding a conflict between the pin type specified in LEF file and the net type specified in DEF file (POWER/GROUND/SIGNAL). For example, the PIN could be defined as of type POWER in LEF section. But this pin be connected to a net which is defined as of TYPE GROUND in the DEF section. Note that the NET type defined in both PINS section and NETS/SPECIALNETS section in the DEF should match with the PIN type defined in the LEF file.

SEVERITY & IMPACT
RedHawk will ignore the logical connectivity. This instance will be dis-connected from the PG network in the analysis.

ACTIONS
Correct the PIN/NET type attributes in LEF/DEF and proceed.

! PIS-605
PinInst<name:name> is already connected to net<name>!
Replace its connected net with new net<name>.

DESCRIPTION
An instance's pin is connected to more than 1 net in the design.
The latest input net always overrides the previous ones.

SEVERITY & IMPACT
Warning. Could be a DEF netlist mistake and become critical.

ACTIONS
A user should check the DEF files to make sure there is no conflicting net assignments for the instance and pin shown.

! PWR-020
Pin, CLK , of cell, <cell> , is reset to clock pin.

DESCRIPTION
The mentioned pin is identified as a non-clock pin from def ,but its liberty definition says it is a clock pin .

SEVERITY & IMPACT
Since .lib defines the pin as a clock ,RH will reset it to clock.

ACTIONS
Check why a clock pin is defined as non-clock signal pin in def and correct it.

#----------
! PWR-021
Pin added to LEF of the cell during power calculation

DESCRIPTION
Redhawk compares cellview in LEF and cellview in GDS2DEF data for all cells to ensure that pin definitions are consistent. If it finds a pin missing, it will add that pin before starting power calculations and give this message. Difference in cellview can happen due to many reasons such as improper configuration during GDS2DEF data preparation or if there is difference in LEF used during GDS2DEF data preparation and LEF read through GSR using LEF_FILES. Ideally only GDS2DEF data should be read-in for any cell.

SEVERITY & IMPACT
Warning.

ACTIONS
Check if LEF/DEF data is being read in twice for a cell and if it is inconsistent investigate why it is so. Also review the configuration file used during GDS2DEF data preparation.

! PWR-060
Cell, AND2_F10_LL , defined as 1 power pin 1 ground pin in LEF, does not match 3 power/ground pin in cell.current.

DESCRIPTION
RedHawk expects same number of pins in the APL data for a cell as its LEF. It gives this WARNING when it sees multiple P/G pins in APL data for a 1p1g cell.

SEVERITY & IMPACT
In the above case of a mis-match, RedHawk does not use the APL profiles for the cell if POWER_MODE APL/MIXED is used. It uses the .lib data only.

ACTIONS
Please check the APL profiles to confirm the cause of mismatch in the number of pins as compared to LEF.

#----------
! PWR-100
reactivated non-switching instances that were not covered in STA

DESCRIPTION
RedHawk by default treats the instances that are not covered in STA file as non-switching instances (with zero frequency). But there are a few cases that such an instance will be reactivated as a switching instance.

The scenarios include:
1) the instance is active in VCD file so it should switch,
2) the instance has power specified BLOCK_POWER_FOR_SCALING in GSR,
3) data mismatch in STA file. One example is the driver has zero frequency but its fanouts have running frequency. The incurred power will be annotated to the driver and force it to be a switching instance.

SEVERITY & IMPACT
This is a data integrity issue that should be fixed. The reactivation process potentially makes the static power more accurate but does not help the dynamic analysis because the timing windows are still not available. For these instances, the timing windows are set to the whole cycle.

ACTIONS
Users may check out adsRpt/reactivate.inst for the list of instances that are reactivated. For these instances, the following steps can be taken to nail down the data issues:
1) If VCD files were used to generate the toggle file, users may verify if these instances are switching in the VCD files. If yes, the problem is found and the inconsistency between VCD and STA files must be fixed.
2) If BLOCK_POWER_FOR_SCALING is used, users may check if this instance or its master cell is specified with a fixed power. If yes, the problem is found and the inconsistency between BLOCK_POWER_FOR_SCALING and STA files need to be fixed.
3) Users may check the STA files to verify if any fanouts of these instances are switching but the driving pins of these instance are not. If yes, the problem is found and the data inconsistency in STA files must be fixed.
4) If none of the above applies, please submit your data to Apache Design Solutions for further investigation.

! PWR-106
Vdd (VDDx) for instance (<>) (cell <>), not defined in GSR.

DESCRIPTION
This error code implies that a set of instances in a design belonging to a particular cell master does not have its VDD pin/net defined in the GSR file.

SEVERITY & IMPACT

Warning. This error is important if connectivity is missing from the power net(s) specified in the GSR file to the pin/net highlighted through this error message. If however, there is physical and logical connectivity, this error message can be ignored.

ACTIONS

To verify, user should check if logical and physical connectivity exists for the specified instances to the power grid networks specified in the GSR.

#----------

! PWR-107

instance_name_pin = xx/Z has unreasonable small input slew, < 0.005 ns; set it to 0.005 nsec.

DESCRIPTION

This error message implies that for a particular instance, the input slew as specified through the STA data is less than 5ps which may be incorrect. In this case, the input slew is re-set to 5ps.

SEVERITY & IMPACT

If this only happens to a small number of input pins, it may have little impact on the static IR or DvD results.

ACTIONS

If this is the case for large number of input pins, the user should verify the correctness of the STA information especially pertaining to the slew data.

#----------

! PWR-113

Power for this block is zero.

DESCRIPTION

No power information is available for the block highlighted by the error message. This error message is typically seen for higher level blocks that are made up of standard cells or other leaf cells and would not typically have a power specification. It is also seen for other cells in the design which typically do not have any power consumption like blocks made up of antenna cells or blocks in the scribering, etc. For these cases, this error message is not critical and can be ignored.

SEVERITY & IMPACT

If the highlighted block should have power like an IP block and the block participates in the VDD/VSS nets for the analysis, it will impact the DvD drop surrounding the IP block.

ACTIONS

If the specified block should have power, please examine why there is no power in the provided .lib or use BLOCK_POWER_FOR_SCALING in GSR to specify the block power.

#----------
! PWR-114
Following cells are not defined in .lib

DESCRIPTION
The cells specified through this error message are not found in any of the libraries provided to RedHawk through the import_lib_dir command.

SEVERITY & IMPACT
This would cause a loss of information for this cells and errors in the RedHawk analysis if the cells highlighted in the error message are present in the design.

ACTIONS
If the cells highlighted by the error message are not present in the design, then this error message can be ignored. If it does and is not filler cells, antenna cells, etc, its corresponding .lib should be provided or use custom lib file to specify its property.

#----------
! PWR-116
For instance, <instance_name> (cell <cell_name>), the difference between APL-based and .lib-based power is more than 100%. .lib-based leakage and instance power, 0 , 0
APL-based leakage and instance power, 5.3541e-10 , 5.3541e-10

DESCRIPTION
This message indicates that power difference is considered large between .lib-based and APL-based power calculation.

SEVERITY & IMPACT
If the total power difference between .lib-based and APL-based power calculation is large (say, more than 25%), users may need to investigate whether it is caused by no sufficient power information in .lib or a mismatch between the PVT conditions used in APL and .lib.

ACTIONS
The most often case is that the original .lib does not have power information for the cell and therefore 0 leakage or internal power for the instance that uses the cell. If this is the case, users can examine the corresponding .lib file and find out why there is not power specification for the cell. If there are a large number of instances issued for PWR-116, users may need to examine the conditions used for APL characterization if they are the same as used in .lib files. The conditions include

----
! PWR-122
IPF file has invalid instances

DESCRIPTION
The IPF file given has instances which are not a part of the design

SEVERITY & IMPACT

Its a warning during power calculation stage.Redhawk will ignore this warning and proceed with IPF file reading.

ACTIONS

Check the IPF coverage reported.Check your IPF file for errors and correct it.

! PWR-123

Warning message showing low IPF coverage

DESCRIPTION

If user has used an Instance Power File in the design , then after importing the file , if the IPF coverage is less than 50%, Power Calculation Engine will show the percentage of instances in the design covered by IPF file.

SEVERITY & IMPACT

Its a warning message.Redhawk will assume zero power for the instances in the design unless POWER_MISSING_IPF_POWER is set to 1

ACTIONS

Check the IPF file for low coverage and correct it

! PWR-124

cell, KM1CKI1DU , with 2 power/ground pins in cell.cdev does not match the number of power/ground pins, 3 , in LEF.

DESCRIPTION

RedHawk expects APL to be generated for all the power/ground pins of a cell, which are defined in the LEF. If for a cell, there is a mismatch between the number of pins in LEF and APL, then tool issues out this warning.

SEVERITY AND IMPACT

If the domains connected to the pins, which are not present in APL are analyzed, then the analysis will be inaccurate.

ACTIONS

Re-characterize APL for those cells after including all the LEF pins.

! PWR-125

WARNING(PWR-125): cell, Tercam_256x106 , with 1 power/ground pins in cell.current with non-pin-associated format does not match the number of power/ground pins, 4 , in LEF.

DESCRIPTION

RedHawk expects same number of pins in the APL data for a cell as its LEF. It gives this WARNING when it sees only 1p1g in APL data for a multirail cell.

SEVERITY & IMPACT
In the above case of a mis-match, RedHawk does not use the APL profiles for the cell if POWER_MODE APL/MIXED is used. It uses the .lib data only.

ACTIONS
Please check the APL profiles to confirm the cause of mismatch in the number of pins as compared to LEF.

#----------
! PWR-126
Intended final power from BLOCK_POWER_FOR_SCALING generates a toggle rate greater than 1 for some non-clock-network instances.

DESCRIPTION
If the power specified for an instance using BLOCK_POWER_FOR_SCALING settings in GSR results in an unrealistic toggle rate (ie larger than 1.0, for a non-clock-network instance),this warning is flagged.

SEVERITY & IMPACT
Major warning. The power setting for instances may be unrealistic.

ACTIONS
Please check & correct the power settings before the next run.

! PWR-131
WARNING(PWR-131): In imported instance power file, instance or cell name, lvds/adsU1, has invalid pin name defined, VDD!

DESCRIPTION
This WARNING states that for the instance lvds in the design, there is a pin-name VDD! used in the BPFS specification which did not match any pin-name in the LEF for this cell.

SEVERITY & IMPACT
RedHawk gives these warning messages and ignores the BPFS for the above instamce in power calculation.

ACTIONS
Please sepcify the correct pin-name in the BPFS power file and proceed.

#----------
! PWR-136
cell, cell_name , with 5 power/ground pins in cell.cdev

DESCRIPTION
This warns the user to check the data given in the LEF and APL file matches for the given cell, as there is a mismatch in the number of pins given in both files

SEVERITY & IMPACT
It is a critical message, as the leakage current for the mismatching pins for

such cells will be considered as 0 if POWER_MODE is set to APL

ACTIONS
Please make sure the data given in APL and LEF file, matches with respect to power/ground pins

----
! PWR-138
cell, <>, with <> power/ground pins in cell.current

DESCRIPTION
The number of power/ground pins in the LEF doesn't match the number of PG pins in APL data.

SEVERITY & IMPACT
The pin whose current profile is not captured in APL, its current magnitude will be set to 0 if POWER_MODE APL/MIXED is specified.

ACTIONS
Please verify that APL has current profile for all critical PG pins.

! PWR-139
Instance <inst> is not in the design or is a floating instance.

DESCRIPTION
Either the instance name mentioned through user inputs cannot be found in the design or the instance is a floating instance. It is not connected to any PG domains.

SEVERITY & IMPACT
This instance will be excluded from the analysis

ACTIONS
Check whether the instance name is correct in the user inputs. If yes, check whether the instance is connected to PG domains.

#----------
! PWR-157
For block, DDR6A23Q1S1, intended final block power, 0.03 , is smaller than the block constant power, 0.0457748.

DESCRIPTION
RedHawk by default computes some power based on the frequency, toggle rate, load and slew. This warning is displayed when the RedHawk computed power is different from the user-assigned power.

SEVERITY AND IMPACT
The warning allows user to verify whether the power assignment specified by him is taken into account by the tool or not.

ACTIONS
User can change the power value, if it doesn't match his desired value.

! PWR-158
For block, <>, intended final block power is,<>.However, the
original block power, <>, is equal to the block constant power, <>.
No scaling is possible.

DESCRIPTION
Unreasonably high power target is specified for the block in
BLOCK_POWER_FOR_SCALING.

SEVERITY & IMPACT
The user specified power number will be discarded and the original block
power will be used in analysis.

ACTIONS
Please verify the power number of the block.

! PWR-161
No matched net name <> in design for instance <>.

DESCRIPTION
The instance doesn't have any Power pin connected to net <>.

SEVERITY & IMPACT
The power number specified for this instance for net <> in the
INSTANCE_POWER_FILE will be discarded.

ACTIONS
Please specify a net name which connects to a power pin of that
instance in INSTANCE_POWER_FILE.

! PWR-163
BLOCK_POWER_FOR_SCALING given for unknown instances/cells

DESCRIPTION
If BPFS is given for instances or cells which do not exist in the design,
then this warning message is given

SEVERITY & IMPACT
Its a warning.Redhawk will ignore this warning and proceed with power calculation

ACTIONS
Check your BLOCK_POWER_FOR_SCALING settings for this mismatch and correct it.

! PWR-164
In BLOCK_POWER_FOR_SCALING, instance or cell name, <>,
has an invalid pin defined, <>.

DESCRIPTION
The pin specified for that instance or cell doesn't exist in LEF of that cell.

SEVERITY & IMPACT
The power/current specified for that pin will not be used in analysis.

ACTIONS
Please correct the pin name for that instance/cell.

! PWR-201
Assigned Power is less than leakage power for an instance

DESCRIPTION
If user has assigned power for an instance through IPF or BPFS and it the
power value is less than the leakage power of the instance obtained through
.lib , then this warning is issued

SEVERITY & IMPACT
Warning, the leakage power for the instance is reset to the intended power

ACTIONS
Check the IPF/BPFS settings and correct it

! PWR-202
Assigned Power is less than leakage power for an instance power pin with
multiple power pins

DESCRIPTION
If user has assigned power for an instance pin through IPF or BPFS and it
the power value is less than the leakage power of the instance pin obtained
through .lib , then this warning is issued

SEVERITY & IMPACT
Warning, the leakage power for the power pin is reset to the intended power.

ACTIONS
Check the IPF/BPFS settings and correct it

! PWR-211
For instance, <>, output pin load capacitance, <>, is larger than cell max_cap, <>.

DESCRIPTION
The pin load capacitance for that cell is larger than the max_cap defined in lib.

SEVERITY & IMPACT
Higher load cap will have impact on power calculation & voltage drop analysis.

ACTIONS
Please verify if the load cap is correct.

! PWR-300
Time duration in BLOCK_TOGGLE_FILE, xyz, is not consistent with the time span, abc, specified for VCD file(s).

DESCRIPTION
Time duration specified in BLOCK_TOGGLE_FILE created using vcdtrans utility is not consistent with time span specified in VCD_FILE(s) in GSR.

SEVERITY & IMPACT
Time duration in BLOCK_TOGGLE_FILE is used for analysis.

ACTIONS
It is strongly recommended to avoid using BLOCK_TOGGLE_FILE and instead use VCD_FILE, BLOCK_VCD_FILE GSR keywords to achieve the same.

! SIM-301
No STA data is read in the vector-less simulation

DESCRIPTION
In simulation, all instances have output timing windows of whole cycles, which indicates that no STA timing window information is available.

SEVERITY & IMPACT
This is an INFO message. If STA is provided in early design set up, this message shows something is wrong in STA processing.

Without a timing window defined for an instance, the whole cycle will be assumed. Hence, the output of the instance could switch at any time. Since switching time is important to simulation result, no valid timing window information could reduce the result accuracy.

ACTIONS
1. Make sure no error in STA parsing.
2. Try to create valid STA.

! SIM-302

None of instances have timing windows

DESCRIPTION
In the vector-less simulation, none of instances have output timing windows, which indicates that they are all in CONST state (or will not switch) according to STA.

SEVERITY & IMPACT
This is an INFO message. If the STA provided has valid timing windows for some instance, this message shows something is wrong in STA processing.

ACTIONS
Make sure no error in STA parsing.

! SIM-322
Intrinsic decap of cell is larger than 10pF

DESCRIPTION
The intrinsic decap of the cell is larger than 10pF. This large decap will be considered in the vector-less simulation for each instance of the cell.

SEVERITY & IMPACT
This is a warning message. Normally, a cell should not have its intrinsic decap value larger than 10pF unless it is a large macro block or a large memory. If the cell is not, this message shows something is wrong in decap characterization or data processing.

ACTIONS
Verify the cell.cdev for correctness especially for standard cells.

! SIM-325
High power cell XXX has no current profile

DESCRIPTION
The cell, which consumes a large power according to its power table in .Lib, does not have current profile characterized. A .Lib-based profile will be used in the vector-less simulation for the instances of the cell.

SEVERITY & IMPACT
This is a warning message. If the cell is characterized, this message shows something is wrong data processing.

It is important to have current profiles for high power cells, especially memories to achieve higher accuracy.

ACTIONS
1. Make sure no error in characterizing the cell.

2. Characterize the cell.

! SIM-401
No STA data is read in the VCD-based simulation

DESCRIPTION
In simulation, all instances have output timing windows of whole cycles, which indicates that no STA timing window information is available.

SEVERITY & IMPACT
This is an INFO message. If STA is provided in early design set up, this message shows something is wrong in STA processing.

Without a timing window defined for an instance, the whole cycle will be assumed. Hence, the output of the instance could switch at any time. Since switching time is important to simulation result, no valid timing window information could reduce the result accuracy.

ACTIONS
1. Make sure no error in STA parsing.
2. Try to create valid STA.

! SIM-402
None of instances have timing windows

DESCRIPTION
In VCD-based simulation, none of instances have output timing windows, which indicates that they are all in CONST state (or will not switch) according to STA.

SEVERITY & IMPACT
This is an INFO message. If the STA provided has valid timing windows for some instance, this message shows something is wrong in STA processing.

ACTIONS
Make sure no error in STA parsing.

! SIM-422
Intrinsic decap of cell is larger than 10pF

DESCRIPTION
The intrinsic decap of the cell is larger than 10pF.
This large decap will be considered in the VCD-based simulation for each instance of the cell.

SEVERITY & IMPACT
This is a warning message. Normally, a cell should not have

its intrinsic decap value larger than 10pF unless it is a
large macro block or a large memory. If the cell is not, this
message shows something is wrong in decap characterization or
data processing.

ACTIONS
Verify the cell.cdev for correctness especially for standard cells.

! SIM-425
High power cell XXX has no current profile

DESCRIPTION
The cell, which consumes a large power according to its power table in
.Lib, does not have current profile characterized. A .Lib-based profile
will be used in VCD-based simulation for the instances of the cell.

SEVERITY & IMPACT
This is a warning message. If the cell is characterized, this
message shows something is wrong data processing.

It is important to have current profiles for high power cells, especially
memories to achieve higher accuracy.

ACTIONS
1. Make sure no error in characterizing the cell.
2. Characterize the cell.

! SLB-015
Default pad cell to input pad cell.

DESCRIPTION
If a pad have no direction specified, assume it is an input pad.

SEVERITY & IMPACT
Warning. No impact for RedHawk result but may have an impact on
global IOSSO simulation.

ACTIONS
Users should specify pad pin direction in .lib.

! SLB-016
Pin cannot be unate on function.

DESCRIPTION
This message reports cell vector unidentified.

SEVERITY & IMPACT
Warning. APL may not run on this cell. Users may need to provide vector
for custom APL.

ACTIONS
Users can specify cell vectors in custom lib file.

! SLB-017
rise_power and fall_power should be both defined in internal_power group at line 134161. Skip the error.

DESCRIPTION
Input .lib have only rise or fall power tables under internal_power section.
Ideally both rise_power and fall_power should be defined in .lib

SEVERITY & IMPACT
Analysis may have inaccuracy.

ACTIONS
Try to get correct/complete libs and rerun.

! SLB-019
LIB pin doesn't exist in the corresponding LEF macro.

DESCRIPTION
This message reports pin inconsistency between LEF and LIB.
Lib cell pin doesn't exist in the corresponding LEF macro.

SEVERITY & IMPACT
Warning. LIB pins that are not defined in LEF are ignored.

ACTIONS
Users should make sure that LEF and LIB are consistent.

! SLB-025
File ".apache/apache.import.libs" is empty.

DESCRIPTION
Liberty files were not imported correctly.

SEVERITY & IMPACT
Critical warning. This will impact analysis.

ACTIONS
Please make sure that the lib files(if any) are successfully imported. This may have dependencies on other error/warning.

! SLB-026
Failed to access the directory.

DESCRIPTION
The directory specified in .libs file doesn't exist or the user has no permission to access it.

SEVERITY & IMPACT
Critical. RedHawk will stop after lib import.

ACTIONS
Users should check the existence or permission of the directory.

#----------
! SLB-027
There are no lib files present in the mentioned path in gsr

DESCRIPTION
Synopsys-format library files contain logical descriptions of library cells, with power and timing tables, and are used for power calculation. RedHawk is not able to find any lib files in the mentioned path

SEVERITY & IMPACT
This warning is important and it is advised to provide correct .lib files.
RH will not calculate power for those cells with missing libs.

ACTIONS
Cells with missing libs will be reported in <adsRpt/apache.refcell.noLib >.
Warning can be avoided by providing correct .lib path

! SLB-033
Lib file failed semantic check.

DESCRIPTION
A suite of semantic checks are run against lib files.
The checks supported by Liberty parser include group/attribute names and ownership, attribute type and range, etc.

SEVERITY & IMPACT
Critical. RedHawk will stop for semantic errors.

ACTIONS
Users can reference Synposys Liberty User Guide to fix the errors.
It is possible that some attributes or groups are not supported by Liberty parser, in that case please contact Apache for support.

! SLB-035
Attribute nom_voltage is not defined. Use default 1.2V.

DESCRIPTION
If nom_voltage is not defined in .lib, set it to 1.2V.

SEVERITY & IMPACT
Warning. Power can be affected if nom_temp is not 25 for this .lib file.

ACTIONS
Users can define the attribute nom_voltage in .lib file or in custom lib file.

! SLB-037
Attribute nom_temperature is not defined. Set it to 25.

DESCRIPTION
If nom_temperature is not defined in .lib, set it to 25.

SEVERITY & IMPACT
Warning. Power can be affected if nom_temperature is not 25 for this .lib file.

ACTIONS
Users can define the attribute nom_temperature in .lib file or in custom lib file.

! SLB-040
Missing cell power info.

DESCRIPTION
This message reports LIB cells that have no power info.

SEVERITY & IMPACT
Warning. Missing power may affect Power, IR, and EM results.
If the reported cells are referenced in the design, cell instances won't have power in .lib-based power calculation.

ACTIONS
Users should check the power for the reported cells.
The missing power cells are listed in adsRpt/apache.noPWR.

! SLB-041
The cells in .lib are not defined in LEF.

DESCRIPTION
This message reports LIB cells that have no LEF definitions.

SEVERITY & IMPACT
Warning. Since only cells with physical geometry are considered by RedHawk, the reported cells are ignored.

ACTIONS
Users can define the reported cells in LEF.
The missing LEF cells are listed in adsRpt/apache.noLEF.

! SLB-042
The cells in .lef are not defined in LIB.

DESCRIPTION
This message reports LEF cells that have no LIB definitions.

SEVERITY & IMPACT
Warning. The reported cells and their pins may have incorrect attributes.

ACTIONS
Users can define the reported cells in LIB or in custom lib file.
The missing LIB cells are listed in adsRpt/apache.noLIB.

! SLB-043
Set LIB cell pin direction to input if it is not defined.

DESCRIPTION
If pin direction is not defined in LIB, set it to input.

SEVERITY & IMPACT
Warning. Pin direction may be set incorrectly.

ACTIONS
Define pin directions in LIB files or define them in custom lib file.

! SLB-048
Sanity check on cell vectors.

DESCRIPTION
Report errors in cell vectors.

SEVERITY & IMPACT
Warning. Apl may not operate on the specified cell.

ACTIONS
Users may do this cell via custom apl flow, or define vectors in custom lib file.

! SLB-052
Report duplicated cells defined in LIB.

DESCRIPTION
For cells that have multiple LIB definitions, cell names are logged in <adsRpt/apache.dupLIBCell>.

SEVERITY & IMPACT
Warning. Cells and their pins may have wrong attributes.

ACTIONS
Users should remove duplicated cell definitions.

! SLB-065
Report cells that are defined in LIB and have triggered edges unidentified.

DESCRIPTION
The reported cells are listed in <adsRpt/apache.noTriggerEdge>.
See also ITG-008.

SEVERITY & IMPACT
Warning. No impact if cells are not referenced in design.

ACTIONS
Define cell triggered edges in custom lib file.

! SLB-066
Consistence check on LEF and LIB pins. Report pins that are defined
in LIB cells but not defined in the corresponding LEF cells.

DESCRIPTION
LEF pins should be consistent with LIB's. The LEF undefined pins are
listed in <adsRpt/apache.noLefPins>.

SEVERITY & IMPACT
Warning. No impact if cells are not referenced in design.

ACTIONS
Correct LEF or LIB data.

! SLB-067
Consistent check on LEF and LIB pins. Report pins that are defined
in LEF cells but are not defined in the corresponding LIB cells.

DESCRIPTION
LEF pins should be consistent with LIB's. The LIB undefined pins are
listed in <adsRpt/apache.noLibPins>.

SEVERITY & IMPACT
Warning. The reported pins may have wrong attributes. No impact if
cells are not referenced in design.

ACTIONS
Correct LEF or LIB data.

! SLB-068
Failed to generate cell state table.

DESCRIPTION
Errors occur while generating state tables.

SEVERITY & IMPACT
Warning. Apl may not operate on the reported cells.

ACTIONS
Users can run these cells via custom apl flow, or define
vectors in custom lib file.

! SLB-070
Set default power rail "<pin>" to 1.200.

DESCRIPTION
RH is not able to find power rail info for the given pin in .lib. The
"rail_connection" define all multiple power supply and ground domain pins
used in a cell. It maps the library power supply name to the cell LEF
power pin name to which it is connected.

SEVERITY & IMPACT
if power_rail information is missing RH assumes a default value .

ACTIONS
Correct the .lib and add power_rail information

#----------
! SLB-077
Specified instance not defined in lef

DESCRIPTION
Redhawk will give this warning when a cell pgarcs have been defined in a custom
LIB, but the cell has not been defined in LEF. In such cases, Redhawk will ignore
the pgarc information specified in the LIB. Review if the cell figures in
adsRpt/noLefLib, adsRpt/noLef, adsRpt/noLib, adsRpt/noPwr reports.

SEVERITY & IMPACT
Warning. Cell will be excluded from analysis.

ACTIONS
Input the lef view of the cell.

! SLB-078
Invalid pin name "" in cell "" pgarcs. Cell pgarcs ignored.

DESCRIPTION
The pin name <> doesn't exist in the LEF file of that cell. The

pgarc definition of that cell will be ignored.

SEVERITY & IMPACT
Critical warning. This may impact the analysis.

ACTIONS
Please correct the pgarc definition for that cell.

! SLB-080
Invalid pin name <pin_name> in cell <cell_name> rail _connection.

DESCRIPTION
The <pin_name> is not specified in the LEF as USE POWER/GROUND pin for the specified cell <cell_name>,but used in the the rail_connection statement .lib section for the cell.

The 'rail_connection' attribute must define all multiple power supply and ground domain pins used in a cell. It maps the library power supply name to the cell LEF power pin name to which it is connected.

SEVERITY & IMPACT
Warning generated during setup design while importing libraries.
Rail connection will be ignored.

ACTIONS
Specify the PIN as USE POWER/GROUND in DEF and then use in rail_connection.

! SLB-082
PG-arc definition contain a non power/ground pin.

DESCRIPTION
The PG-arc definition should be from a power pin to ground pin.

SEVERITY & IMPACT
APL for cells with wrong PG-arc will be ignored from the simulation.

ACTIONS
Provide correct PG-arc through custom lib.

! SLB-083
LIB File has negative leakage power number in its tables

DESCRIPTION
Redhawk expects the leakage power numbers in LIB files to be positive.If any leakage power table in the lib files have negative power , this warning will be given during import library files stage

SEVERITY & IMPACT
Its a warning.Redhawk will ignore this warning and proceed with LIB file reading.
The negative leakage power will be ignored and leakage number from that table will be reset to zero

ACTIONS
Check if your LIB file is having bogus values and correct it.

! SLB-091
Incomplete PG-arc definition in UPF .lib.

DESCRIPTION
RedHawk derives the PG-arc for the cells from the UPF pg pin definition. For the above cell the PG arc is incomplete in the UPF .lib

SEVERITY & IMPACT
APL for the cells with incomplete PG arc will be ignored in the simulation.

ACTIONS
Use correct UPF .libs and rerun RedHawk.

! SLB-099
Duplicated library definition for cell "<cell_name>".

DESCRIPTION
More than one library definition exists for the same cell. This can happen say if multiple .lib files are being imported and different definitions for the same cell exist in different .lib files.

SEVERITY & IMPACT
Warning.The last definition would be accepted and rest ignored.

ACTIONS
Please see adsRpt/apache.dupLIBCells to see the cells having multiple definitions.

! SNO-100
TOP SPICE file done.
Please check the control settings and input waveforms before starting your SPICE simulations.

DESCRIPTION
An information telling users that spice netlisting is complete.

SEVERITY & IMPACT
No impact.

ACTIONS
As described in the message.

! SPF-104
driverPinInst of net is NULL

DESCRIPTION
This error may happen during importing SPEF/DSPF files. It indicates a net covered in the SPEF/DSPF file does not have a driver. Usually this is due to some data error in LEF/DEF or SPF.

SEVERITY & IMPACT
The parasitics of this net will be ignored. Because the parasitics is a local property, the impact to the final analysis will be insignificant as long as the nets having this problem are few.

ACTIONS
Users may ignore this error if there are only a few but need to pay extra attention if there are many. Since this usually is due to data error, it is recommended to take a few nets and verify why they do not have drivers.

! SPF-115
Failed to extract C1/R/C2 for cell:net.

DESCRIPTION
RedHawk extracts C1/R/C2 from the RC trees defined in SPEF/DSPF. This error reports the RC trees can not be converted to C1/R/C2 model. Usually this is due to some data errors in the RC trees.

SEVERITY & IMPACT
The drivers to this net will not have C1/R/C2 computed from the SPEF/DSPF. Instead, RedHawk will estimate the parasitics using other models like steiner tree. Because the parasitics is a local property, the impact to the final analysis will be insignificant as long as the nets having this problem are few.

ACTIONS
Users may ignore this error if there are only a few but need to pay extra attention if there are many.

! SPF-117
The nets/lines in <xx.spef> cannot be mapped to design.

DESCRIPTION

This message reports (1) the number of nets that do not exist in the corresponding DEF (or the full design for a flattened SPF file), and (2) the number of lines that contain missing design objects like nets, pins, and pinInsts. The detailed list of all the missing objects is provided in adsRpt/apache.rcBogus for reference.

SEVERITY & IMPACT

Warning. Important if the percentage of nets/objects that are missing are large. The parasitics associated with these design objects will be ignored. For the nets that are not annotated from the SPF files, RedHawk will estimate their parasitics using Steiner Tree.

ACTIONS

Users need to check out adsRpt/apache.rcBogus to verify the consistency between SPF files and DEF files, including hierarchical dividers, pin/bus delimiters, and name case sensitivity. Also, users need to check the completion of DEF files.

! SPF-118

The nets in <xx.spef> have incomplete drivers.

DESCRIPTION

This message reports the number of nets which have inconsistent drivers between SPF and DEF. Most likely this applies to a hierarchical net whose sibling nets are not fully covered by SPF files. Another often scenario is the confusing pin directions due to missing .lib, as a consequence of NET-412. The inconsistency is reported in adsRpt/apache.rcMismatch.

SEVERITY & IMPACT

In the case of only partial parasitics is available, the partial parasitics (from SPF files) will be ignored, and RedHawk will estimate the parasitics using Steiner trees instead.

In the case of confusing pin directions, the SPF file is honored.

The severity of this warning depends on the percentage of nets that are having problems. Also, in general, Steiner tree estimation on a hierarchical net is less accurate than the estimation on a local net. So there is some accuracy penalty when the SPF files are not complete.

ACTIONS

Users need to check out the nets listed in adsRpt/apache.rcMismatch and investigate what cause the mismatch between SPF and DEF files. Most likely the SPF files are not complete or .lib is not complete, but it could also because the design objects in SPF cannot be mapped to DEF properly due to inconsistent name case sensitivity, mismatched hierarchical divider, or inconsistent pin/bus delimiter.

! SPF-119

The nets in <xx.spef> failed in calculating RC.

DESCRIPTION
RedHawk internally calculates C1/R/C2 for each driver of a net using the corresponding RC networks specified in SPF files. This message reports the number of nets whose RC networks cannot be successfully converted into Pi-models for the drivers. This often happens due to incomplete or disjointed RC networks.

SEVERITY & IMPACT
The parasitics of these nets (from SPF files) will be ignored, and RedHawk will estimate the parasitics using Steiner trees instead. The severity of this warning depends on the percentage of nets that are having problems.

ACTIONS
Users need to check out the nets listed in adsRpt/apache.rcFail and investigate what may cause the failure. One checking that users can do is investigating if the RC networks are disjointed. Otherwise users may need to submit the data to Apache Design Solutions for more detailed checking.

! SPF-121
The number of nets in the design have no drivers.

DESCRIPTION
This message summaries the number of nets in the design that have no drivers. This info is provided for data integrity checking.

SEVERITY & IMPACT
The impact to the final analysis should be insignificant as long as the number is relatively small.

ACTIONS
Users may check adsRpt/apache.rcNoDriver and verify why the nets dont have drivers. If the number is small, users however may just ignore this message.

! SPF-122
The data errors during RC annotation.

DESCRIPTION
RedHawk internally calculates C1/R/C2 for each driver of a net using the corresponding RC networks specified in SPF files and annotate the C1/R/C2 to the corresponding drivers. This message reports the number of errors during the annotation. Usually this happens due to the data inconsistency between SPF and DEF.

SEVERITY & IMPACT
The severity of this warning depends on the percentage of nets that are

having problems. All the drivers that cannot be annotated successfully will have the estimated parasitics using Steiner tree.

ACTIONS
Users may check out the errors listed in adsRpt/apache.rcLog for the details, or submit the data to Apache Design Solutions for further checkup.

! SPF-123
The nets that have zero capacitance values.

DESCRIPTION
This message reports the nets that have zero capacitance values. This most likely is due to incomplete SPF files. The nets that have zero capacitance will get the estimated parasitics using Steiner tree model.

SEVERITY & IMPACT
The severity of this warning depends on the percentage of nets that are having zero cap values. All the nets that have zero capacitance will will get the estimated parasitics using Steiner tree model.

ACTIONS
Users may check out the nets listed in adsRpt/apache.rc0Net for the details and verify they are not correctly covered in SPF files.

! SPF-411
Spef is provided for a block which is not there in the design.

DESCRIPTION
The CELL_RC_FILE definition in the gsr has a block which is not instantiated in the design, This could be due to wrong block name provided in the GSR.

SEVERITY & IMPACT
RedHawk will not read signal parasitics from this spef.

ACTIONS
Make sure the spef is provided for a block which is instantiated in the design.

! SPF-452
The cap value is unusually small.

DESCRIPTION
RedHawk performs the data integrity checking on C values in SPF files. If a C value is less than 1e-24 and the associated C_UNIT is less than 1,

RedHawk would issue this warning message to flag a potential data error. For example, sometimes users may modify the SPF files without noticing the C_UNIT and assign the real C value to the field, which caused the combined value way too small.

SEVERITY & IMPACT
This is a warning. If the reported value indicates a real error, it will affect power calculation and downstream analysis, especially if the complete SPF is wrongfully scaled.

ACTIONS
Users need to verify the C_UNIT and the values.

! STA-106
STA File has information for PG pins for an instance

DESCRIPTION
STA_FILE expects the timing window and slew to be present for I/O pins. If any PG pin is found to be having timing info , then this warning message will be given

SEVERITY & IMPACT
Its a warning.Redhawk will ignore this warning and proceed with STA_FILE reading.The timing info for this pin will be ignored

ACTIONS
Check if your pin has wrong type in LEF/DEF and correct it.

! STA-107
Net in the STA file is a PG net.

DESCRIPTION
This error message identifies a TW line in the STA file associated with a pinInst which is connected to a P/G net. Usually P/G net is for tie-high (or tie-low) and stays constant. There should not be a TW associated.

SEVERITY & IMPACT
This TW will be ignored.

ACTIONS
Users should verify why the pinInst connecting to the P/G net has TW line in the STA file.

! STA-108
illegal format - <>

DESCRIPTION

Redhawk found incorrect syntax in STA_FILE.

SEVERITY & IMPACT
Redhawk will report error and stop running.

ACTIONS
Please check the reported format in input STA file, correct it, and rerun.

! STA-117
Pin <pin_name> dangling.

DESCRIPTION
The pin <pin_name> does not have any connectivity specified in DEF, although slew is assigned for the same in the STA file.

SEVERITY & IMPACT
Warning generated during setup design while importing STA file. Instance won't be assigned any slew.

ACTION
Please provide the connectivity in DEF.

! STA-411
Failed in processing the STA timing file.

DESCRIPTION
One or more errors are encountered when parsing and processing the STA timing file specified in GSR.

SEVERITY & IMPACT
This is a serious error and the RedHawk will exit. The errors encountered will compromise the correctness and the accuracy of the subsequent power analysis and simulation.

ACTIONS
Users should review the error message or messages immediately preceeding the exit to isolate the reason of fail. The potential causes are, but not limited to:
(1) The specified STA file does not exist or have the correct permission.
(2) There are more than one STA file specified.
(3) No write permission to adsRpt subdirectory.
(4) Inconsistent EXTRACT_CLOCK_NETWORK specification between GSR and STA file.
(5) Serious syntax errors.
(6) Too many clock domains (limitation is 253).
(7) Internal error in propagating timing data into GDS blocks.

! STA-452

ClockRoot specified in STA timing file cannot be found in the design.

DESCRIPTION
The design object specified in ClockRoot does not exist in the design. The possible causes include name case mismatch, incorrect/missing hierarchical path, hierarchical divider or pin/bus delimiters mismatch, incomplete DEF files, or simply a typo.

SEVERITY & IMPACT
Important to fix or this clock root will be ignored. The direct impact is to a wrong clock network extraction and frequency propagation.

ACTIONS
Do the following checking:
(1) any typo?
(2) the specified design object does exist in DEF
(3) the name case is consistent to DEF. Also, check if NAME_CASE_SENSITIVITY is set to 0 in GSR. In this case, all design object names are remapped to uppercase.
(4) correct hierarchical path name with consistent hierarchical divider.
(5) consistent pin delimiter and bus delimiter, if the clock root is a pin or pinInst.

! STA-453
ClockRoot<> in timing file is not connected to a net.

DESCRIPTION
The clock defined by clockRoot is not connected to any net.

SEVERITY & IMPACT
Clock network identification will be impacted when EXTRACT_CLOCK_NETWORK is used.

ACTIONS
Please verify if the STA file and DEF are in sync.

! STA-461
<count> insts in the design are not covered in the timing file(s)

DESCRIPTION
The timing window vaues are not specified for some instances in the timing file given as input to Redhawk. Redhawk cannot hence get the exact switching details of the instance from timing file.

SEVERITY & IMPACT
Static results will not be affected, but dynamic simulation results will be affected by absence of timing window. Redhawk will assume that the instances, for which timing window is absent, are switching at active edge of the clock. Hence more drop will be seen during clock edge if many instances are missing this data.

ACTIONS
Please include the timing window details of these instances in timing file
and rerun analysis

! STA-462
<count> insts in the design have clock pins not covered in the timing file(s)

DESCRIPTION
The timing window values for clock pins are not specified for some instances
in timing file given as input to Redhawk. Redhawk finds the timing window for
clock transistion of each instance from the timing file specified.

SEVERITY & IMPACT
Redhawk will assume the clock transition for these instances to occur at the
starting of simulation time for the fequency value specified. This will affect
dynamic simulation results. More drop obtained at the starting of simulation or
at each clock edge might be caused due to this issue.

ACTIONS
Please include the clock pin timing window details for these instances in
timing file and rerun the analysis

! STA-463
(<count>) insts in the design have clocks arriving after the outputs are ready

DESCRIPTION
Some instances in the design are having timing window values of clock and
output such that clock arrives after the output is ready. This is because
the design is not timing closed.

SEVERITY & IMPACT
The design is not timing closed. Redhawk will be giving output values based
on this design.

ACTIONS
Please make the design timing closed and rerun the analysis using Redhawk

----
insts in the design have min arrival larger than max.
! STA-466
DESCRIPTION
Certain instances have minimum TW arrival time defined after the maximum
TW arrival time. These cells are listed in <adsRpt/apache.twMinMax>.

SEVERITY & IMPACT
In STA file, the min arrival time should be smaller than or equal to the max
arrivial time. When a min arrival time is larger than the paired max arrival
time, it would trigger STA-466 and asim internally would swap the min/max
pair to process.

ACTIONS
Please check <adsRpt/apache.twMinMax> for details about such instances.

! SUB-045
User requested freq too small, reset to 250MHz. Please increase simulation time.
The lowerest freq cannot be lower than freq_resolution

DESCRIPTION
While generating the frequency domain color map, tool expects user requested frequency to be greater than 1/(simulation-time). If user-requested frequency is lesser than 1/(dynamic simulation time) of the analysis, then it issues out this warning.

SEVERITY & IMPACT
Graph will not be generated.

ACTIONS
Either increase the dynamic simulation time in the analysis using the GSR keyword "DYNAMIC_SIMULATION_TIME" or specify a bigger frequency while generating the color map.

! SUB-046
User requested freq too big, reset to 25000MHz. To increase the freq upper bound, please reduce dynamic simulation time step.

DESCRIPTION
While generating the frequency domain color map, tool expects user requested frequency to be lesser than 1/(time-step) of analysis. If user-requested frequency is greater than 1/(time-step) of the analysis, then it issues out this warning.

SEVERITY & IMPACT
Graph will not be generated.

ACTIONS
Either reduce the dynamic time step in the analysis using the GSR keyword "DYNAMIC_TIME_STEP" or specify a smaller frequency while generating the color map.

! SUB-083
Specifying pwell_config/nwell_config data in SUBSTRATE_TECH_FILE is deprecated.

DESCRIPTION
User has mentioned pwell_config/nwell_config in SUBSTRATE_TECH_FILE. nwell_config/pwell_config blocks should be in the SUBSTRATE_LIBRARY_FILE, since they provide information that is more library than technology related.

SEVERITY & IMPACT
This is not critical since the function of the tool is the same regardless of where the nwell_config/pwell_config blocks appear.But it is advised to keep this information in SUBSTRATE_LIBRARY_FILE

ACTIONS
Move pwell_config/nwell_config data from SUBSTRATE_TECH_FILE to SUBSTRATE_LIBRARY_FILE

#----------
! TBL-000
Hashtable size is too small, may slow down RH.

DESCRIPTION
Some hashtables used in RH have too small bucket size.

SEVERITY & IMPACT
WARNING. It may slow down RH.

ACTIONS
Contact Apache Design Solution's RD with test case.

! TCL-008
Specified net not connected to any pad

DESCRIPTION
You will get this warning when the net specified in the tcl command is not connected to any pad mentioned in pad.current file. This might happen in cases where the net is being driven by internal switch cells.

SEVERITY & IMPACT
Warning. Plot will not be drawn for the specified net

ACTIONS
Remove either the net or -pad from the tcl command

! TEC-015
Attribute 'EM or WIDTH_BASED_EM or POLYNOMIAL_BASED_EM_DC or POLYNOMIAL_BASED_EM_RMS or POLYNOMIAL_BASED_EM_PEAK' is not specified.

DESCRIPTION
None of the EM parameters is specified for that metal/via.

SEVERITY & IMPACT
EM calculation for that particular metal/via will not be performed.

ACTIONS
Please specify EM calculation parameters for that metal/via.

! TEC-019
Dielectric should be defined for METAL/DIELECTRIC.

DESCRIPTION
If ABOVE attribute pecified for METAL/DIELECTRIC, the ABOVE should be defined as dielectric layer in the technology file.

SEVERITY & IMPACT
Error. This error will cause technology file parsing fail.

ACTIONS
Define ABOVE as dielectric in the technology file.

! TEC-023
Metals overlap

DESCRIPTION
The height of two layers is same in the layer stack-up.

SEVERITY & IMPACT
This may impact the PG grid extraction.

ACTIONS
Please verify that the height of two layers is correct.

! TEC-025
Layer should be defined in Technology file.

DESCRIPTION
The layer is defined in the lef file, but not in the technology file.

SEVERITY & IMPACT
Error. This error should be fixed.

ACTIONS
Define layer in the technology file.

! TEC-029
EM limit of metal '<name>' is set to zero.

DESCRIPTION
The EM limit for the specified metal is set as 0.

SEVERITY & IMPACT
Major warning.This would result in incorrect analysis as all EM violations > 0 would be flagged.

ACTIONS
Please specify the correct EM limit in TECH file

! TEC-031
Layer does not exist in technology.

DESCRIPTION
The layer is defined in the technology file, but not in the lef file.

SEVERITY & IMPACT
Error.

ACTIONS
Define the layer in the lef file or remove definition of the layer in the technology file.

! TEC-032
via models do not have their parameters set correctly.

DESCRIPTION
The parameters of VIA are not set correctly in the technology file, such as negative EM current limit or negative resistance. The via model name can be found in the file adsRpt/tech.viaModel.error.

SEVERITY & IMPACT
Error. This error should be fixed.

ACTIONS
Correct VIA parameter settings in the technology file.

! TEC-040
RH C extraction does not handle conformal dielectric layer

DESCRIPTION
RH C extraction does not handle conformal dielectric layer. Typically conformal dielectric will have some impact to PG C extraction , but the impact is very small (<10% of PG caps). For RH IR/EM analysis, PG caps only counts for less than 10% of total caps, so the impact is less than 1% of the total cap, this impact should be negligible

SEVERITY & IMPACT
This warning is not critical since it imapct is only less than 1% .

ACTIONS
It can be avoided by modifying the tech file to get rid of the conformal layers

----
! TEC-056
EM nominal temperature "Tnom_em" is not specified inside tech file

DESCRIPTION

For all layers and vias in design, Redhawk needs EM Nominal Temperature to be defined inside the Rehawk Tech file which is used for accurately calculating EM values

SEVERITY & IMPACT
Redhawk will ignore EM Temperature derate effect during calculation. This affects the accuracy of EM values.

ACTIONS
Define the EM nominal temperature in tech file and rerun the analysis

----
! TEC-063
Layer <>: both WIDTH_BASED_EM and EM are specified, the former will be used for EM calculation!

DESCRIPTION
Both WIDTH_BASED_EM and EM values are specified in the tech file for that layer. WIDTH_BASED_EM will be used for EM calculation

----
! TEC-065
Nominal temperature "Tnom" is not specified inside tech file

DESCRIPTION
For all layers and vias in design, Redhawk needs Nominal Temperature to be defined inside the Rehawk Tech file. This Nominal temperature value is required for calculating the temperature effect on resistance.

SEVERITY & IMPACT
Redhawk will not considerTemperature effects during resistance calculation and the resistance values will be calculated ignoring this effect.

ACTIONS
Define the nominal temperature in tech file and rerun the analysis

----
! TEC-066
Layer <xyz>: actual temperature is not specified (either through "T" in tech file or "TEMPERATURE" in gsr file),

DESCRIPTION
Redhawk did not find analysis temperature specified in GSR or tech file. It will use the tech file resistance values as it is. Temperature coefficients specified in tech file for deration purpose will not be used.

SEVERITY & IMPACT
No impact unless you want to do analysis at a different temperature than the nominal temperature used for tech file creation and also stored in it as Tnom.

ACTIONS

Specify temperature in GSR if you want to do analysis at a different temperature than the nominal temperature used for tech file creation and also stored in it as Tnom

----
! TEC-069
Invalid negative resistance computed

DESCRIPTION
If the resistance of a metal/via computed turns out to be negative , then the error is reported, mostly due to invalid temperature coefficent values in tech file

SEVERITY & IMPACT
Its a warning message.The temperature effect for these geometries will be ignored and nominal temperature resistance will be assigned.

ACTIONS
Check the tech file and correct it

! TEC-072
The conformal effect of top dielectric nitride_passiv at top metal AP is ignored

DESCRIPTION
This message is to remind that RH does not handle conformal type dielectric.

SEVERITY & IMPACT
RedHawk C extraction does not handle conformal dielectric layer. Typically conformal dielectric will have some impact on PG C extraction, but the impact is very small(<10% of PG caps). For RH IR/EM analysis, PG caps only count for < 10% of total caps, so the impact is less than 1% of the total cap. Hence this impact is negligible.

ACTIONS
Currently, you can remove the message by modifying the techfile to get rid of the conformal layers

----
! TEC-110
No via size information found on via layer 'RV' for EM calculation in EM_MODE peak. Please set via AREA, WIDTH or EMV_VS_AREA!

DESCRIPTION
For EM computation, RedHawk expects size related info for the via layers to be present in the tech file.

SEVERITY & IMPACT
RedHawk will ignore these vias in the EM analysis

ACTIONS
Add any of the following information in the tech file:

1.WIDTH {<via_width>}
2.AREA <via_area>
3.EMV_VS_AREA {
{<area1> <EM1>} {<area2> <EM2>} ...
}

! TEC-118
Incomplete VIA definition in the technology lef.

DESCRIPTION
The VIA size and other properties should be predefined in the technology lef file. For the above mentioned via the definition in the technology lef is incomplete.

SEVERITY & IMPACT
The PG extraction may be affected.

ACTIONS
Provide correct technology LEF file.

! TEC-414
Incomplete clock path in the STA file.

DESCRIPTION
STA file should capture the complete clock network from the clock root to all branches. For the above mentioned clock STA clock network information is incomplete.

SEVERITY & IMPACT
Analysis results may be affected.

ACTIONS
Clock network assumed in the analysis may be wrong.

! VCD-111
instance <inst>/adsmem_U0 should be defined first in adsPower.

DESCRIPTION
instance <inst>/adsmem_U0 should be defined first in adsPower.

SEVERITY & IMPACT
Error. This error should be fixed.

ACTIONS
Need fix.

! VCD-114
instance <inst> referenced in memory input but not defined in power.

DESCRIPTION
instance <inst> referenced in memory input but not defined in power.

SEVERITY & IMPACT
Error. This error should be fixed.

ACTIONS
Need fix.

! VCD-121
net is not defined in VCD

DESCRIPTION
net is not defined in VCD

SEVERITY & IMPACT
Warning

ACTIONS
Warning

! VCD-123
net <net>, defined in VCD, does not have driver information

DESCRIPTION
net <net>, defined in VCD, does not have driver information

SEVERITY & IMPACT
Warning

ACTIONS
Warning

! VCD-131
clock net <net> is not defined in VCD

DESCRIPTION
clock net <net> is not defined in VCD

SEVERITY & IMPACT
Error

ACTIONS
Error

! VCD-132
clock instance <inst> is not defined in <file>

DESCRIPTION
clock instance <inst> is not defined in <file>

SEVERITY & IMPACT
Error

ACTIONS
Error

! VCD-133
VCD name <name> index <index> out of bound in <file>

DESCRIPTION
VCD name <name> index <index> out of bound in <file>

SEVERITY & IMPACT
Warning

ACTIONS
Warning

! VCD-301
DEF name <name> in mapping file not used in reading <file>

DESCRIPTION
DEF name <name> in mapping file not used in reading <file>

SEVERITY & IMPACT
Error

ACTIONS
Error

! VCD-302
VCD name <name> in mapping file unmatched. Could be invalid.

DESCRIPTION
VCD name <name> in mapping file unmatched. Could be invalid.

SEVERITY & IMPACT
Error

ACTIONS
Error

! VCD-311
Events occuring too close in VCD

DESCRIPTION
Usually in a VCD , its expected to have one event per cycle.If two events in the VCD occur close within 0.25 times the cycle period , then this

SEVERITY & IMPACT
This is a warning.The power calculation with proceed and only the first event will be read.The second event will be ignored.

ACTIONS
Check your VCD for this mismatch

! VGC-001
Vectorless gated-clock analysis supports only dynamic mode and proceed with dynamic mode.

DESCRIPTION
RedHawk allows user to specify certain keywords to analyse clock-gated designs. However these keywords are useful only if user is doing a dynamic analysis. So, user must make sure that the analysis done is dynamic, whenever clock-gated keywords are used.

SEVERITY AND IMPACT
RedHawk will not consider the clock gating aspect while doing the analysis

ACTIONS
In the command file, add the keyword "set analysis_mode dynamic" in the beginning. And finally, perform a dynamic analysis.

! WIR-017
Dielectric constant definition in tech file is missing or incorrect.

DESCRIPTION
The tech file should have the dielectric layers defined in between the metal layers and these dielectric layers should possess a positive dielectric constant.

SEVERITY & IMPACT
The PG capacitance extraction may be affected.

ACTIONS
Provide correct tech file with dielectric constants.

#----------
! LIB-142
Pratio file error

DESCRIPTION
ERROR(LIB-142) is the error message identified during reading Pratio file.
They are a set of messages sharing the same message ID: LIB-142, inlcuding:
1). "number of fields <num> is not 2 or 3"
Each line of Pratio file is supposed to have 2 or 3 fields.
2). "Cell <cell_name> doesn't exist"
The cell_name specified after "#cell" does not exist in RedHawk db.
3). "no cell specified"
No cell line "#cell cell_name" in Pratio file.
4). "pin <pin_name> doesn't exist"
Cell doesn't contain a pin named "pin_name"
5). "first field <> is not an integer"
First field of each line(except the #cell and #pin line) must be an integer.
6). "second field <> is not an integer"
Second field of each line(except the #cell and #pin line) must be an integer.
7). "second field <> is negative"
Second field of each line(except the #cell and #pin line) must be a positive integer.
8). "no pin specified"
No pin line "#pin pin_name" in Pratio file.
9). "invalid pin index <num> (valid index range [0, num])"
Pin index(first field of each line) is not valid.

SEVERITY & IMPACT
RedHawk fails to read the Pratio file.

ACTIONS
Fix the data errors in Pratio file according to the messages.

! LEF-081
line <>: When there are 2 layers, VIARULE <> GENERATE must have 2 RoutingLayer and 0 CutLayer..

DESCRIPTION
This error is issued when Via Rule Generate is specified with only 2 routing layers and no cut layer.

SEVERITY & IMPACT
This Via Rule Generate will be discarded.

ACTIONS
Please fix the imported LEF.

! LEF-016
Nondefault Rule <> not having layer <>.

DESCRIPTION

Layer specified in LEF/DEF NONDEFAULTRULES is not defined in tech or lef.

SEVERITY & IMPACT
NONDEFAULTRULES definition for this layer will be discarded.

ACTIONS
Please fix the imported LEF/DEF

! GEN-128
Data corrupted in file<>:line<>!.

DESCRIPTION
The file specified is corrupted.

SEVERITY & IMPACT
Major error. Simulation results would be incorrect.

ACTIONS
Please check and correct the file.

! FAO-86
Hot inst <> rail layer mismatches with <> decap(s).

DESCRIPTION
The message indicates rail layer mismatches when trying to place decaps in hot inst region.
The number of decap cells with height not matching with the hot instance's height is reported

SEVERITY & IMPACT
Decap wont be placed if its height does not match with the hot instance's height

ACTIONS
Please provide decap cell with correct height for placing.

! ECO-021
Via<> to be deleted cannot be found in the ECO file!.

DESCRIPTION
This error is issued when the via specified in the ECO file is not found in design.

SEVERITY & IMPACT
RedHawk ignores the ECO command.

ACTIONS
Please check the ECO file.

! DEF-044

Layer<> specified in GSR:IGNORE_ROUTE is not available

DESCRIPTION
The layer specified under IGNORE_ROUTE GSR keyword is not defined in tech.

SEVERITY & IMPACT
IGNORE_ROUTE keyword will be skipped.

ACTIONS
Please check and modify your GSR file

! LEF-083
line : VIARULE <> GENERATE has 2 layers.

DESCRIPTION
Via Rule Generate is specified with only 2 layers.

SEVERITY & IMPACT
This Via Rule Generate will be discarded.

ACTIONS
Fix the imported LEF.

! DEF-118
Ignore pad pin <>, because it is on a internal net.
Try set "IGNORE_PLOC_INTERNALNETS 0" to force RH accept it

DESCRIPTION
This message is issued when a pad pin specified is on an internal net.

SEVERITY & IMPACT
This pad pin will be ignored by RedHawk.

ACTIONS
User can set the GSR keyword 'IGNORE_PLOC_INTERNALNETS 0' to force RedHawk
to accept pad pins on internal nets.

! STA-498
Skipping C1-R-C2 data since GSR::USE_SIGNAL_LOAD_FROM_STA is not set.

DESCRIPTION
Signal load information (that is, C1-R-C2 data) from the STA file is ignored
by RedHawk.

SEVERITY & IMPACT
Signal load information is available in ATE-generated STA files.
If 'USE_SIGNAL_LOAD_FROM_STA 1' is not used, this data will be ignored.

ACTIONS

User may set USE_SIGNAL_LOAD_FROM_STA to 1 to use the signal load available in the STA file. When this keyword is set, RedHawk reads C1-R-C2 data from the STA file and skips SPEF reading.

! DEF-008
Inst <> already exists in cell <> now using cell <>.

DESCRIPTION
This message is issued when the same instance appears twice in a def OR incremental PG def import.

SEVERITY & IMPACT
The instance's cell master will be replaced by the current one.

ACTIONS
Please correct the imported DEF.

! FAO-168
-window { x1 y1 x2 y2 } not provided. try fao_region gsr value.

DESCRIPTION
This warning appears if the area for FAO operation is not specified.

SEVERITY & IMPACT
If -window is not specified,then FAO uses the FAO_REGION GSR keyword definition as the window. If FAO_REGION is not defined, then the whole chip is used.

ACTIONS
Either user -window switch or use GSR keyword FAO_REGION for defining the region for FAO operation.

! PAD-006
Pad Cell<> does not have pin<>.

DESCRIPTION
If pad file lines are in "<pad_cell_name> <pin_name> [layer_name]" format, and RedHawk can't find pin <pin_name> in cell <cell_name> this message is printed out.

SEVERITY & IMPACT
This line is ignored and RedHawk continues reading the next pad file line.

ACTIONS
Please fix the data errors in pad file according to the messages.

! FAO-255
*** Dynamic analysis NOT performed ***

DESCRIPTION

Dynamic analysis is not performed before FAO 'decap' operations. No hot instance could be identified.

SEVERITY & IMPACT
FAO 'decap' operation won't be performed.

ACTIONS
Please perform dynamic analysis before using FAO 'decap' commands.

! FAO-256
*** Use lastest simulation results ***

DESCRIPTION
Latest simulation results available will be used for identifying hot instances.

SEVERITY & IMPACT
Results may not be accurate.

ACTIONS
Please perform dynamic analysis before using FAO commands for accurate results.

! FAO-252
both noise_limit and noise_reduction are zero.

DESCRIPTION
In FAO 'decap' operations, both targets noise_limit and noise_reduction are 0. Hence there is no available target for decap operations.

SEVERITY & IMPACT
FAO 'decap' operation won't be performed.

ACTIONS
Please provide non-zero target noise_limit or noise_reduction.

! FAO-110
Ignore hot instance <> (memory/block).

DESCRIPTION
Instances defined by GDS_CELL won't be identified as hot instances.

SEVERITY & IMPACT
Such instances are ignored.

ACTIONS
Use normal cell to define instance for FAO operations.

! STA-451
block<> in BLOCK_STA_FILE is not a compCell!.

DESCRIPTION
This warning is issued when a block specified under the gsr Keyword "BLOCK_STA_FILE" is actually not a compCell in RedHawk.

SEVERITY & IMPACT
RedHawk will ignore the sta file of this block.

ACTIONS
Please check the completion of DEFs.

! SPF-124
Cell <> has no enhanced LEF! DSPF Ignored.

DESCRIPTION
This is an obsolete message, no longer used in RedHawk.

SEVERITY & IMPACT
None.

ACTIONS
None.

! SPF-111
Failed to find net<> in cell<>!.

DESCRIPTION
This is an obsolete message, no longer used in RedHawk.

SEVERITY & IMPACT
None.

ACTIONS
None.

! PWR-312
Block <block_name> can't be found, treat this block as top level.

DESCRIPTION
While processing VCD, tool did not find this block in DEF specified either at the top or at block level. So treat this block as Top level and do Power Calculation.

SEVERITY & IMPACT
Warning. Could be a Typo in block name in GSR file.

ACTIONS
User should check the DEF and GSR file to check if the Design name specified for VCD path is really present in DEF.

! EVP-164
VCD symbol '<wire_name>' is defined more than once, ignored.

DESCRIPTION
This is a VCD file syntax problem. VCD file has ASCII value in bit index field for that wire. Index field cannot have ASCII characters.
For example: suppose we have two wires in VCD file like below.

```
$var wire     1 Fr!  TEST_TAP[CTRL]  $end
$var wire     1 Fr!  TEST_TAP[OBSERVE]  $end
```

So tool will issue this error: VCD symbol 'core/inst1/TEST_TAP[0]' is defined more than once, ignored.

SEVERITY & IMPACT
Error. Power Calculation will get affected as PowerStream will not find the activity for that net.

ACTIONS
Check the VCD file and search for the wire if that is even present or not.
If present, check if the syntax is ok.

! SLB-055
Syntax error at line <line number>

DESCRIPTION
The file that has been read just before this error, has syntax error at the line number mentioned.

SEVERITY & IMPACT
Error. Tool will generate another ERROR (SLB-053). File will not get read.
It could affect power calculation

ACTIONS
Check the file, correct the syntax and run again.

! EVP-163
DEF net '<net_name[index]>' cannot be mapped to VCD symbol bit because its bit index is out of range.

DESCRIPTION
A net is present in DEF, that is having multiple bits. But VCD file is having only single bit for that wire. So tool will not be able to map DEF net to VCD symbol

SEVERITY & IMPACT
Error. Tool will not be able to assign activity to this net from the VCD file.
It will impact power calculation.

ACTIONS
Make sure that DEF net names are matching with the VCD net names.
So that accurate mapping can be done.

! SLB-051
In file "<lib_file>", "nom_process" is not defined. Set it to 1.

DESCRIPTION
No process value is specified for the library. So set it to 1, that is default.
Every library file is characterised for a particular PVT condition.

SEVERITY & IMPACT
Warning. Power calculation will not be affected.

ACTIONS
Check the library files and make sure that PVT conditions are present.

! PWR-319
FREQ value(<freq>) does NOT match any clock in STA file.

DESCRIPTION
This is the dominant frequency specified in the GSR file, but this frequency value
is not matching with any of the clock in STA file.

SEVERITY & IMPACT
Warning. Frequency will not affect power calculation because Redhawk
power calculation is based on Energy.

ACTIONS
User should check the clocks present in STA file first and then specify the
FREQ in GSR file.

! GSR-131
BLOCK VCD <block_name> START_TIME <time> ps is too short for
DYNAMIC_PRESIM_TIME < >ps.

DESCRIPTION
DYNAMIC_PRESIM_TIME is the time to initialize capacitance charge and
inductor current before the actual simulation starts. So the VCD simulation
start time should be greater than DYNAMIC_PRESIM_TIME.

SEVERITY & IMPACT
Warning. RedHawk automatically takes care of pre-sim start time depanding upon the
VCD sim start time. This warning will not impact power calculation.

ACTIONS
USER can ignore this warning.

! PWR-320
GSR FREQ value(<value>) may be NOT appropriate, since the frequency domain power
is less than 10% of total power

DESCRIPTION
GSR FREQ value should be the frequency at which more than 10% of the chip power is consumed and hence this warning is issued.

SEVERITY & IMPACT
Warning. Frequency will not affect power calculation because Redhawk power calculation is based on Energy.

ACTIONS
User should check all the clocks present in STA file first and then specify the FREQ in GSR file.

! GSR-136
BLOCK VCD <block_name> END_TIME <time> ps is replaced by <modified_time> ps (START_TIME + DYNAMIC_SIMULATION_TIME).

DESCRIPTION
With GATE VCD/RTL VCD(VCD_DRIVEN 0) RedHawk does power calculation only for DYNAMIC_SIMULATION_TIME. So if END_TIME is set, RedHawk modifies the END_TIME as (START_TIME + DYNAMIC_SIMULATION_TIME).

SEVERITY & IMPACT
Warning. User will think that power calculation results are for the specified START and END time in the GSR file. But actually the results are for START time and (START + DYNAMIC_SIMULATION_TIME)

ACTIONS
User do not have any control on this.

! PWR-012
No library information provided.

DESCRIPTION
No library files are specified in the GSR file.

SEVERITY & IMPACT
Major warning. Whatever cells are referenced in the design will not report any power because cell libraries are not present. So it will drastically affect the power calculation. Redhawk will assign 0 power for all the cells.

ACTIONS
Make sure all the referenced cell libraries has been passed to RedHawk.

! VCD-308
VCD name {<net_name>} in mapping file can't be found in vcd.

DESCRIPTION
DEF net is successfully mapped with mapping file but mapped VCD name in

mapping file is not found in VCD.

SEVERITY & IMPACT
Warning. Tool will not be able to assign activity to this net from the VCD file.
It will impact power calculation.

ACTIONS
Make sure you are using the correct VCD file.

! PWR-166
Net <net_name> is covered in more than one VCD/FSDB file.
Toggle rate derived from the first VCD/FSDB file is used.

DESCRIPTION
When we do Heirarchical analysis with multiple Block VCDs and if one unique net is covered in more than one VCD, then this Warning is issued and RedHawk assignes the activity from the first VCD file.

SEVERITY & IMPACT
Warning. If the activity is different in those VCD files for that net,
then power calculation will be affected, otherwise it is not a problem.

ACTIONS
Make sure that you are setting correct block/top level design name for each VCD.

! PWR-322
User specified end time (<time>) is larger than end time in VCD/FSDB (<time>),
use end time in VCD/FSDB.

DESCRIPTION
If the user specified end time is larger than the actual VCD/FSDB end time,
then RedHawk will do power calculation till end of the VCD.

SEVERITY & IMPACT
Warning. User will think that power calculation results are from start and
end time specified in the GSR file. But results will be till end time of the VCD

ACTIONS
User has to make sure that he is providing the start and end time within
the available VCD time span.

! SLB-113
Pin <pin_name> is recognized as an asynchronous pin in cell <cell_name>
power tables defined for the pin will be ignored.

DESCRIPTION
Generally we get this warning for memory cells. In memory there are some pins that are asynchronous.

SEVERITY & IMPACT
Warning. Power calculation for those asynchronous pins will not happen.

ACTIONS
User can ignore this warning.

! SLB-073
Undefined power rail {"<rail_name>"} in cell {"<cell_name>"} rail_connection attribute. rail_connection ignored.

DESCRIPTION
In library, for each cell rail_connections are defined. Example:
rail_connection (vss, vss)
Here one 'vss' is cell pin name and other 'vss' is power rail name.
power_supply() section in library defines all the power rails present.
So, if the 'vss' rail is not defined in the power_supply() section,
this error will appear.

SEVERITY & IMPACT
Error. rail_connection will be ignored.

ACTIONS
User has to make sure that all the rails are properly defined in library file.

! PWR-332
Instance <instance_name>is reactivated, power will be calculated based on default freq and toggle rate.

DESCRIPTION
When instance has zero frequency or clock index in not defined for that instance in timing file, this warning occurs.

SEVERITY & IMPACT
Warning. Power will be calculated for that instance based on default freq and toggle rate value.

ACTIONS
User can ignore this warning.

! LIB-128
Failed to find cell<cell_name> in library that was referenced at file<file_path>:line<line_no>

DESCRIPTION
Cell in not present in the library file.

SEVERITY & IMPACT
Error. Power will not be calculated for all the instances of this master cell.

ACTIONS
User should make sure that all the master cells library is present.

! SP-401
Toggle rate or duty cycle out of range (<toggle_rate>/<duty_cycle>) for net <net_name>. Protecting to <toggle_rate>/<duty_cycle>

DESCRIPTION
Generally we get this warning message with State Propagation mode enabled. When SP receives toggle_rate / duty_cycle from VCD scan, it does a sanity check. In the Warning message toggle_rate and duty_cycle reported is half of the actual value. For the most part, SP checks if the T/D is out of range. SP limits the toggle rate to 1.0 for signal net and 2.0 for clock net.

SEVERITY & IMPACT
Warning. State Propagation engine will protect the toggle_rate and duty_cycle for the net and power calculation will be performed considering those values.

ACTIONS
User can ignore this warning.

! DEF-012
CELL < > is not defined in lef or def, but is referenced by others in def. Please define this cell first!

DESCRIPTION
This message warns the user about the CELL XXX not present in the lef or def, but it's instantiated in one of the def file. These cells are figured in the adsRpt/apache.refCell.noLefLib reports.

SEVERITY & IMPACT
This will create lef def mismatch error. No physical view for this cell in GUI.

ACTIONS
Add the missing lef or def file for the CELL XXX and rerun the setup analysis

----
! DEF-055
Ignore cell <name> doesn't exist in design.

DESCRIPTION
This cell <name> is ignored by gsr keyword IGNORE_CELLS.

SEVERITY & IMPACT
RedHawk ignores this cells for all analysis.

ACTIONS
User can ignore this error, if user set this cells intentionally under IGNORE_CELLS keyword in GSR file.

! LEF-044
Decap cell <name> is not defined in LEF. Ignore it.

DESCRIPTION
A decap cell specified in the DECAP_CELL gsr keyword is not included in the input LEF files.

SEVERITY & IMPACT
This error is important for dynamic IR drop analysis and user must specify LEF files for this decap cells. The given decap cell will not be treated as a decap if LEF file is missing.

ACTIONS
Provide LEF file for the decap cell to remove this error.

! STA-104
<sta_file_path>.timing__<line_number>: clockIndex(X) is not defined in the timing file.

DESCRIPTION
clockIndex(X)'s related CLOCK is not defined in the timing file under CLOCK SECTION.

SEVERITY & IMPACT
This error won't have big impact on the run, but gives wrong power value. Due to this issue some insts in the design are not covered in the timing file. This missing instances are reported in <adsRpt/apache.tw0>.

ACTIONS
Provide proper STA timing file to remove this error.

! ITG-021
.pratio file is missing for temporal cell <name>.

DESCRIPTION
Missing of .pratio for temporal cell <name>.

SEVERITY & IMPACT
Usually .pratio files specify factors for the current distribution for P/G pins in temporal cell <name>. Missing of this file causes RedHawk to distribute the current as in lef pins.

ACTIONS
Provide the missing .pratio file.

! FAO-155
FAIL to get via model at (X Y)

DESCRIPTION
Reqired via model parameter (size) is not found in technology lef file.

SEVERITY & IMPACT
RedHawk won't create a via model. It leads to open circuit.

ACTIONS
Please make sure
1, Enough strap width is available to create via model.
2, Check suitable via model is available in techology lef file.

! LN-002
inst <name> is not fully logic connected. 1 input edge NULL.

DESCRIPTION
The specified instance's 1 input pin is not logically connected to net.

SEVERITY & IMPACT
Warning. 1 input pin is not connected to any net. This will affects power numbers.

ACTIONS
Please dump the def from LVS cleaned design. Make sure input pins should not be floating.

! AP-102
<inst_1_name>/<pin_name> and <inst_2_name>/<pin_name> are mapped to the same symbol in mapping file!

DESCRIPTION
<inst_1_name>/<pin_name> & <inst_2_name>/<pin_name> are mapped to the same RTL symbol in mapping file!

SEVERITY & IMPACT
Warning. Creates mapping problem.

ACTIONS
Please provide proper mapping file (or) user can add "GenerateMapfile" command in command file before "perform pwrcalc" step. This command will generate mapping file automatically.

! PWR-128
For instance <name> intended final block/instance power from BLOCK_POWER_FOR_SCALING, <X> Watt,generates final toggle rate X, larger than 2.0, for a clock network instance.

DESCRIPTION
User specified BLOCK_POWER_FOR_SCALING <X> Watt for the instance <name> generates final toggle rate X (greater than 2.0). Clock network instance's

toggle rate should not greater than 2.0.
Unreasonably high power target is specified for the block/instance in
BLOCK_POWER_FOR_SCALING.

SEVERITY & IMPACT
Warning. User specified power number will be used for analysis.

ACTIONS
Please correct the power number of the block/instance provide through BPFS.

! PWR-147
Instance/cell <name> is an 1V1G cell but Vdd/Vss pin VDD is defined
in BLOCK_POWER_FOR_SCALING or BLOCK_POWER_FOR_SCALING_FILE.
Only one of the power will be recognized.

DESCRIPTION
This message indicates that the single VDD/GND cells has VDD pin definition
in BPFS. We no need to specify the VDD pin under BPFS.

SEVERITY & IMPACT
Warning

ACTIONS
Please check the instance/cell in gsr under BPFS keyword and remove
VDD pin definition

! CCL-073
Inconsistent vdd value (1.320000 vs 1.080000) in power file
<adsPower/TOP.nets.adsLib.power>:<line_4>

DESCRIPTION
Library voltage & voltage value set in gsr are not matching.

SEVERITY & IMPACT
In such cases RH will take lib volatage for power calculation.
This warning won't affect power numbers.

ACTIONS
User should specify proper vdd voltage in gsr file.

! DCS-009
Some vcd is true time but some are not.

DESCRIPTION
Some VCD provided is set true time 1 but some are not set in the analysis.

SEVERITY & IMPACT
If TRUE_TIME = 0 , Uses STA timing data and assumes no glitches.
If TRUE_TIME = 1 , Uses VCD switching and timing data.

This will have impact on power values.

ACTIONS
Please make sure the TRUE_TIME attribute specified in gsr is correct for
all the block vcd filess.

! SP-406
STA output net frequency <> larger than instance frequency <>.
Protect toggle rate from <> to <> on instance <name> output net <name>.

DESCRIPTION
Frequency of Instance provided in GSR as FREQ_OF_MISSING_INSTANCES is lesser than
the FREQUENCY derived from the driver. So There are two frequencies for this
inst <name>. At same time toggle rate is also greater than a protect limit

SEVERITY & IMPACT
Warning. For power calculation, RedHawk will take the frequency value from
FREQ_OF_MISSING_INSTANCES keyword. So there will be optimistic power.

ACTIONS
Please make sure that inst <name> is present in STA timing file.

! PWR-329
The instance <name> is covered by VCD. BPFS power is ignored.

DESCRIPTION
Priority of the BPFS will be less if we provide VCD. Automatically tool
will ignore the BPFS power.

SEVERITY & IMPACT
This is not critical warning. RedHawk will honor VCD for power caluculation.

ACTIONS
User can ignore this warning. Check gsr to see if this is what you want
(VCD or BPFS)

! ESD-042
ESD pin pair defined in clamp_info_file does not exist for the clamp instance.

DESCRIPTION
i) The clamp cell pin could be disconnected from the PG/signal net domain.
ii) For the net, there could be only logical connectivity and no physical
connectivity.

SEVERITY & IMPACT
Error: ESD pin pair of the clamp instance will not be conisdered for
ESD analysis.

ACTIONS

Check the connectivity of the clamp cell.

! ESD-038
Invalid pin location in the clamp cell.

DESCRIPTION
i) Clamp cell does have any instances in the design.
ii) Pin location/name given in clamp info file is wrong.

SEVERITY & IMPACT
The clamp cell is not considered for ESD analysis.

ACTION
i) Check the top def for cell instantiation.
ii) Check the pin definition in clamp info file.

! GSR-110
File specified with BLOCK_POWER_ASSIGNMENT_FILES is not readable.

DESCRIPTION
File pointer given in BLOCK_POWER_ASSIGNMENT_FILES could be wrong.

SEVERITY & IMPACT
ERROR: Setup design will not read tihs file while importing gsr.

ACTION
Check the path of file given with BLOCK_POWER_ASSIGNMENT_FILE

! ESD-083
Ignore isolated clamp points in PG grid of clamp instances for
C2I resistance checks.

DESCRIPTION
i) The clamp instance could be disconnected from the related PG/Signal net domain
ii) There is no connectivity between the two clamp instances

SEVERITY & IMPACT
i) The PG/signal nets would not be considered for ESD analysis

ACTIONS
Check the connectivity of the net.

! ESD-022
ESD pin of the clamp instance is not connected to any PG/Signal net.

DESCRIPTION

The clamp instance is disconnected with respect to the concerned PG/signal net.

SEVERITY & IMPACT
Warning: Connectivty of the the clamp cell is wrong. ESD pin of the clamp cell will not be conisdered for ESD analysis.

ACTIONS
Check the connectivity of the clamp cell.

! ESD-103
Failed to find clamp point in the pin for the clamp instance.

DESCRIPTION
The clamp instance is disconnected from the corresponding PG/signal net.

SEVERITY & IMPACT
Warning: The connectivity of the cell is not proper. ESD pin of the clamp cell will not be conisdered for ESD analysis.

ACTIONS
Check the connectivity of the clamp cell.

! ESD-081
ESD pin pair defined in clamp_info_file does not exist for the clamp instance.

DESCRIPTION
i) The clamp cell pin could be disconnected from the PG/signal net domain.
ii) For the net, there could be only logical connectivity and no physical connectivity.

SEVERITY & IMPACT
Warning: ESD pin pair of the clamp instance will not be conisdered for ESD analysis.

ACTIONS
Check the connectivity of the clamp cell.

! ESD-024
Clamp instance is not connected to P/G network.

DESCRIPTION
Clamp instance is disconnected from P/G.

SEVERITY & IMPACT
Warning: Clamp instance will not be considered for ESD analysis.

ACTIONS
Check the connectivity of the clamp instance.

! ESD-021
Clamp instance is not connected to P/G net

DESCRIPTION
Clamp instance is disconnected from P/G net

SEVERITY & IMPACT
Warning: Clamp instance will not be considered for ESD analysis.

ACTIONS
Check the connectivity of the clamp

! ESD-137
No bump-to-bump pair found in net.

DESCRIPTION
The clamp instance is not connected to the net. Logical connectivity is present physical connectivity could be absent.

SEVERITY & IMPACT
Warning: Clamp instance will not be considered for ESD analysis.

ACTIONS
Check the connectivity of the clamp.

! TEC-126
Missing via definition between metal layer: <layer1> and <layer2>

DESCRIPTION
No via between <layer1> and <laye2> in tech file. Via definition is wrong in tech file.

SEVERITY & IMPACT
Via will not be extracted.

ACTIONS
Check the definition of via in tech file and correct it.

! PWR-121
No match in design for instance <instance name> found in imported instance file

DESCRIPTION
Instance given in IPF file is not present in the design.

SEVERITY & IMPACT
Instance power will be zero and will not be part of analysis.

ACTIONS
Check the instantiation of the instance.

! APL-015
No cell from <apl file> is matched in design.

DESCRIPTION
Cell is present in APL file, but missing in the design.

SEVERITY & IMPACT
Warning: The APL data for the cell will not be part of analysis.

ACTIONS
Check the definition of cell in def file.

! ACK-075
Aplchk limit error exists in leakage current (output high).

DESCRIPTION
Inexplicably high leakage current for cell mentioned in Info (ACK-107) in apl file.

SEVERITY & IMPACT
Occurs during import apl. Analysis will go wrong as the leakage current assigned to the cell is not reasonable.

ACTION
Correct the apl file with proper leakage curent for the cell.

! GEN-135
PowerStream Failure

DESCRIPTION
PowerStream cannot calculate power due to missing input data or user input error in gsr file.

SEVERITY & IMPACT
Power Calculation fails.

ACTION
Check the log file for missing input data or the gsr keyword error.

! ESD-133
<clamp cell name> is not a clamp cell name, ignored for CLAMP_CELL.

DESCRIPTION
Clamp cell name listed by 'CLAMP <cell name>' in the rule file is not a clamp cell.

SEVERITY & IMPACT
Warning: 'CLAMP <cell name>' statement will be ignored in the rule file.

ACTION
Check the clamp cell name given in the rule file under 'CLAMP <cell name>' and correct it.

! ESD-026
<instance name> is an instance of clamp cell <cell name>.

DESCRIPTION
For C2I/C2M, instance specified under INST_FILE in the esd_rule_file is actually a clamp instance, and not a standard cell/hard macro.

SEVERITY & IMPACT
Warning: Issues the warning and continue the analysis.

ACTION
Check the instance name under INST_FILE given in the esd_rule_file and correct it.

! GSR-026
gsr set" is not allowed after stage "import tech".

DESCRIPTION
generate_cpm: setting value is not allowed. after setup design.

SEVERITY & IMPACT
Error: Simulation stops after setup design.

ACTION
Include "generate_cpm 1" in gsr and do setup design

! GSR-007
<n> parameters expected at <gsr file name>:<line number>

DESCRIPTION
Syntax error in the GSR keyword. Number of parameters given for the GSR keyword is wrong.

SEVERITY & IMPACT
The keyword for which error is reported is not honoured properly.

ACTION
Check the syntax of the keyword and correct the usage.

! ESD-165
Missing clamp I-V data for ESD_PIN_PAIR in the clamp cell file.

DESCRIPTION
The <clamp_IV_name> is not defined in I-V section in the clamp file.

SEVERITY & IMPACT
ESD Check will fail for the clamp cell.

ACTION
Provide CLAMP_IV definition in the clamp cell file.

! CCL-069
All VDD and GND pad pins are floating. Design does not have any connected VDD nor GND sources! Redhawk extraction later may fail!

DESCRIPTION
Design does not have any valid power/ground pads

SEVERITY & IMPACT
Major error. RedHawk analysis may fail as no voltage sources are present.

ACTIONS
Please verify that the defined pads are valid or use FAO commands to createa proper VDD and GND sources.

! SEM-004

DESCRIPTION
The signal EM flow used is different from the standard signal EM flow.

SEVERITY & IMPACT
Error. The command flow used for SignalEM analysis is different from the expected flow.

ACTIONS
Please review signal EM flow used.

! TEC-129

DESCRIPTION
Tech file provided to RedHawk does not have EM rules for average mode for a layer.

SEVERITY & IMPACT
Error. RedHawk does not have EM data for averaged mode EM analysis.

ACTIONS
Please include Tech file with averaged EM rules for the layer.

! SEM-001
The keyword KDTREE_GRID_ENABLE is obsolete. Use 'NX_SEM 1' instead.
For compatibility purpose, Signal EM analysis will proceed with 'NX_SEM 1'.
This keyword will be unavailable starting from RH-SEM version 11.2

DESCRIPTION
Starting from 11.2, the GSR keyword KDTREE_GRID_ENABLE is obsolete.
For compatability, Signal EM will proceed with 'NX_SEM 1'.

SEVERITY & IMPACT
Warning. Keyword KDTREE_GRID_ENABLE is obsolete.

ACTIONS
No action required from the user

! MOD-008
Invalid resistance is created on layer< > due to rpsq/rho unspecified please check the tech file entry for the layer.

DESCRIPTION
TECH file does not have any resistance parameters defined for the layer.

SEVERITY & IMPACT
Major error. Resistance extraction cannot be done without resistance values in the TECH file.

ACTIONS
Please include resistance parameters for the layer in the TECH file.

! TEC-050
ERROR(TEC-050): Unreal height or thickness setting for dieletric '<layer name>'.

DESCRIPTION
The height/thickness in the TECH file is suspiciously high.

SEVERITY & IMPACT
Error. TECH file data may be incorrect.

ACTIONS
Please review TECH file data to confirm taht the dimensions of layer are valid.

! NET-406
Ignoring the connectivity to Net<net name>::Pin<pin name> which has no physical geometry.

DESCRIPTION
Pin does not have any physical geomtery defined.

SEVERITY & IMPACT
Warning. The connectivity will be ignored.

ACTIONS
Please ensure that all the pins have physical geometry in the LEF/DEF.

! CCL-104
All VDD and GND pad pins are floating. Design can only be connected to VDD or GND sources through CPs defined in .tsv file!

DESCRIPTION
All VDD/GND pins are floating. The only power source for the design is the CPs defined in .tsv file

SEVERITY & IMPACT
Warning. TSVs are the only power source.

ACTIONS
Please ensure that VDD/GND connections are expected only through CPs.
If expected, make sure that TSVs provide connections to VDD and GND.

! TEC-011
Unsupported 'VIA' attribute '<attribute name>' at '<tech_file path:line number>'

DESCRIPTION
TECH file contains some unsupported parameters.

SEVERITY & IMPACT
Warning. TECH file contains unsupported attributes.

ACTIONS
Please ensure that the syntax and attributes in the TECH file are correct.

! TEC-152
WARNING(TEC-152): DEF/LEF_SCALING_FACTOR has been set, keyword 'HALF_NODE_SCALE_FACTOR' at <tech_file:line_number>'' is duplicate.

DESCRIPTION
Scaling factors have been defined multiple times.

SEVERITY & IMPACT
Major Warning. The scaling factor may not be applied correctly.

ACTIONS
Please check DEF/LEF_SCALING_FACTOR and HALF_NODE_SCALE_FACTOR definitions and make sure that no duplication is occuring.

! SEM-202

WARNING(SEM-202): Total <number of nets> nets are dropped from Signal EM, please check the file adsRpt/SignalEM/SE.droppedSignalNets for more details

DESCRIPTION
One or more nets were dropped from Signal EM analysis. Please refer to adsRpt/SignalEM/SE.droppedSignalNets for list of dropped nets.

SEVERITY & IMPACT
Warning. All nets have not been considered in Signal EM analysis.

ACTIONS
Please check why the nets were dropped from analysis. Possible causes include missing driver, disconnects, STA defined the net as CONST, missing SPEF, missing frequency/slew.

! CCL-067
No GND pad pins connected to wires in the design!

DESCRIPTION
No GND pad pins connected to wires in the design

SEVERITY & IMPACT
Warning. Ground nets will not be analysed in the design.

ACTIONS
Please check your top DEF and .ploc/.pad/.pcell files for GND pad pins.

! SEM-203
For Signal EM it is highly recommended to import power or perform pwrcalc

DESCRIPTION
For Signal EM it is highly recommended to import power or perform pwrcalc

SEVERITY & IMPACT
Major warning. Power calculation has to be performed for Signal EM analysis.

ACTIONS
Please perform power calculation or import power for RedHawk Signal EM analysis.

! CON-117
Net <net name> connects both EXT_PIN (<exit pin name>) and INT_PIN (<inst_pin_name>) in switch instance <switch name>. This instance will not be considered as switch cell

DESCRIPTION
Both internal and external pins of a switch are connected to same net.

SEVERITY & IMPACT
Major Warning. Switch instance will not be considered a switch.

ACTIONS
Please check net names and connectivity in the DEF to ensure that switch connections are proper.

! ECO-022
Wire to be deleted can not be found in the ECO file!

DESCRIPTION
ECO_Wire<$s> is not present in the design, which is mentioned to delete.

SEVERITY & IMPACT
Minor error. Wire mentioned to delete not found in the design.

ACTIONS
Please verify the ECO changes, and check whether 'wire name' mentioned is correct or not.

! DEF-018
DEF layer is neither defined in the Apache tech file nor in the LEF files.

DESCRIPTION
Definition for the corresponding layer is neither present in the techlef nor in Apache tech file.

SEVERITY & IMPACT
It effects during extraction(resistence calculation) and EM for the particular layer will not be calculated.

ACTIONS
Please define the layer in tech LEF and apache.tech file.

! CMM-012
Missing original DEF for cell <$cell_name>, since it is used outside and also inside CMM(s), the original DEF must be imported

DESCRIPTION
DEF file is missing for some cell(s) in the design which is referenced inside CMM model and present in top level as well.

SEVERITY & IMPACT
Major error. Missing cell will not be included in any type of analysis which will affect results.
Make sure to import the original DEF for this cell, otherwise some of the design content may be missing.

ACTIONS
Please include the DEF file for missing cell in GSR.

! STA-414
STA critical path report does not include full clock path.

DESCRIPTION
STA critical path report which is mentioned in the GSR under
STA_CRITICAL_PATH_FILE does not include full clock path.

SEVERITY & IMPACT
Tool can not perform full clock path analysis hence that affects FAO on timing
of clock instances in those critical paths.

ACTIONS
While generating the critical timing paths report from timing tool please
include the option of generating full clock path report.

#----------
! STA-112
Pin <pin_name> have inside/outside frequency mismatch

DESCRIPTION
Same Pin is getting different frequency values from block and top-level
STA file, this is because the same pin is referenced by two different clocks
in different STA files.

SEVERITY & IMPACT
TWs in STA files are not correct. Conflict in the assigning frequency value
during power calculation.

ACTIONS
Please verify the frequency value assigned to the pin in block level and
top level. STA file correct value for each pin in the STA file.

#----------
! PAD-043
P/G pin<> of pad cell<> has an incorrect direction.

DESCRIPTION
Mismatch in the Power/Ground pin definition of the cell which is mention in
the lef and lib.

SEVERITY & IMPACT
Conflict for tool, whether which direction to consider for the mentioned pin
and overwrite with the .lib direction.

ACTIONS
Please check your lef and lib files for the cell.

#----------
! PAD-013
Undefined type/net for source in ploc file.

DESCRIPTION
In ploc file, defined type or net for corresponding source
(e.g. type: POWER/GROUND/CLOCK etc. or valid net name) is not correct.

SEVERITY & IMPACT
Major error. Here Pad is ignored for further analysis, hence that affects
the extraction and IR Drop analysis.

ACTIONS
Please use proper type/net_name in the ploc file.

#----------
! BPA-140
Self-intersecting rectilinear region boundary at ($x $y) for
cell <region_name/cell_name>

DESCRIPTION
Rectilinear region specified under BLOCK_POWER_MASTER_CELL in GSR is not
complete, may be $x or $y location is missing.

SEVERITY & IMPACT
Minor warning.

ACTIONS
Please specify complete rectilinear coordinate for rectilinear region in GSR

#----------

! BPA-108
REGION <Region_name> overlaps with REGION <Region_name>.

DESCRIPTION
One region is overlapping with another region

SEVERITY & IMPACT
Major warning. This will affect the IR drop of the instances which are
present in the common region.

ACTIONS
Please verify the coordinates mentioned for those two region in GSR,
and make them correct or Specify the keyword OVERLAP_OK for that region.
#----------

! PAD-073
<Line > in file <File_path> has wrong number of argument(s).

DESCRIPTION
Wrong number of arguments are entered for some keyword(s) in .tsv file.
Note: .tsv(Through Silicon Via) file is used in case of 3D IC.

SEVERITY & IMPACT
Major warning. Data missing might have some effect on the analysis

ACTIONS
Please correct your tsv file.

#----------

! GSR-021
Primary net <net_name> does not exist in top design!

DESCRIPTION
Primary net which is mention under voltage domain nets in gsr, does not exist in top design!
Please check your GSR and make sure post-flattening net names are used.

SEVERITY & IMPACT
Minor warning. Tool will skip equivalent net merging.

ACTIONS
Please mention proper net name under VDD_NETS or GND_NETS in GSR.

#----------

! FAO-259
Decaps are removed during Fix and Optimization before Dynamic simulation based on imax_less_then <limit>.

DESCRIPTION
Ineffective Decaps are removed in fao_region, fix_window or on fao_nets(which ever GSR keyword mentioned) before Dynamic simulation based on the option mentioned in the tcl (like user_deap or hot_area etc).

SEVERITY & IMPACT
Major warning. Effects the Dynamic IR drop analysis.

ACTIONS
Please confirm that exclusion of the decaps and its imax limit during FAO.

#----------

! FAO-194
DRC issue due to via presence @ location <$x,$y>

DESCRIPTION
During FAO mesh creation, 'net created' is overlapping with same metal layer segment(or via layer) hence cretion of strap in that region will not happen.

SEVERITY & IMPACT
Minor warning. FAO is not possible in that region.

ACTIONS
Please check IR drop in the region where net is not created, if that is not ok then specify offset such that, commmand properly creates net in that region

#----------

! ITG-022
Cells with Reference name <master_name> has no apl sampling.

DESCRIPTION
Cells with APL DI data but APL sampling for those cells is not generated due to instance output pin connectivity problem.

SEVERITY & IMPACT
Minor warning. Dynamic IR drop will be affected.

ACTIONS
Please verify the output connectivity of those cell or assign load for those pins

#----------

! ANL-302
Instance <inst_name> is not controlled by a power gate but specified to be POWER-UP through GSC

DESCRIPTION
Instance is not controlled by a power gate but specified to be POWER-UP through GSC in GSR, this might be due to missing of the Power/Ground pin definition in Lib/Lef of it's master cell or absence of the referenced cell.

SEVERITY & IMPACT
Major warning. That cell is completely excluded from the Rampup analysis.

ACTIONS
Please verify the presence of the inst's master cell and its PG pins in Lib/Lef

#----------

! LIB-430
The output voltage of LDO instance <inst_name> at pin <pin_name> is defined differently in GSR and LDO model

DESCRIPTION
Mismatch in the voltage value mentioned in GSR and LDO model file.
Here GSR value will be over written.

SEVERITY & IMPACT
It affects the Static/Dynamic simulation.

ACTIONS
Please correct the voltage vaule associated with the VDD/VSS net connecting to pin of the LDO instance in GSR as in the LDO model file or Remove the net from GSR as it is taking value from the LDO model file.
To see the net, which is connected to the LDO instance, refer SPECIALNETS

section in DEF.

#----------
! FLW-003
Performing ramp-up analysis but no blocks are POWER_UP in GSC.

DESCRIPTION
Inconsistent flow: Performing ramp-up analysis but no blocks are POWER_UP in GSC

SEVERITY & IMPACT
Major warning. Ramp-up analysis will be affected.

ACTIONS
Please verify the keyword and block mentioned in the GSC file is correct

#----------
! EQV-320
Failed to create thread of state table entry " state" for internal node "<cell name>" pin

DESCRIPTION
There is some error when building cell lookup table for logic purpose. State table in .lib of "<cell>" and pin definition in .apache/adsLib.output are not matching

SEVERITY & IMPACT
State dependent leakage and internal power will be affected

ACTIONS
In this case, mostly is that State table does not match with cell pin definition. Please make sure state table is correct

#----------
! SLB-053
Errors occur while parsing custom library file

DESCRIPTION
This message is shown due to the syntax error in custom library file

SEVERITY & IMPACT
Critical message as Reading of custom library failed & this error will not allow to complete setup design

ACTIONS
Please fix the syntax error in custom libary file and try again

#----------
! LNW-022
Toggle rate is bigger than maximum value of 2

DESCRIPTION

Maximum toggle rate should be 2 in state propagation for any signal
This is the corner case where output net toggle rate is more than imput
toggle rate due to the feedback from ouput to input or reconvergence of net

SEVERITY & IMPACT
It is not critical message , it warns that toggle rate if more than 2 will be
reduced to 2
There can be small variation in Switching and Internal power calculated but
State propagtion is purely probability based so toggle rate 2 itself
sufficent to derive the probability of switching

ACTIONS
No action is required from user

#----------
! AP-104
Couldn't find cell in liberty file

DESCRIPTION
This cell is not present in the provided library.

SEVERITY & IMPACT
Since there is no library for the cell , No leakage and internal power
computation info. This will lead optimistic power calculation and analysis
will not account the impact of this cell on overall

ACTIONS
Please add the library for the cell

#----------
! SP-403
Large Fanin/Fanout net found

DESCRIPTION
This is the information that there is a large fan in / fan out net present
in the design

SEVERITY & IMPACT
It will affect the power calculation since the capacitance load seen by the
driver net will be more which in turn will lead to high switching power and
internal power

ACTIONS
Please check the driver net to make sure its intended high fanin/fanout net

#----------
! SP-402
Clock gate doesn't have clock ouput

DESCRIPTION
This instance is a clock gating cell, but it's output net is not marked as
"CLOCK" in STA file

Since no clock output Instance will not be considered as clock gate please
check for the adsRpt/state_propagation.GCControls for more info

SEVERITY & IMPACT
This instance will not be considered as clock gate. GOP applied in
static/dynamic will not be applicable

ACTIONS
Please mark the output net of clock gate as "CLOCK" net in STA

#----------
! PWR-313
There is no event detected in vcd so apache.scenario is empty

DESCRIPTION
This wanring will be issued when there is no event in the VCD file or
" 0%" coverage of Instance and Net in the design
The nets in block not covered by VCD/FSDB file please refer to
adsRpt/vcd_missing_nets

SEVERITY & IMPACT
Critical warning. Dynamic voltage drop is completed based on the switching
scenario i.e apache.secario. There is no internal and switching power for the
block. Only leakage power will be reported so results will be incorrect

ACTIONS
Please check the BLOCK_VCD or VCD_FILE provided in the GSR.Make sure VCD is
proper. Make sure there should be at least 90% coverage for best results

#----------
! AP-110
Instance have toggle rate 2.0 Its frquency and toggle count may not match

DESCRIPTION
Maximum toggle expected is 2 for clock nets only.If there is any net or
instance toggle rate more than 2 Warning will be issued

SEVERITY & IMPACT
Depending on the toggle rate derived from VCD or state propagation only
switching and internal power is calculated. When toggle rate is more than 2
It will be limited 2

ACTIONS
No actions required from user

#----------
! LNW-072
SP Template is automatically generated by Power Calculation in MIXED_MODE.

DESCRIPTION
After Power calculation PowerStream generates a file containing the default
toggle rates assumed at propagation origins. Please refer this file for more

information adsRpt/state_propagation.template.
User can edit the default toggle rates in state_propagation.template file
and pass it back to Powerstream using the following GSR option

```
STATE_PROPAGATION {
  PROPAGATION_MODE probability
  CONSTRAINT_FILE <modified constraint file>
}
```

SEVERITY & IMPACT

Primary port and Flip flop will take default toggle from GSR to propagate the toggle and derives the output toggle rate of combinational instances based on that. Since toggle rate is the key factor in power Power will be scaled based on that

ACTIONS

No need to dump the sp template separately since its already dumped automatically by power calculation

#----------

! LNW-034

"<net name>" hasn't visited by state propagtion after inserting propagation Origin

DESCRIPTION

Nets haven't visited in state propagation after inserting propagation origin or Clock nets not visited by state propagation
Please check <adsRpt/propagation.unvisited.edges.gz> for details.

SEVERITY & IMPACT

During the propagation of toggle rate, Probability of TR01 & TR10 will be considered as 0 and TR11 and TR00 are 0.5 respectively So switching power of net will be "0"

ACTIONS

Please check propagation.unvisited.edges.gz for details and incase if you want to assign the toggle rate please include in adsRpt/state_propagation.template

#----------

! PWR-314

Multiple entries for the instance & the first one will be used.

DESCRIPTION

Incase there is multiple entry for same cell in INSTANCE_POWER_FILE we will see this warning

SEVERITY & IMPACT

Since there are multiple entry tool will not take other than first entry.
By default all other entries will be ignored so intended power assignment will not happen

ACTIONS

Please make sure to provide single entry for each cell in INSTANCE_POWER_FILE

#----------
! RPM2AVM-205
Can not find cell '<cell name>' in standard cell.cdev '.apache/cell.cdev'.
Ignored this cell

DESCRIPTION
The cell cdev charcterization is not done or cdev charcterization file will
be missing in the APL FILES section

SEVERITY & IMPACT
Impact of intrinsic capacitance of the cell is not taken into
account for dynamic analysis since cell is ignored without "cell.cdev"
which will affect simulation results.

ACTIONS
Please check the APL_FILES section in GSR and add the "<cell>".cdev

#----------
! LNW-065
Power_Transient has 40 instances of clock toggle rates greater than 2.0.

DESCRIPTION
Maximum toggle expected is 2 for clock nets only.If there is any net or
instance toggle rate more than 2 then Warning will be issued

SEVERITY & IMPACT
Depending on the toggle rate derived from VCD or state propagation switching
and internal power is calculated.
When toggle rate is more than 2 It will be limited 2 since to avoid the
pessimistic power calculation

ACTIONS
No actions required from user

#----------
! PCO-2
DRC Short between VDD and VSS pin

DESCRIPTION
Due to Def issue VDD and VSS pins have got shorted

SEVERITY & IMPACT
Moderate warning. This will be ignored and RedHawk will continue .

ACTIONS
Please correct your Def files

#----------
! NET-162
Net should connect to pin of instance

DESCRIPTION
Net is not connected to any of the pin of the instance

SEVERITY & IMPACT
Major warning . This net will be dropped from RedHawk SignalEM analysis and no current will be allocated to it .

ACTIONS
Please correct your Def files .

#----------
! BPA-010
Region out of bounds

DESCRIPTION
Region defined is out of design bound . BBox defined is incorrect .

SEVERITY & IMPACT
Moderate warning.

ACTIONS
Give proper cordinates for region. The units might be an isssue .

#----------
! ANL-405
Invalid content in current file

DESCRIPTION
Current file is not valid

SEVERITY & IMPACT
Major error, may effect dynamic results.

ACTIONS
Give proper current file for simulation

#----------
! ANL-402
Instance not found

DESCRIPTION
Required instance cannot be found in the user provided cuurent file .

SEVERITY & IMPACT
Moderate warning. The instance will be ignored for the RedHawk analysis .

ACTIONS
Provide a correct current file .

#----------
! ANL-401

Invalid current for instance in EM analysis

DESCRIPTION
Invalid current due to some internal nets not fully connected

SEVERITY & IMPACT
Major warning.

ACTIONS
Check DEF and ensure to have proper connectivity .

#----------
! ANL-400
Invalid current for net in EM analysis

DESCRIPTION
Invalid current due to net is not fully connected

SEVERITY & IMPACT
Major warning.

ACTIONS
Check DEF and ensure to have proper connectivity .
#----------

! SEM-206
Cell doesnot have certain pin
DESCRIPTION
Missing pin defination for the cell in lef file

SEVERITY & IMPACT
Major warning.

ACTIONS
Please correct your Lef files

#----------
! SEM-205
Cell is not found in lib
DESCRIPTION
Missing cell from the lib specified

SEVERITY & IMPACT
Major warning.

ACTIONS
Check your lib file

#----------
! BPA-115
<> is not assigned to ground net in BLOCK_POWER_ASSIGNMENT
DESCRIPTION

The ground net is not provided any power in Block Power Assignment

SEVERITY & IMPACT
Major warning.

ACTIONS
Provide a power number to ground net

#----------
! TSV-002
Die <topdie>: VDD bump is NOT connected to POWER/GROUND
DESCRIPTION
VDD bump is not connected to Power domain .

SEVERITY & IMPACT
Major warning.

ACTIONS
Check TSV file for top die's position and orientation

#----------
! NET-601
Pin of a cell physically touches both net<VDD_INT> and net<VDD>
DESCRIPTION
Pin touches both external and internal domain of power net .

SEVERITY & IMPACT
Major imapact . In extraction it will only be hooked to the former net
RLC network and may cause the latter net's network disconnected!

ACTIONS

#----------
! NET-115
Pin<VDD> of cell<top> has inconsistent type(POWER) as the net<VSS>(GROUND)
DESCRIPTION
Lef pin defination of net is type power but net is ground

SEVERITY & IMPACT
Major imapact .

ACTIONS
Correct the Lef file

#----------
! NET-114
Pin<VDD> of net<VSS> has inconsistent type(POWER) as the net in
cell<top>(GROUND)
DESCRIPTION
Lef pin defination of net is type ground but net is power

SEVERITY & IMPACT

Major imapact .

ACTIONS
Correct the Lef file

#----------
! NET-113
Via of net has inconsistent type(CLOCK) as the net in cell(SIGNAL)
DESCRIPTION
Via of net is type clock but net is signal type .

SEVERITY & IMPACT
Major imapact .

ACTIONS
Correct the Lef file

#----------
! NET-112
Wire of net has inconsistent type(CLOCK) as the net in cell(SIGNAL)
DESCRIPTION
Wire of net is type clock but net is signal type .

SEVERITY & IMPACT
Major imapact .

ACTIONS
Correct the Lef file

#----------
! CMM-117
Automatically creating layer mapping for cell

DESCRIPTION
Due to difference in CMM creation tech file and Top level tech file and if a
layer map file is not provided RedHawk automatically creates a layer mapping
file based on layer index

SEVERITY & IMPACT
RedHawk will generate layer mapping file and continue .

ACTIONS
Layer mapping can be provided using Gsr keyword CMM_LAYER_MAP_FILES
#----------
! SLB-089
"nom_temperature" in .lib is out of the range specified by gsr
"TEMPERATURE_DEVICE".

DESCRIPTION
The GSR keyword TEMPERATURE_DEVICE defaults on with T=TEMPERATURE and delta=5
Syntax: TEMPERATURE_DEVICE <T> [<delta>]
Without TEMPERATURE_DEVICE being specified, RedHawk checks the nominal temperature

(defined by GSR:TEMPERATURE) against
- the value of the APL configuration file keyword "TEMP", and
- the value of the .lib parameter "nom_temperature"

SEVERITY & IMPACT
Major error. The .lib/APL file will be rejected to import with an error message if the temperature of .lib/APL is out of range [T-delta, T+delta].

ACTIONS
Please verify the values of GSR keyword "TEMPERATURE" and .lib parameter "nom_temperature".

#----------
! DEF-112
CELL "<cell_name>" is not defined in LEF or DEF but is referenced by others in DEF

DESCRIPTION
The cell is instantiated in the DEF but its definition is absent in LEF

SEVERITY & IMPACT
Error is reported since the definition for the cell is absent in LEF file.

ACTIONS
Please define the cell in LEF

#----------
! PPI-109
<VDD> net and <GND> net are shorted.

DESCRIPTION
The VDD net and GND net are found to be shorted at some positions whose co-ordinates are provided in the error message.

SEVERITY & IMPACT
Major warning. Inaccurate results are generated.

ACTIONS
Check the connectivity and fix the issues.

#----------
! LEF-085
Neither "MINWIDTH" in Apache Tech file nor "WIDTH" in LEF Tech file are defined.

DESCRIPTION
If the value of the width specified in any of the input files for the routing layers, the tool cannot perform the calculations accurately. MINWIDTH in Apache Tech file and WIDTH in LEF Tech File should be defined.

SEVERITY & IMPACT
Critical warning as this may result in inaccurate calculations.

ACTIONS
Provide the values for width in Apache Tech File and Tech LEF File.

#----------
! DEF-119
No NETS section in this DEF file. The power number may not be reasonable.

DESCRIPTION
Since the DEF File does not have NETS section, there may be connectivity issues.
The nets can be floating and thus the power numbers may be unreliable.

SEVERITY & IMPACT
Warning is generated. May result in inaccurate calulations

ACTIONS
Check the DEF File and provide NETS section.

#----------
! GSR-034
Same keyword is defined twice in the GSR file.

DESCRIPTION
Any GSR keyword should be defined only once in a GSR File. If it is defined
more than once, the previous values will be overridden and latest value is used.

SEVERITY & IMPACT
A warning is generated. Only the latest value is used.

ACTIONS
Check the GSR file and remove the repetitions of keyword definitions.

#----------
! CCL-115
There are instances that are not in the power network.

DESCRIPTION
There are connectivity issues for the instances and they are not connected to the
power network. Such instances are not considerd in the calculation of power and
will result in inaccurate results.

SEVERITY & IMPACT
Minor warning. It warns the user that some instances are not there in the power
network.

ACTIONS
Please refer the file adsRpt/<file_name>.power.unused

#----------
! LEF-053
The RECT's coordinates on layer<metal> used in PIN<name> are converted to (x,y)
co-ordinates.

DESCRIPTION
The rectangular coordinates for a pin should be in the format (llx,lly),(urx,ury) where ll implies lower left and ur implies upper right. The lower left coordinates should always be less than the upper right cordinates. Else RedHawk will swap the coordinates automatically and warning is generated.

SEVERITY & IMPACT
Warning is generated .

ACTIONS
Please check the pin definitions in DEF File.

#----------
! SLB-116
Clock pin '<pin_name>' of comb cell '<cell_name>' is reset as a signal pin.

DESCRIPTION
SLB-116 is issued when there is conflict about clock pin between LEF and LIB. The pin is not declared as clock pin in LEF, but defined as clock pin in LIB. Since in RH flow, LEF is honored, so the pin is reset as signal pin, even it's defined in LIB as clock pin.

SEVERITY & IMPACT
Warning is generated and the clock pin will be reset as signal pin.

ACTIONS
Please declare the pin as clock pin in LEF also.
#----------
! PWR-367
For instance <INSTANCE_NAME> the frequency is inconsistent between STA file and VCD file.

DESCRIPTION
The frequency of an instance in VCD file should be same as that shown under the Timing Window section of the STA File. When RedHawk reads VCD data to select cycles, it uses the clock frequency defined in the STA file to cut cycle frames from VCD

SEVERITY & IMPACT
If both the frequencies are different, a warning is shown.

ACTIONS
To fix the warning, STA_VCD_FREQ_RATIO parameter can be included under the GSR keyword BLOCK_VCD_FILE. If this option is used, VCD data is processed according to the STA frequency and the specified ratio.

#----------
! AP-123
Lib cell missed for instance <instance_name>

DESCRIPTION
LEF file didnot contain the information for the cell

- In RH flow, the order of processing lef/def/lib is
1. LEF : that instance's lef cell doesn't exist
2. LIB : because the lef cell doesn't exist, so we don't import the lib cell with same name
3. DEF : here RH create a lef cell with that name

So from powerstream's perspective, that lib cell doesn't exist

SEVERITY & IMPACT
Error is reported. The lib cell is not imported

ACTIONS
Include the cell definition in LEF file.

#----------
! AP-100
Feedback pins are not converged for 3 iterations

DESCRIPTION
In state propagation feedback methodology is used for converging the toggle rates after many iterations. This message only warns the user that even after 3 iterations of the feedback, the toggle rate value has not converged.

SEVERITY & IMPACT
Minor warning is generated.

ACTIONS
Please make sure all the inputs used in the AVM config file have correct values

#----------
! PWR-376
FRAME_SIZE is obsolete in MIXED-MODE ignored.

DESCRIPTION
The usage of FRAME SIZE in MIXED MODE is no longer made in the current versions and has become obsolete.

SEVERITY & IMPACT
Warning is generated

ACTIONS
Please provide the keywords as per the latest versions. Refer manual.
#----------
! GSR-103
Files under GSR keywords INSTANCE_TOGGLE_RATE_FILES and VCD_FILE are empty

DESCRIPTION
These keywords specify the instance toggle rates file which provide the toggle rates for instances on the chip and the Global Switching Configuration File respectively. If the files are empty it sends out a warning.

SEVERITY & IMPACT

Minor Warning. The values not specified will not be used for calculation and will yield inaccurate results..

ACTIONS
Review the files specified under the GSR keywords.

#----------
! DSG-317
Only 0% instances are recognized as standard cell instances.

DESCRIPTION
As per the latest build only those instances defined with ROWS in DEF File are considered as standard cells. If this keyword is not specified , those instances are ignored and will not be considered as standard cell instances.

SEVERITY & IMPACT
WARNING. There instances are not considered as standard cell instances

ACTIONS
Inside DEF provide the standard cell instances with keyword 'ROWS'

#----------
! ESD-208
ESD_CLAMP_PIN_FILE or ESD_CLAMP_FILE/ESD_SHORT_CLAMP_PIN gsr keyword is not specified for clamp cell/pin shorting.

DESCRIPTION
ESD_CLAMP_FILE allows you to specify one or more ESD clamp cell definition files. When executing 'perform esdcheck', if a clamp file is not specified in the command, RedHawk picks up the clamp definition file from this ESD keyword and executes 'perform esdcheck' on the clamp cells listed in the file.

SEVERITY & IMPACT
WARNING. Without these commands, RedHawk cannot get the clamp files needed to perform esdcheck

ACTIONS
Include the keyword ESD_CLAMP_PIN_FILE or ESD_CLAMP_FILE/ESD_SHORT_CLAMP_PIN in the GSR file.

#----------
! PWR-358
For instance "<inst_name>" the toggle rate is larger than 1.0 after BPFS setting

DESCRIPTION
The maximum value for toggle rate of signal is 1.0. After doing BLOCK_POWER_FOR_SCALING, the toggle rate should not exceed this value 1.0. The power value following this keyword should be provided accordingly in GSR.

SEVERITY & IMPACT
Warning is generated. The power calculation is performed with toggle rate value as 1.0

ACTIONS
Check the GSR file. Provide the power value in BPFS keyword accordingly so that the toggle rate does not exceed 1.0

#----------
! DEF-060
Ignore DEF viaModel<via_name>. There is no CUT metal layer specified for this VIA.

DESCRIPTION
The VIA definition in the DEF should contain the top and bottom metal names and also the cut metal layer dimensions. This warning informs that the cut metal of the via is not defined in the DEF.

SEVERITY & IMPACT
Warning is generated.

ACTIONS
Provide the VIA definitions as per the syntax in the DEF file.

#----------
! DEF-091
DEF block '<cell>' dimension <> does not match the same-named LEF MACRO dimension <>.

DESCRIPTION
This message warns the user that the DEF block dimension does not match the same named LEF MACRO's dimension.

SEVERITY & IMPACT
Warning. RedHawk will ignore LEF definition for the block.

ACTIONS
The dimension of a block in DEF & LEF should be matched.

#----------
! CMM-120
Automatically creating layer mapping for cells.

DESCRIPTION
Automatically creating layer mapping by tech stack order or layer height if CMM_LAYER_MAP_FILES option is not used in the gsr.

SEVERITY & IMPACT
Warning generated while importing CMM/ERV cell models. If neither a layer map file nor AUTO is specified for a block, its layers are merged automatically based on layer height.

ACTIONS
Include keyword CMM_LAYER_MAP_FILES in the GSR file:
Syntax:
CMM_LAYER_MAP_FILES

```
   {
    <CMM_cellname> [<filename> | AUTO | UNUSED]
            ...
   }

#-------------------------------------------------------------------------------
! PWR-359
   For an instance '<name>' clock toggle rate is larger than 2.0
   after BPFS setting for a clock-network instance.

DESCRIPTION
   BPFS keyword by default scales  the signal toggle rate and clock toggle rate
   uniformly to reach specified total power consumption. This warning lists the
   instances with clock toggle rate scaled by more than 2.0.

SEVERITY & IMPACT
   Warning. User specified power number will be used for analysis.

ACTIONS
   Use keyword SCALE_CLOCK_POWER 0 to avoid clock net scaling.
   Otherwise change the BPFS value of that instance.

#-------------------------------------------------------------------------------
! ITG-025
   There are '<count>' cells having inconsistent multistate and APL
   specifications.

DESCRIPTION
   The reported cells with inconsistent multistate and APL specification
   are listed in < adsRpt/apache.refCell.msAplErr >. Check whether APL is
   there corresponding to GSC states of the cell .

SEVERITY & IMPACT
   Warning. It reduces the accuracy of run. If GSR keyword
   IGNORE_MULTISTATE_ERROR is true, the instances of these cells
   will be disabled.

ACTIONS
   The following actions can be done :
   (1) Make sure that there is an  APL curresponding to each cell in GSC file.
   (2) Make sure that every cell in GSC has corresponding state in APL.
   (3) To ignore this warning 'IGNORE_MULTISTATE_ERROR 1' keyword is used in GSR,
      which disables that particular instance having error.

#-------------------------------------------------------------------------------
! TEC-175
   Width based EM rule for via layer '<name>' does not specify width or area.

DESCRIPTION
   The reported vias have width or area information in tech file. The warning
   is shown when the via area is missing in the tech file.
```

SEVERITY & IMPACT
Warning. This shows the modifications of the tech file required.

ACTIONS
Include the area information in the tech file for the via showing the error.

#----------
! SP-317
Clock roots special toggle rate 0.0 defined by user: '<clock_net_name>'

DESCRIPTION
This warning is issued when the clock roots has a user defined toggle rate from VCD,SAIF, NET_TOGGLE_RATE other than the default toggle rate from GSR.

SEVERITY & IMPACT
Warning.This makes the user honors the setting on this clock root. It does not take others action.

ACTIONS
No action required.

#----------
! PWR-364
For block '<name>' the sub blocks have been scaled or covered by VCD, BPFS cannot be met, ignored.

DESCRIPTION
If VCD and BPFS comes toghether, higher priority should go to the VCD based power calculation. Thus BPFS cannot be met and ignored.

SEVERITY & IMPACT
Warning. Which increases accuracy of the result.

ACTIONS
BPFS should be avoided for blocks scaled or covered by VCD.

#----------
! AP-107
Net '<name>' has frequency 0.0, toggle counts tc01 <> tc10 <>,reset frequency to <>.

DESCRIPTION
If the frequency in STA file is 0.0 and it toggles in VCD this warning generated. The frequency is reset.

SEVERITY & IMPACT
Warning. This warning resets the frequency to '<value>'.

ACTIONS
The STA and VCD file should be synchronized. Use the keyword 'FREQUENCY_OF_MISSING_INSTANCES' to assign reset value of

frequency.

#----------
! EVP-152
Unhandled VCD variable type at line <>, ignored.

DESCRIPTION
If the variable inside vcd is real, it is unhandled and ignored while
building signal hierarchy

SEVERITY & IMPACT
Warning. The variable unhandled and ignored.

ACTIONS
Check with the front end team.

#----------
! SPP-029
Package Node <> is floating

DESCRIPTION
The nod in the package is floating.

SEVERITY & IMPACT
Warning, This indicates a floating node in the package subckt
and it reduces the accuracy.

ACTIONS
Check with the package design team and correct the floating node.

#----------
! AMG-218
Temperature '<>' in file '<APL_file_name>' is different from '<>' in
file '.apache/cell.cdev'

DESCRIPTION
This is caused by the difference of temperature in APL files and in
gsr.

SEVERITY & IMPACT
Warning. Shows the user whether any difference in temperature is there.
Reduces the accuracy of results.

ACTIONS
Check with the APL generation team whether the GSR and APL are matching.

#----------
! SIM-04W
Found short between nodes in different P/G domains.

DESCRIPTION
This warning is generated during analysis stage of RedHawk run when

some shorts between nodes in different P/G domain was seen.

SEVERITY & IMPACT
Warning. Incorrect analysis.

ACTIONS
Check the power grid for shorts and use FAO for fixing it if required.

#----------
! PWR-222
Clock pin toggle rate of '<count>' instances are greater than 2.0

DESCRIPTION
This warning gives the number of clock pins having toggle rate greater than 2.0.

SEVERITY & IMPACT
Warning. In some condition clock toggle rate may be larger than toggle rate. The reason may be of higher clock net frequency compared to instance frequency.

ACTIONS
Check for the frequency mismatches in STA and VCD file.

#----------
! SIM-324
Leakage power of cell '<name>' '<leakage power>' is larger than 0.001...

DESCRIPTION
This warning is created when during simulation stage when the lekage power is greater than 1mW.

SEVERITY & IMPACT
Warning. This makes the user recheck the the cell properties if such a leakage is un expected. This doesn't make changes in the simulation.

ACTIONS
Check whether the lekage is expected for that cell.

#----------
! ACK-249
No interpolation is performed on '<filename>', since none of the cells in the design is found in the APL file.

DESCRIPTION
This warning is given when APL is contains cell which are not present in the design. The '<adsRpt/apache.refCell.noAPL*>' show cells in the design which have missing APL data. Include the APL files which contains the missing cells.

SEVERITY & IMPACT
Warning. The APL having which none of the cells are in the design are

avoided from interpolation.

ACTIONS
Check the APL file mentioned in the warning using aplreader and analys
why non of the cells are present in the design.

#----------
! SIM-447
Cell '<name>' has negative charge of pin VDD for output switching
(the <> th sample).

DESCRIPTION
This alerts the user if any of the pin has negative charge. It can
be due to an APL error.

SEVERITY & IMPACT
Warning. This makes inaccurate result in simulation stage.

ACTIONS
Check the APL file and correct the files having error.

#----------
! MOD-007
Ignore POLYNOMIAL_BASED_THICKNESS_VARIATION effect in resistance calculation.

DESCRIPTION
The thickness variation is > +- 25% of the base thickness of metal '<M>'.
Then value of thickness variation will be clipped to 25%.

SEVERITY & IMPACT
The warning is introduced to reduce the inaccuracy.

ACTIONS
User can investigate on thickness variation in the specified region.

#----------
! SIM-613
Ignoring custom state for cell '<name>'.

DESCRIPTION
Ignoring custom state for cell because it has no pg-arc definition in lib.

SEVERITY & IMPACT
The warning generated during dynamic simulation.

ACTIONS
Make sure the pg-arc definition for all cells defined in .lib/custom.lib.

#----------
! SLB-072
Undefined signal level '<power_pin>'.

DESCRIPTION
Power pin specified for related_power_pin is not defined in the LIB.

SEVERITY & IMPACT
The Error generated during import lib by PowerStream. The cell with multiple power pins, output pin switching power may not added to the correct power net in this case.

ACTIONS
Verify voltage_map() construct is present in .lib for the specified pg pin.

#----------
! SIM-443
Unbalanced pin power of APL '<cell>'.

DESCRIPTION
Vdd/vss pin power of APL cell is unbalanced.

SEVERITY & IMPACT
It's issued when total_vdd_pins_charge > 2* total_vss_pins_charge (or the other way) for certain APL cell. This may lead to unbalanced current source between power and ground.

ACTIONS
User should be aware of this and do sanity check on APL characterization and cell pins.

#----------
! SIM-32E
APL for cell '<cellname>' is ignored.

DESCRIPTION
APL for cell is ignored due to inconsistent pgarc definition.

SEVERITY & IMPACT
The warning is generated during simulation stage. APL for this cell ignored during simualtion may cause accuracy of result.

ACTIONS
Make sure that apl current profiles are present for all the cell's pins that are mentioned in the .lib/custom.lib

#----------
! ANL-033
Negative current value in VDD pad '<pad>'.

DESCRIPTION
The VDD pad has negative current value during simulation.

SEVERITY & IMPACT
The warning is generated during simulation.

ACTIONS
Check sanity of package.

#----------
! SIM-372A
Inconsistent cell type from RH and APL for cell '<cell>'.

DESCRIPTION
If APL cell definition doesn't provide standby-trig or ntrig (C00, C11) current profiles for flip-flop or memory, then ASIM will identify as this as a combination one. So the RH and APL cell type conflict each other.

SEVERITY & IMPACT
This warning is generated during simualtion stage, which confuse the simulation and may lead to unexpected result.

ACTIONS
Check if the imported apl is from the same design/version/flow as RH run and make sure APL current profile should provide standy-trig or ntrig for flip-flop and memory cells.

#----------
! CCL-076
Inconsistent P/G current for instance '<instance>'.

DESCRIPTION
Inconsistent P/G current for instance('<instance>') in the file '<adsPower/<design>.nets.adsLib.power.gz>'.

SEVERITY & IMPACT
This warning is generated after power calcualtion. The power pins current is not consistent with the ground pins. This may affect result.

ACTIONS
Check the instance pins current is consistent.

#----------
! SIM-686
Two power domains are shorted at one node.

DESCRIPTION
The two power domains '<VDD_1>' and '<VDD>' are shorted at node '<node>'.

SEVERITY & IMPACT
This ERROR happens during the simulation, the shorted domain may result in unrealistic results.

ACTIONS
Rectify the short in the design.

#----------
! SIM-444

Instance '<instance>' cycle is less than the current profile duration.

DESCRIPTION
The duration of APL current is larger than the instance real cycle applied in design(i.e, faster frequency).

SEVERITY & IMPACT
The warning during simulation, the current profile of APL cannot be fully applied in the simulation; which may cause unreliable result(i.e, miss the peak current; power target cannot be met, etc).

ACTIONS
User can verify the instance frequency higher than the apl waveform duration.

#----------
! DEF-088
Layer '<via>' is not routing layer.

DESCRIPTION
In DEF for regular wiring statement specified for nets, the specified layer on which wire lies must be metal layer, but reference to a cut layer generates an error.

SEVERITY & IMPACT
Error during DEF reading. RH will discard this wire path definition and continue running.

ACTIONS
Do sanity check on DEF and correct DEF syntax.
#----------
! PAD-009
Multiple definition of same ploc name in '<Ploc file name>' at lines 'line number1, line number2....'

DESCRIPTION
There are two or more plocs defined in ploc file which have the same name.

SEVERITY & IMPACT
Error. Ploc will be placed only at the location specified by first definition.

ACTIONS
Use different ploc names for each definition.

#----------
! PIN-016
Pin '<Pin name>' of cell '<cell name>' has been simplified because the number of rect statements is more than 5000.

DESCRIPTION
If a pin has more than 5000 rect geometries, then RH will discard pin

geometries which have area less than 0.05 um2.

SEVERITY & IMPACT
Error. Very small difference in simulation results

ACTIONS
To turn-off this behavior, use the keyword COMPLEX_PIN_MAX_RECT_NUM 0 in gsr

#----------
! GSR-012
Error at or before <GSR FILE>:<Line Number>

DESCRIPTION
This error message appears when there is a wrong entry in GSR file
It also points the line number where the error occurred

SEVERITY & IMPACT
Critical Error, RedHawk run will terminate

ACTIONS
Go to the line number specified and correct the GSR file

#----------

! STA-115
GSR BLOCK_STA_FILE definition syntax error. No top level STA defined.

DESCRIPTION
BLOCK_STA_FILE allows hierarchical STA files to be used for analysis.
This warning appears when top level STA is not defined in BLOCK_STA_FILES
section in GSR.

SEVERITY & IMPACT
Warning message - RedHawk will continue without the top level STA file.
Instances in the top level will not have STA coverage.

ACTIONS
Provide top level STA file

#----------
! GSC-019
END_EXACT_INSTANCE_NAME_MATCH is expected before the end of file

DESCRIPTION
EXACT_INSTANCE_NAME_MATCH in GSC file instructs RH to do faster and
direct name mapping. So If you specify EXACT_INSTANCE_NAME_MATCH as
a header in your GSC file, tool expects END_EXACT_INSTANCE_NAME_MATCH
before the end of the file

SEVERITY & IMPACT
Non Critical warning - RedHawk will continue analysis

ACTIONS
Provide the END_EXACT_INSTANCE_NAME_MATCH appropriately in your GSC file.
#----------
! DEF-097
Ignore invalid ViaModel '<via model>' since bottom layer '<>', cut layer '<>', and top layer '<>' can't constitute a legal viaModel.

DESCRIPTION
The VIA defined in tech file might be between different pair of metals than one defined in def.

SEVERITY & IMPACT

ACTIONS
Please check the consistancy of VIA definition in tech file and DEF

#----------
! GSR-001
Unrecognised or absolete keyword.

DESCRIPTION
The keyword used in the GSR file may be erroneous or absolete for the current version of RedHawk.

SEVERITY & IMPACT
This is a major error.The GSR file parsing will be aborted.

ACTIONS
Please correct the keyword or refer manual for latest keywords
#----------

! NCU-002
Net <net_1> is shorted to net <net_2> in i'th psu by wire_and_pin_instance at (x1 ,y1 <metal>)

DESCRIPTION
This warning is flagged when there is a short between a net and a pin of an instance.

SEVERITY & IMPACT
It is a major warning.The instance who's pin is shorted will not be included for analysis.Hence can impact the results.

ACTIONS
This may be occuring due to issues in DEF or the nets might not be defined as equivalent nets in GSR if that is intended.

#----------
! SLB-117
The custom cell "<cell name>" is not referenced in the design.

DESCRIPTION

The cell name defined in the custom lib file is not defined in
DEF/LEF/LIB

SEVERITY & IMPACT
It is a major warning.The instance will not be included in the analysis.

ACTIONS
Make sure that the cell name defined in custom lib is present in
the DEF/LEF/LIB
#----------

! CMM-107
net '<net name>' is skipped in CMM model generation.

DESCRIPTION
The nets will be skipped in generated CMM model.

SEVERITY & IMPACT
It is major error.The net will not be considered for analysis and
hence not used in the generated CMM model.

ACTIONS
Add the net names in the vDD_NETS section of the GSR file to include
the nets in the CMM model generation.

#----------

! AMG-219
Nominal Vdd value '<voltage value>' in file '<file name1>' is different
from '<voltage value>' in file '<file name2>'

DESCRIPTION
This is flagged when the voltage value used in GSR is different from
what is used for cdev generation.

SEVERITY & IMPACT
It is a warning message.

ACTIONS
Make sure that the voltage value is consistant for analysis in RedHawk
and cdev generation.
#----------
! SIM-350
PWLCAP is missing

DESCRIPTION
PWLCAP is missing for standard cells.Please refer
adsRpt/apache.refCell.noAplPwcap for details.

SEVERITY & IMPACT
Warning

ACTIONS
Include missing PWLCAP cells in APL_FILES section in gsr.

#----------
! AMG-220
Corners used for generating cdev are not consistent

DESCRIPTION
Spice files used for generating cdev have different corners and while RH merges these cdev's it finds inconsistency .

SEVERITY & IMPACT
Warning

ACTIONS
Ensure that the spice files used for cdev generation are consistent for all cells in the design. Use IGNORE_APL_PROCESS_CORNER 1 for runs to proceed

#----------

! VIA-614
NO corresponding EM_POLYNOMIAL for VIA

DESCRIPTION
Custom via whose EM_POLYNOMIAL defn are not defined in APACHE Tech life.

SEVERITY & IMPACT
Critical Error.EM analysis will not be accurate

ACTIONS
Align with Foundary to get the EM_POLYNOMIAL for this custom via and update the tech file.

#----------
! NCU-001
Shorts seen in design

DESCRIPTION

Short seen in PDN n/w 2 PG nets

SEVERITY & IMPACT
Error.Shorted nets segments will be ignored for simulation.

ACTIONS
Resolve shorts seen in the design.
#----------

! SIM-05W
Nodes disconnected to all P/G domains

DESCRIPTION

PG network is disconnected.

SEVERITY & IMPACT
Error

ACTIONS
Resolve all power opens seen in the design.Ensure that PDN is connected to respective plocs

#----------
! SIM-355
Switches missing timing window.

DESCRIPTION
STA fle does not include timing window for the reported switches.

SEVERITY AND IMPACT
Major warning. Rampup analysis will not be performed for the domain whose switches do not have timing window.

ACTION
Use STA file with better coverage or include the missing switches during STA file generation.
#----------

! CPM-IO-415
IO cells not included for CPM generation.

DESCRIPTION
While generating CPM using "perform powermodel", when -io option is not specified IO cells will be ignored for CPM generation.

SEVERITY AND IMPACT
Not a critical warning. IO cell PG geometries and current characterization are not considered during the CPM generation.

ACTION
Include -io switch with "perform powermodel" command during CPM generation.
#----------
! STA-118
Memory output pin is defined as part of clock network in STA file.

DESCRIPTION
Some output pin of memory is defined as part of clock network in STA file. But, Memory instance can't be treated as a clock instance. So, RH will ignore this info in STA file.

SEVERITY & IMPACT
WARNING. Can have impact on power calculation and average activity.

ACTIONS
Please verify if the celltype definition in lib/custom lib and output pin info in sta file are correct.

#----------
! PWR-348
Instance pin <pin_name> is connected to net <net_name> but they are mapped to different VCD symbols. Events from VCD symbol mapping to the instance pin are ignored and events from VCD symbol mapping to the net are used.

DESCRIPTION
This can be due to some VCD mapping issues because of which the connected instance pin and net are mapped to different VCD symbols. This is internally dealt such that the events from VCD symbol mapping to the instance pin is ignored and events from VCD symbol mapping to the net are used.

SEVERITY & IMPACT
Warning is generated

ACTIONS
Check the VCD file for symbol mapping issues.
#----------

! PWR-221
Signal pin toggle rate of <count> non-clock network instances are greater than 1.0 and <count> clock network instaces are greater than 2.0

DESCRIPTION
This warning gives the number of signal pin toggle rate of non-clock network instances greater than 1.0 and the number of clock network instances greater than 2.0

SEVERITY & IMPACT
Warning is generated. This may be because of any BLOCK_POWER_FOR_SCALING keyword setting in GSR

ACTIONS
Check for frequency mismatches in STA and VCD file. Also check the BPFS setting.

#----------
! DEF-096
The value of cut spacing (X Y) plus cut size (X Y) defined in DEF viaModel <model_name> doesn't match the value of cut spacing (X Y) defined in the specified LEF VIARULE <VIAGENTOP>.

DESCRIPTION
There is mismatch between the via cut size specified in DEF viamodel and that specified in the LEF viarule.

SEVERITY & IMPACT
Warning is issued.

ACTIONS
Please turn on GSR keyword "USE_DEF_VIARULE" if DEF definition is needed.

#----------
! PAD-083
Pad pin <PIN_NAME> is not connected to any wire in the design. Please check the pad file.

DESCRIPTION
This error says that the specified pad pin is not connected to any wire in the design. This error is generated in the extraction stage while building connectivity.

SEVERITY & IMPACT
Error is reported

ACTIONS
Please check the pad file

#----------